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Platform For Analyzing and Managing Collaborations Between Advertisers and Content Creators

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Dedications

To my beloved parents and siblings,

No words can truly express the depth of love, gratitude, and respect I hold for you.
Your endless support, kindness, and sacrifices have guided me through every step of this journey.
May God always bless and protect you.

To my beloved grandmother, Amity,

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I love you more than words can express, and I will always carry your love with me.

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my path. Thank you.

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With all my heart,

Salah

Dedications

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General Introduction

In recent years, the explosion of social media platforms like Instagram, TikTok, and YouTube has transformed the way brands connect with their audiences. At the center of this shift is **User-Generated Content (UGC)**, which offers a level of authenticity and engagement that traditional advertising often lacks. In response, companies are increasingly turning to content creators to amplify their message and build stronger connections with consumers.

However, managing these collaborations, identifying the right creators, coordinating content, and measuring impact, can be complex and time consuming. Despite the growing importance of influencer marketing, many advertisers still lack the tools needed to effectively organize and analyze their partnerships with creators.

To address this gap, our end-of-studies project, developed during our internship at **ACCESS**, introduces **Colab**, a smart full-scale web platform that reinvents the way advertisers and creators collaborate. Powered by a ReactJS frontend and a NodeJS/ExpressJS backend, Colab integrates web scraping techniques to discover and track creator performance in real time, and leverages OpenAI's natural language processing to auto generate campaign briefs, summarize performance metrics, and recommend ideal creator matches based on brand goals.

By combining automation, data intelligence, and a clean centralized workflow, Colab eliminates manual overhead and enables brands to launch, manage, and analyze influencer campaigns with unprecedented efficiency and insight. It is more than a tool; it is a smarter way to collaborate in the creator economy.

The present report is structured into five chapters. The description of each chapter is as follows.

- **Chapter 1 : General Context** introduces the host organization, analyzes current practices and challenges in the management of influencer campaigns, reviews existing market solutions, and presents the proposed platform and development methodology adopted for this project.
- **Chapter 2 : Analysis and Specification of Requirements** defines the functional and non-functional requirements of the system and describes the adopted technical architecture.
- **Chapter 3 : Sprint 1 — User Onboarding & Profile Management** covers the development of foundational features such as authentication, profile handling, and basic admin tools.
- **Chapter 4 : Sprint 2 — Brief Lifecycle & Platform Coordination** presents the implementation of brief creation and management, brand dashboards, and admin coordination.
- **Chapter 5 : Sprint 3 — Campaign Execution & Performance Tracking** details campaign tools, creator scoring, notifications, and creator-facing dashboards.

Finally, the report concludes with a general conclusion summarizing the outcomes, reflecting on the initial objectives, and suggesting potential improvements and future extensions.

Chapter 1 : General Context

1 Introduction

This chapter serves to introduce the general context of the project. It begins with a presentation of the host company where the internship was carried out. Then, it highlights the main problem that the project aims to solve, along with an overview of current solutions and the innovative approach proposed. Finally, the methodology followed throughout the different stages of the project will be detailed.

2 Host Organization

2.1 Presentation of the Host Company

Access [1] Group is a global technology and consulting company specializing in digital transformation, outsourcing, and technological innovation. With over 25 years of cumulative experience, a workforce of more than 120 skilled professionals and a portfolio of over 1,000 partners, the company has established a strong international footprint across Canada, France, Tunisia, Mauritania, Saudi Arabia and Dubai.

Access is known for its customer-centric approach, which utilizes the latest technologies in AI, data analytics, and software engineering to deliver high-impact business solutions.



Figure 1: Access Group Global Presence and Organizational Structure

2.2 Activities

Access Group operates across several domains, with a particular focus on digital innovation. Its main activities include :

- **Software development** : Custom solutions for web and mobile platforms using modern tech stacks.
 - **Microsoft cloud services** : Implementation and support for Dynamics 365, Azure AI, and Power Platform.
 - **Artificial Intelligence & Data** : Advanced analytics, machine learning, and data engineering.
 - **Digital marketing** : A 360° marketing agency supporting brands in visibility and branding.
 - **IT outsourcing** : Providing businesses with expert development and operations teams.
- Providing expert development teams, ACCESS also supports sustainability, diversity, and employee well-being through CSR efforts like hybrid work, community support, and professional growth.

3 Study of the Existing Situation

3.1 Current Approach

At Access International, the influencer campaign management process follows a traditional, manual workflow handled by a small, dedicated internal team. The workflow begins when an advertiser submits a campaign request. This request may arrive via email, phone, or in some cases, informal verbal communication. The advertiser usually provides general details, such as campaign objectives, target audience, desired influencer profile, budget, and timeline.

Once the request is received, the internal team begins by interpreting the advertiser's goals and manually translating them into a set of informal selection criteria. The team then relies on past experiences, personal contact lists, and basic online searches (e.g., Instagram, TikTok) to compile a list of potential influencers that match the campaign profile.

The influencer outreach phase is also fully manual. Team members typically contact each influencer individually by phone, WhatsApp, or direct message on social media platforms. During these conversations, they present the campaign details, answer questions, negotiate compensation if necessary, and assess the influencer's interest and availability.

If an influencer agrees to participate, the team continues to manage the exchange manually — confirming deliverables, timelines, and basic expectations. In some cases, media files (e.g., photos, captions) are shared via email or cloud storage links. The campaign status for each influencer is tracked using simple spreadsheets or notes.

3.2 Limitations and Challenges

While Access International’s current manual approach ensures a personal touch and can be sufficient for small-scale influencer campaigns, it presents several major limitations in scalability, efficiency, and performance analysis.

Key challenges include :

- **Lack of automation** : Influencer identification and outreach are entirely manual, consuming valuable time and resources.
- **No centralized database**: Influencer details, campaign briefs, and performance results are scattered across emails, spreadsheets, and personal notes.
- **Limited tracking** : The absence of structured performance monitoring makes it difficult to evaluate campaign effectiveness or apply insights to future efforts.
- **Scalability issues** : As the number of campaigns and creators increases, the current approach becomes impractical and prone to delays or errors.
- **No scoring or ranking system** : There is no mechanism to evaluate creators based on objective performance metrics, making it difficult to prioritize high-performing or relevant influencers.

To assess our platform’s position in the market, we reviewed commonly used tools for influencer campaign management. The table below highlights the main strengths and weaknesses of both digital platforms and manual methods.

Platform	Strengths	Weaknesses
Upfluence[2] / Heepsy [3]	• Large influencer database • Filtering by niche • Campaign tracking tools	• High subscription cost • Limited customization • No AI-powered scoring
CreatorIQ [4] / Tagger [5]	• Strong analytics • Reporting dashboards • Brand collaboration features	• Complex interface • Requires training • Costly for small agencies
Tawa Digital [6]	• Tunisian integrated influencer management • Cross-platform support • User-friendly interface	• Newer platform with smaller user base • Limited third-party integrations • Pricing model unclear

Table 1: Platform Comparative Analysis

3.3 Proposed Solution

We propose a smart and user-friendly platform to improve how influencer marketing campaigns are managed. Right now, brands often face challenges such as spending too much time finding the right influencers, keeping track of ongoing campaigns, and evaluating results. Our platform brings all key steps into one place, making the whole process faster, clearer, and more organized.

The platform allows brands to easily create **briefs**, which are announcements that describe the idea, goals, and requirements of the campaign. Creators can then view these briefs, apply and submit their **campaigns**, which include the content they produce based on the brand's request. This system helps both brands and creators stay aligned and reduces confusion throughout the process.

To support better decision-making, the platform gives each creator a general score based on both their social media activity and the quality of their past content. While the scoring system works in the background, the main focus for users is simplicity—brands can quickly find relevant creators, track which campaigns are live, and monitor performance as it happens.

Overall, the platform saves time, reduces manual work, and improves teamwork between brands, creators, and platform managers. It helps everyone move from planning to results in a much smoother and more professional way.

4 Agile Methodology

The Agile software development methodology is one of the most effective approaches for transforming a business vision into functional software solutions. Agile emphasizes iterative development, flexibility, and continuous improvement, which makes it ideal for dynamic environments like influencer marketing platforms. It encourages rapid delivery of small, working increments and a flexible response to change throughout the project lifecycle.

Agile is based on four core values, which guide the development process:

- Individuals and interactions over processes and tools
- Working software over comprehensive documentation
- Customer collaboration over contract negotiation
- Responding to change over following a plan

4.1 Adopting SCRUM for Agile Project Development

Choosing Agile is not enough on its own; there are various methodologies within Agile, including Extreme Programming (XP), Rapid Application Development (RAD), and Scrum. For this project, we have chosen the Scrum methodology[7], which is one of the most widely adopted Agile frameworks. Scrum is particularly effective for managing complex projects, as it provides simple and

structured solutions to handle difficult tasks.

Scrum focuses on iterative cycles known as sprints, with each sprint resulting in a potentially shippable product increment. This approach allows the team to adapt to changes, avoid unnecessary complexity, and ensure transparency throughout the development process. Scrum's main goal is to help teams focus on key project aspects such as quality, performance, and usability, with regular feedback loops to ensure continuous improvement.

4.2 Identification of the Scrum Team

Scrum assigns clear roles to stakeholders to ensure effective collaboration and accountability :

- **Product Owner** : Represents stakeholders, prioritizes the Product Backlog.
- **Scrum Master** : Facilitates Scrum processes, removes obstacles, coaches the team.
- **Development Team** : Technical experts who build and deliver product increments.

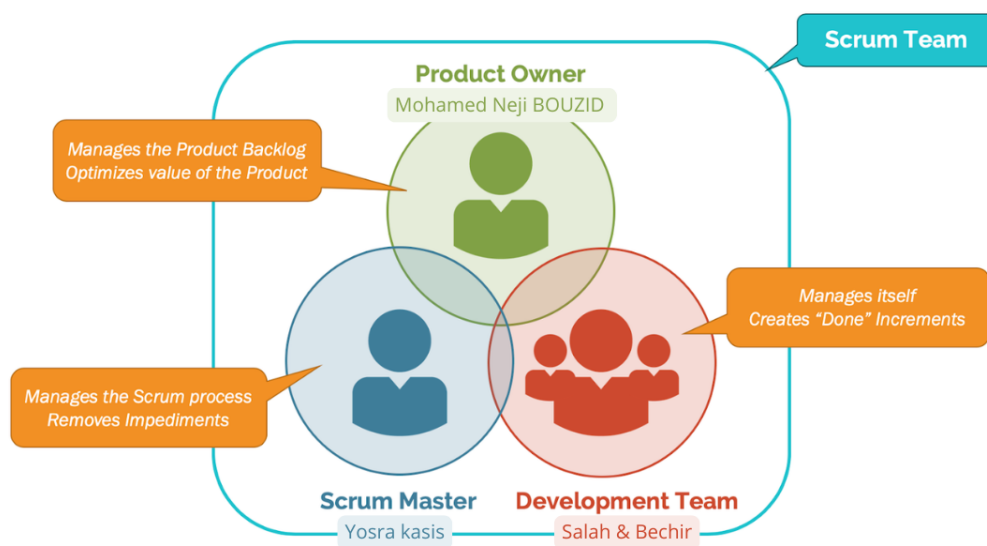


Figure 2: Scrum Team and Roles

4.3 Scrum Events

Scrum is structured around specific events that help the team plan, track, and review the progress of the project :

- Sprint Planning
- Daily Stand-up Meeting
- Sprint Review
- Sprint Retrospective

4.4 Scrum Lifecycle

The Scrum process follows an iterative lifecycle, where each sprint produces a functional product increment. This approach helps ensure that the final product aligns with user needs, maximizing the value delivered at each stage.

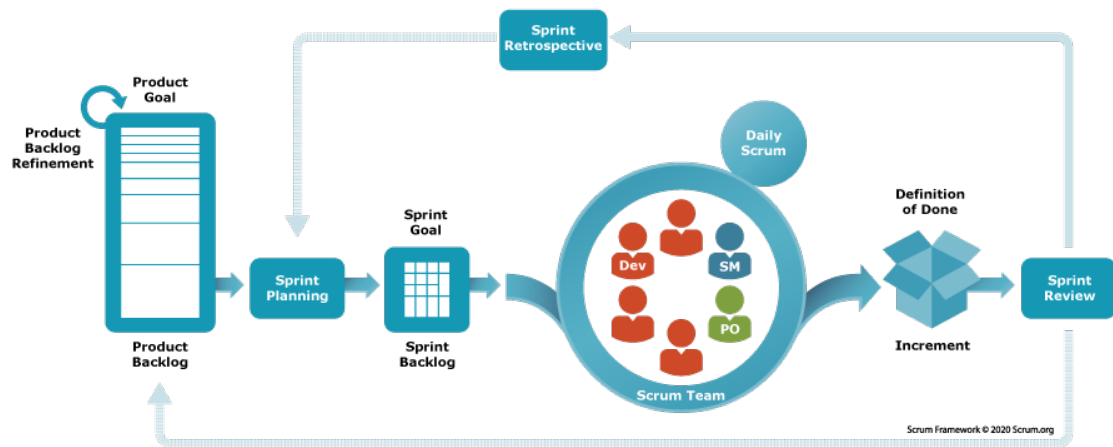


Figure 3: Sprint lifecycle in SCRUM

5 Modeling language "UML"

In this project, we use UML (Unified Modeling Language) [8] as the primary modeling tool. UML is both a visual and textual language created with the aim of understanding, specifying, and documenting all the requirements of a system.



Figure 4: Unified Modeling Language logo

6 Conclusion

The general context and market analysis highlight the need for a centralized influencer campaign management platform. The proposed Colab solution, developed using Agile methodology, offers a streamlined, data-driven approach to improve collaboration, performance tracking, and scalability.

Chapter 2 : Analysis and Specification of Requirements

1 Introduction

This chapter delves into the detailed requirements for the Colab platform. It covers the identification of actors, both functional and non-functional requirements, as well as the technical specifications needed for the project. In addition, it outlines the architecture of the system and the technologies used for development. The chapter also explains the approach to project management, particularly focusing on the SCRUM methodology, product backlog creation, and sprint planning.

2 Specifications of Requirements

2.1 Identification of Actors

The Colab Platform involves multiple types of actors, each interacting with the system based on their role. These actors include both primary users who directly use the platform and secondary actors that enable internal processes.

- **Guest** : A guest is any unauthenticated visitor who can access the home page and initiate account creation. Upon registration, guests transition into authenticated users (Brand or Creator).
- **User (Brand / Creator)** : This is a general role that applies to all authenticated users. It is divided into two specific types :
 - **Brand (Advertiser)** : Responsible for creating briefs, consulting creator profiles, and validating or rejecting submitted campaigns. Brands also access dashboards to monitor campaign performance.
 - **Creator (Content Creator)** : Interacts with briefs by expressing interest or applying, manages their own campaigns, and tracks performance through the scoring system and creator dashboard.
- **Administrator** : Admins ensure the platform operates smoothly. Their responsibilities include managing users, moderating briefs and campaigns, updating statuses, and accessing global statistics via the admin dashboard.
- **Secondary Actor – Scoring System** : This is an automated module that retrieves and dis-

plays creator's social media performance scores (e.g., from Instagram and TikTok). It supports brands in making data-driven decisions during creator selection and campaign evaluation.

2.2 Functional Requirements

Functional requirements define the essential functionalities that the Colab Platform must provide in order to support the goals of its users. These include actions performed by primary actors : guests, brands, creators, and administrators, as well as interactions with secondary system components such as the scoring module.

2.2.1 Guest

Guests are unauthenticated users who interact with the platform before registering. Their capabilities include :

- **Visit Homepage** : Access the public-facing landing page.
- **Create Account** : Register as a Creator or Brand using a sign-up form.
- **Email Verification** : Complete account activation via verification link.

2.2.2 Authenticated User (Brand / Creator)

Once registered and verified, users become either Brands or Creators. They share common functions but also have role-specific capabilities.

Common Functionalities

- **Login / Logout** : Secure session management.
- **Manage Profile** : Edit and update personal or brand details.
- **Receive Notifications** : Stay informed through platform alerts.
- **Access Dashboard** : View activity summaries, brief/campaign status, and performance.

Brand-Specific Functionalities

- **Create and Manage Briefs** : Submit, edit, delete, and view briefs.
- **Search for Creators** : Filter and browse creator profiles based on scoring and criteria.
- **Review Campaigns** : Consult submitted campaigns related to their briefs.
- **Accept/Reject Campaigns** : Validate or reject proposed campaigns.
- **View Brief Performance** : Analyze brief dashboard.

Creator-Specific Functionalities

- **Consult Available Briefs** : Browse briefs created by brands.
- **Express Interest / Apply** : Show interest and submit campaign proposals.
- **Manage Campaigns** : Create, update, and delete campaigns.
- **View Scoring Metrics** : Track social media performance scores.
- **Access Campaign Dashboard**: View real-time performance data.

2.2.3 Administrator

Administrators are responsible for maintaining the system and moderating user-generated content. Their functions include:

- **Manage Users** : Verify, suspend, or delete user accounts.
- **Moderate Briefs and Campaigns** : Validate or reject submissions.
- **Manage Categories** : Create and organize categories for Users or briefs.
- **Consult Platform Statistics** : View high-level performance and usage metrics.

2.2.4 Secondary System Functionality – Scoring Module

The scoring system is an automated service used to assess creator performance :

- **Fetch External Scores** : Aggregate data (scrape) from platforms like Instagram and TikTok.
- **Display Creator Metrics** : Show scoring results to brands and creators.
- **Support Selection Process** : Assist brands in evaluating potential collaborators.

3 Non-Functional Requirements

To support a smooth and secure experience for all users, including brands, creators, and platform managers, our app is built with the following nonfunctional features :

- **Security** : The app uses JSON Web Tokens (JWT) for secure, stateless authentication. When users log in, the backend (Node.js/Express) generates a signed token with user info and permissions. This token is sent to the frontend (React) and included in future requests to access protected routes.

The backend verifies each token and checks user roles (brand, creator, or manager) to control access. All data transfers occur over HTTPS, and tokens are stored securely on the client (HTTP-only cookies or local storage).

User data in MongoDB is encrypted, and passwords are hashed using bcrypt. The system uses email verification, token expiration, and refresh tokens to guard against unauthorized access and session hijacking.

- **Email Verification :** Both creators and brands must confirm their email before using the platform. This helps prevent fake accounts and keeps the platform secure.
- **User-Friendly Interface :** The platform is simple to use for everyone. Brands can easily create briefs, and creators can quickly respond with their campaign content, all in a clear and well-organized system.
- **Fast Performance :** Key actions like posting briefs, submitting campaigns, or checking campaign status are quick, with responses typically in under 2 seconds.
- **Data Backup :** All campaign data and user profiles are regularly backed up. This ensures that important brand or creator content is never lost, even if a technical issue occurs.
- **Reliable Access :** The platform works smoothly on desktop computers and is designed to stay available without interruptions, even when many users are active at the same time.
- **Privacy Protection :** The system respects user privacy. Brand and creator information is kept confidential and only shared when needed for campaign collaboration.
- **Clear Roles and Access :** Brands, creators, and administrators each have access to only the tools and data they need. This helps avoid confusion and protects sensitive content.

4 Technical Specification

4.1 Hardware Environment

The development was conducted using the following machines :

Component	Specification
Processor	Intel Core i5-11400H (6 cores / 12 threads, up to 4.5 GHz, 12 MB Intel Smart Cache)
Graphics Card	NVIDIA GeForce RTX 2050 with 4 GB dedicated GDDR6 memory
RAM	16 GB DDR4
Hard Disk	512 GB SSD M.2 NVMe
Operating System	Windows 11

Table 2: Hardware Specifications : Asus TUF Gaming FX506HF

Component	Specification
Processor	AMD Ryzen 5 7520U (4 cores / 8 threads, up to 4.3 GHz, 6 MB cache)
Graphics Card	Integrated AMD Radeon™ Graphics
RAM	16 GB DDR4
Hard Disk	512 GB SSD
Operating System	Windows 11

Table 3: Hardware Specifications : Asus VivoBook 15

4.2 Technological Environment

This section reviews the key languages and frameworks considered to select the best tools for the project. Table 4 lists the chosen technologies :

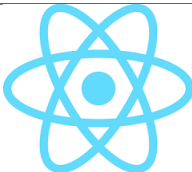
Technology	Description
	Node.js [9] is a cross-platform JavaScript runtime environment that executes JavaScript code outside a web browser. It's built on Chrome's V8 JavaScript engine and is used for developing scalable network applications.
	Express.js [10] is a minimal and flexible Node.js web application framework that provides a robust set of features for web and mobile applications.
	MongoDB [11] is a document-oriented NoSQL database used for high-volume data storage. It stores data in flexible, JSON-like documents.
	React.js [12] is an open-source JavaScript library for building user interfaces, particularly for single-page applications where a fast and interactive experience is required.
	Cloudinary [13] is a cloud service that offers a solution for managing images and videos, including uploads, storage, transformations, optimizations, and delivery.
	Sass [14] is a preprocessor scripting language that is interpreted or compiled into Cascading Style Sheets (CSS). It simplifies writing and maintaining stylesheets.
	Mailtrap [15] is a service for safe email testing, capturing SMTP traffic from staging and development environments without sending emails to real users.
	Vite [16] is a next-generation front-end build tool that improves the development experience by offering faster builds and hot module replacement (HMR).
	OpenAI [17] provides AI models (like GPT) accessible via an API to generate human-like text, assist with tasks, and enable intelligent automation.
	Ollama [18] is a tool that allows you to run, manage, and deploy open-source large language models (LLMs) locally on your machine.

Table 4: Technology and Description

4.3 Software Environment

The following software tools were used during the development :

Software	Description
	Visual Studio Code [19] is a lightweight but powerful source code editor, supporting JavaScript, Node.js, and a rich ecosystem of extensions.
	StarUML [20] is a sophisticated software modeler aimed at supporting agile and concise modeling using the Unified Modeling Language (UML).
	Draw.io [21](also known as diagrams.net) is an online diagramming tool for creating flowcharts, UML diagrams, and other structured visuals.
	Canva [22] is an online platform used to create visual content such as presentations, diagrams, posters, and infographics.
	Git [23] is a distributed version control system used for tracking changes in source code during software development. It helps multiple developers collaborate efficiently.
	GitHub [24] is a web-based platform that uses Git for version control. It provides hosting for software development and tools for collaboration, including repositories, issue tracking, and pull requests.

Table 5: Development and modeling tools

5 Project Architecture

This section provides a detailed overview of the architecture for the Colab Platform, discussing both the logical and physical aspects of the system. The architecture ensures the platform's design is scalable, maintainable, and efficient. We will cover the Model-View-Controller (MVC) architecture, deployment, and class diagrams to illustrate how the system is designed and implemented.

5.1 Logical Architecture

The logical architecture defines the structure and organization of the system's components, focusing on how the various modules interact. In this project, we utilize the **Model-View-Controller (MVC)** pattern to achieve clear separation of concerns, which enhances maintainability and scalability.

The MVC architecture consists of three primary components with distinct responsibilities :

- **Model** : Handles the application's data. It retrieves data from the database, processes it, and organizes it in a way that can be passed to the Controller for further processing.
- **View** : Responsible for presenting the user interface. Its main task is to display data retrieved from the Model, and its secondary role is to capture user actions (e.g., clicks, input) and forward these events to the Controller.
- **Controller** : Serves as an intermediary between the Model and View. It processes input from the View, requests necessary data from the Model, processes that data if needed, and sends the results back to the View for display.

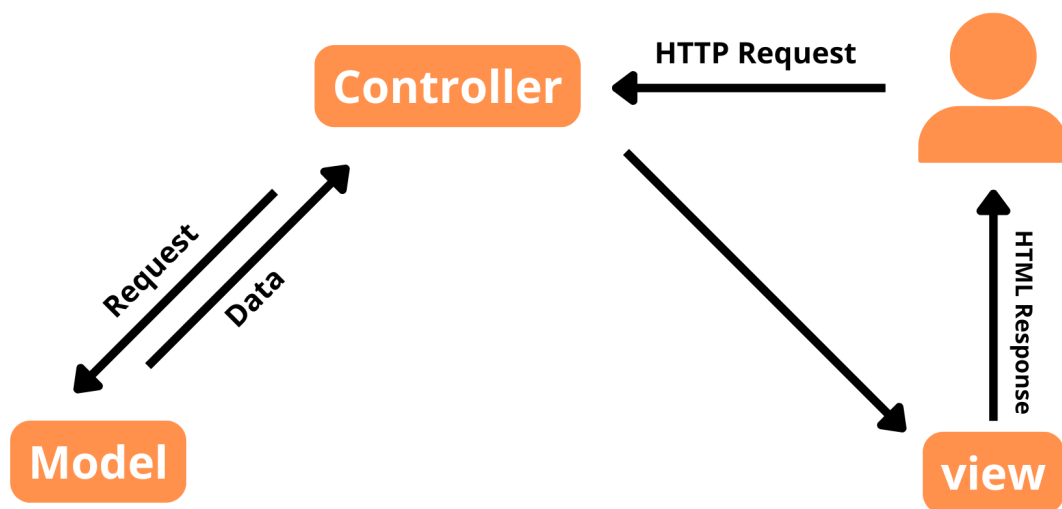


Figure 5: Illustration of MVC architecture

5.2 Physical Architecture

The physical architecture outlines the deployment of the MERN stack application's components across hardware and network infrastructure. This includes client devices running the React frontend, the Node.js/Express server hosted on a backend server, and a MongoDB database managed on a cloud or local database server.

The deployment diagram illustrates how these components are physically distributed and interact over the network. Below is the deployment diagram for our project :

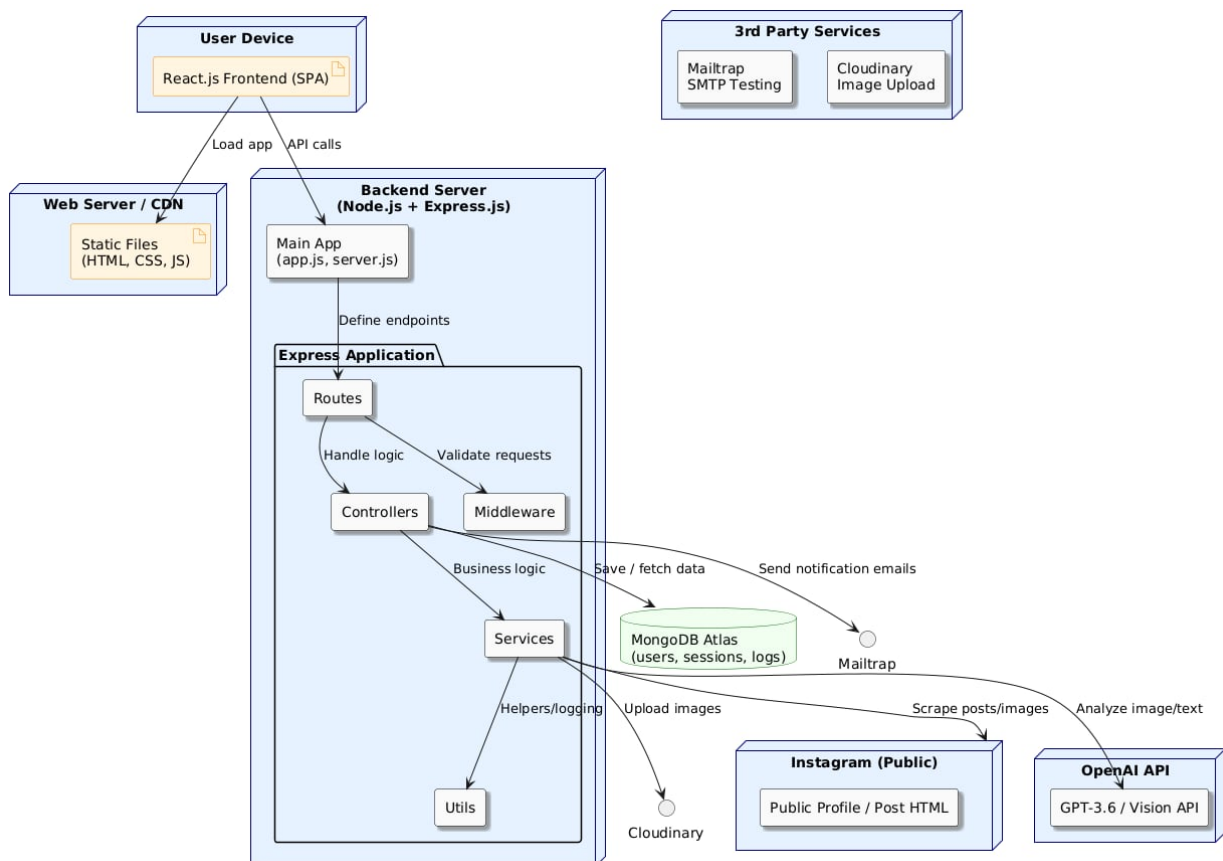


Figure 6: Deployment diagram

6 Global Use Case Diagram

This diagram shows key interactions between users and the *Colab* platform. It outlines actions by five user types and includes an external scoring system to assess creator performance. It serves as a basis for defining functional requirements.

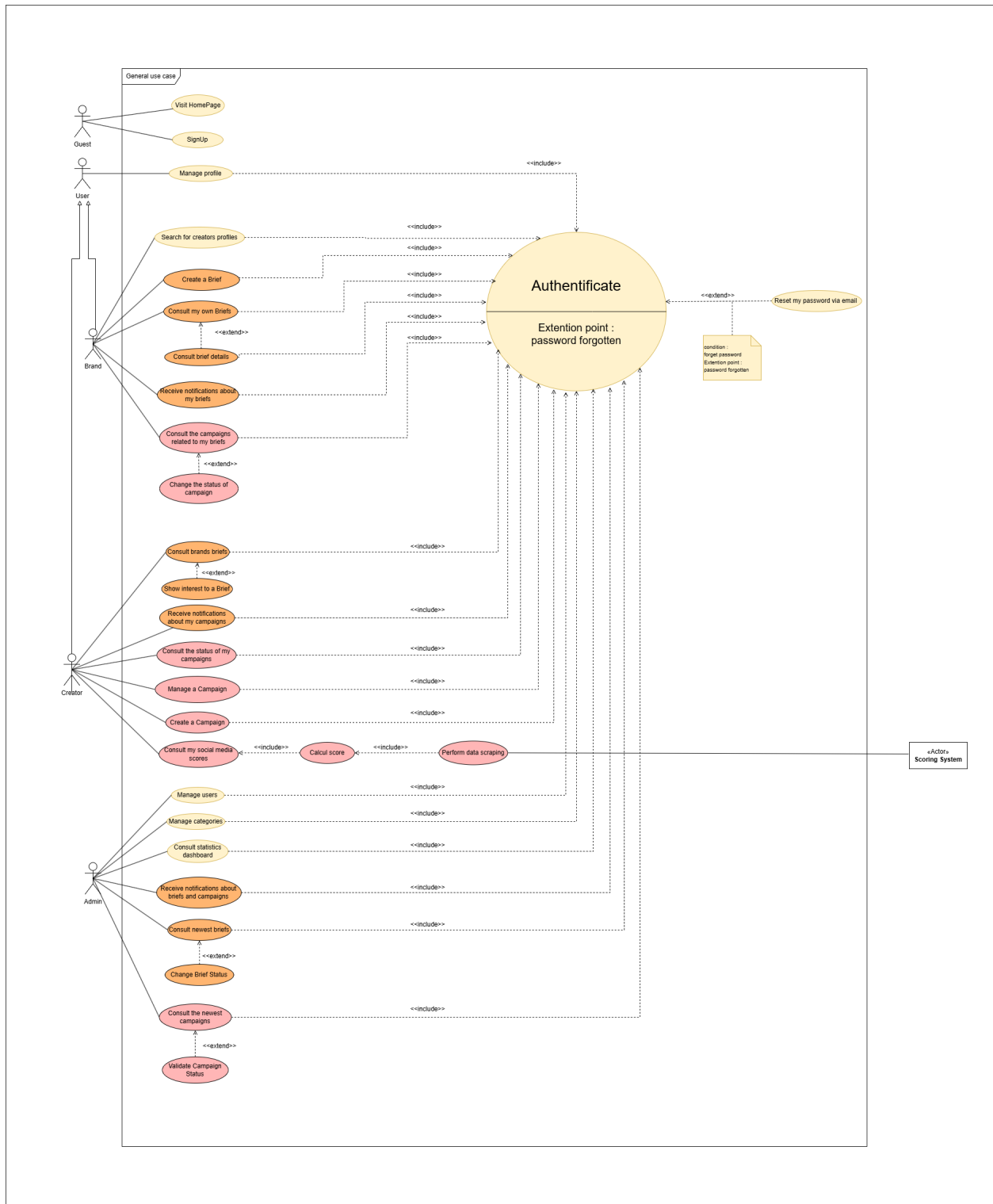


Figure 7: Global Use Case Diagram – Colab Platform

7 Project Management with SCRUM

7.1 Product Backlog

The Product Backlog is the central point of any Scrum project and serves as the primary input for sprint planning. It consists of a list of user stories, which are essential for the development team to understand the features and functionalities that need to be implemented. Each entry in the backlog is prioritized based on its importance and complexity.

Below is an explanation of the fields used in the Product Backlog :

ID : A unique and auto-incremented number for each feature or functionality. It serves as the identifier for tracking purposes.

Feature : This field categorizes the user story under a broader feature or functionality. It helps in organizing the backlog and understanding which feature each user story belongs to.

US ID : Short for User Story ID, this field provides a unique identifier for each user story, ensuring that each task is easily referenced.

User Story : This is a short description of the need of an actor. It typically follows the format :
“As a [role], I want to [goal] so that [reason].” User stories describe what the actor requires from the system.

Priority : This field indicates the importance or urgency of implementing a particular user story. It helps the team decide which features should be tackled first in the sprint.

Complexity : This refers to the effort or difficulty involved in implementing each user story. The Fibonacci sequence (1, 2, 3, 5, 8, 13, etc.) is commonly used to assign a complexity value to a story, with 13 representing medium complexity. The Fibonacci method helps in providing relative estimates of the effort required to complete each task.

ID	Feature	Us ID	User Story	Priority	Complexity
1	User Authentication	1.1	As a visitor, I can access the home page.	Medium	3
		1.2	As a user, I want to sign up and receive a verification code along with a welcome email.	High	21
		1.3	As a user, I want to log in.	High	8

		1.4	As a user, I want to log out.	High	3
		1.5	As a user, I want to reset my password via email when I forget it.	High	5
2	User Profile	2.1	As a user, I want to update my profile.	High	21
		2.2	As a user, I can consult my profile.	High	8
		2.3	As a brand, I can search for creators' profiles.	High	13
3	Brief Management	3.1	As a brand, I want to create briefs to define brief requirements.	High	13
		3.2	As a brand, I can update my brief.	Medium	8
		3.3	As a brand, I can delete my brief.	Medium	5
		3.4	As a brand, I can consult my own briefs.	Medium	8
		3.5	As a user, I can consult all briefs.	High	8
		3.6	As a creator, I can express interest in briefs.	Medium	8
		3.7	As a creator, I can apply for briefs.	High	13
4	Campaign Management	4.1	As a creator, I can create campaigns.	High	13
		4.2	As a creator, I can delete campaigns.	Medium	5
		4.3	As a creator, I can update campaigns.	Medium	5
		4.4	As a creator, I can consult my campaigns.	Medium	5
		4.5	As a brand, I can consult the campaigns related to my briefs.	High	13
		4.6	As a brand, I can accept or reject campaigns related to my briefs.	High	8
5	Scoring System	5.1	As a creator, I can consult my social media scores scraped from Instagram and TikTok.	High	21
6	Notification System	6.1	As a user, I want to receive notifications to stay informed.	High	8
7	Admin Panel	7.1	As an admin, I can manage users.	High	13
		7.2	As an admin, I can manage categories.	High	8
		7.3	As an admin, I can change the status of campaigns.	High	13
		7.4	As an admin, I can change the status of briefs.	High	13
		7.5	As an admin, I can view collaboration statistics.	Medium	8

8	Dashboard Views	8.1	As a brand, I can consult a dashboard with brief statistics and performance.	High	8
		8.2	As a creator, I can consult a dashboard with campaign performance and status.	High	8

Table 6: Product Backlog

7.2 Sprint Planning

By assigning Fibonacci numbers to each user story, we were able to prioritize and distribute the workload effectively. The tasks were grouped into three distinct sprints based on their cumulative Fibonacci scores, ensuring a balanced and manageable workload for each development cycle. Each sprint represents a 4-week period, allowing adequate time for development, testing, and review.

Sprint 1 (Yellow) : User Onboarding & Profile Management (95 points)

Period : February 17, 2025 – March 20, 2025

This sprint focuses on implementing the foundational elements of the platform. It includes user authentication (sign-up, login, logout, password reset), profile management for brands and creators, and basic admin capabilities such as user management. These core features ensure secure access and personalized experiences for all user roles.

Sprint 2 (Orange) : Brief Management & Platform Interaction (92 points)

Period : March 21, 2025 – April 22, 2025

This sprint covers the full lifecycle of briefs, including creation, editing, consultation, and deletion. It also incorporates the notification system, dashboard insights for brands, and administrative functionality for managing the status of briefs. These elements enhance platform interactivity and streamline brand-creator collaboration.

Sprint 3 (Red) : Campaign Management & Performance Monitoring (99 points)

Period : April 23, 2025 – May 24, 2025

This sprint enables creators to manage campaign activities and track performance. It includes campaign creation and review, a scoring system that aggregates social media metrics, performance dashboards for creators, and admin-level access to collaboration statistics. These features provide transparency, accountability, and performance tracking across the platform.

7.3 Sprint Table Breakdown

Below is a Sprint Planning table that maps User Story IDs to their respective complexity points across the four sprints :

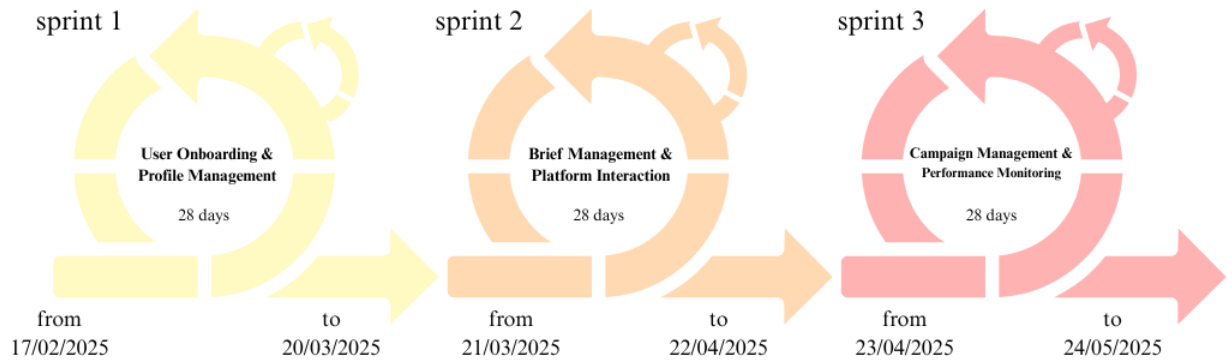


Figure 8: Sprint planinng

Sprint	User Story IDs	Complexity Points	Estimation (Days)
Sprint 1 : User Onboard- ing & Profile Management	US1.1, US1.2, US1.3, US1.4, US1.5, US2.1, US2.2, US2.3, US7.1, Us7.2	103	28
Sprint 2 : Brief Manage- ment & Platform Interac- tion	US3.1, US3.2, US3.3, US3.4, US3.5, US3.6, US3.7, US6.1, US7.4, US8.1	92	28
Sprint 3 : Campaign Man- agement & Performance Monitoring	US4.1, US4.2, US4.3, US4.4, US4.5, US4.6, US5.1, US7.3, US7.5, US8.2	99	28

Table 7: Sprint Planning Table

8 Conclusion

A detailed analysis and well-defined requirements provide a strong base for the development of the **Colab Platform**. The technical specifications and architectural design outlined in the project will serve as a roadmap for the design and implementation stages, ensuring the platform is built efficiently and meets both functional and non-functional expectations.

Chapter 3 : Sprint 1 – User Onboarding & Profile Management

1 Introduction

While following the previously defined project structure, Sprint 1 focuses on delivering the first functional and usable increment of the **Colab Platform**. This sprint includes the implementation of key user management features such as profile management, user access, admin functionalities, and role-based control, forming the foundation for future development.

2 Specification of Requirements

In this section, we will present the Sprint 1 backlog, the corresponding use case diagram, and a selection of user story descriptions relevant to this sprint.

2.1 Sprint Backlog

The Sprint 1 backlog includes the user stories outlined in Table 8. Each user story follows a development process composed of the following steps :

- Design the interface
- Implement the functionality in the backend
- Integrate the frontend with the backend
- Test the feature

For this sprint, we will not detail the internal tasks of each user story. Instead, only the selected user stories and their complexity levels will be listed, as they form the basis for the implementation of user registration, authentication, profile management, and administrative operations.

ID	Feature	US ID	User Story	Priority	Complexity
1	User Authentication	US1.1	As a visitor, I can access the home page.	Medium	3
		US1.2	As a user, I want to sign up and receive a verification code along with a welcome email.	High	21
		US1.3	As a user, I want to log in.	High	8
		US1.4	As a user, I want to log out.	High	3
		US1.5	As a user, I want to reset my password via email when I forget it.	High	5
2	User Profile	US2.1	As a user, I want to update my profile.	High	21
		US2.2	As a user, I can consult my profile.	High	8
		US2.3	As a brand, I can search for creator's profiles.	High	13
7	Admin Panel	US7.1	As an admin, I can manage users.	High	13
		US7.5	As an admin, I can manage categories.	High	8

Table 8: Sprint 1 Backlog

2.2 Use case diagram

Figure 5 shows the use case diagram for Sprint 1. Highlights the core functionalities implemented, including user authentication, profile management, and admin operations such as user and category management. The diagram also includes extension points for password reset and email verification.

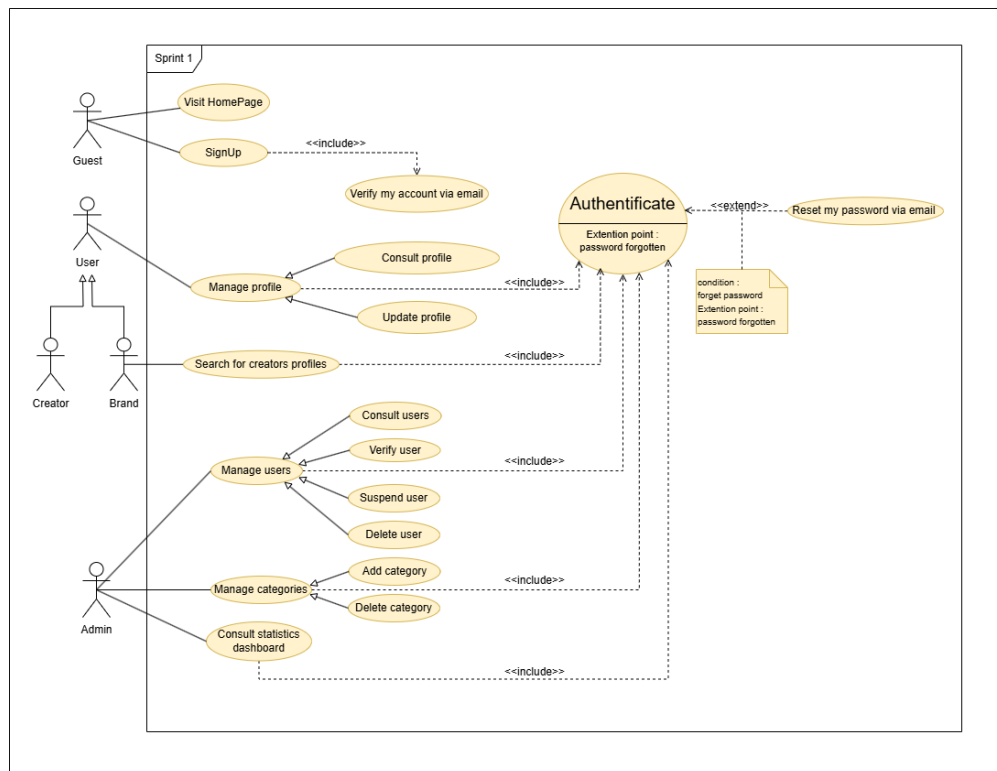


Figure 9: Use Case Diagram – Sprint 1

2.3 Textual description of use cases

In this section, we outline key use cases of the Colab platform, showcasing core features and typical interactions between brands, creators, and administrators. These cases demonstrate how the platform supports collaboration and user engagement in real-world scenarios.

2.3.1 User Authentication

Use Case - Sign Up

Use Case Name	Sign Up
Actor(s)	Guest
Pre-conditions	The user is not already registered. The registration interface is accessible.
Post-conditions	A new user account is created. A verification email is sent.

Nominal Flows	1. The user accesses the sign-up form. 2. The user enters and submits registration details. 3. The system checks if the email already exists. 4. If not found, the system hashes the password and creates the account. 5. A role-specific profile is created. 6. A verification token is generated. 7. A verification email is sent.
Exceptional Flows	1. Email already exists : The system prompts the user to log in instead. 2. Invalid email format : The system displays a validation error.

Table 9: Use Case - Sign Up

Use Case - Log In

Use Case Name	Log In
Actor(s)	User (Brand, Creator)
Pre-conditions	The user must already have an account.
Post-conditions	The user is authenticated. The user is redirected to the appropriate dashboard based on their role.
Nominal Flows	1 The user enters their email and password. 2. The system verifies that the email exists. 3. The system compares the entered password with the stored one. 4. The system checks whether the account is suspended. 5. If valid, the system generates an authentication token and sets a cookie. 6. The user is redirected to their dashboard.

Exceptional Flows	1. Account not found : No account is associated with the entered email. A suggestion to sign up is provided. 2. Incorrect password : The system alerts the user that the entered password is incorrect. 3. Suspended account : Login is denied, and a message informs the user that their account is suspended by an administrator.
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Table 10: Use Case - Log In

Use Case - Log Out

Use Case Name	Log Out
Actor(s)	User
Pre-conditions	User must be authenticated (logged in).
Post-conditions	User is logged out and session token is removed.
Nominal Flows	1. User clicks the logout button. 2. System clears the authentication token. 3. A success alert is shown. 4. User is redirected to the login page.
Exceptional Flows	None explicitly shown in this flow.

Table 11: Use Case - Log Out

Use Case - Reset Password via Email

Use Case Name	Reset Password via Email
Actor(s)	User (Brand, Creator)
Pre-conditions	The user has forgotten their password. The user has access to the registered email account.
Post-conditions	A secure reset link is sent to the user's email. The user can proceed to reset their password.

Nominal Flows	1. The user navigates to the "Forgot Password" form. 2. The user enters their registered email address. 3. The system verifies if the email exists in the database. 4. A reset token is generated and stored. 5. A reset link containing the token is sent via email. 6. The system informs the user that a reset email has been sent.
Exceptional Flows	1. Invalid email format : The user submits an incorrectly formatted email address. The system prompts correction. 2. Email not associated with any account : The system notifies the user that no account matches the entered email. 3. Email delivery failed : The system is unable to send the reset email and informs the user to try again later.

Table 12: Use Case - Reset Password via Email

2.3.2 User Profile

Use Case - Consult Profile

Use Case Name	Consult Profile
Actor(s)	User (Brand, Creator)
Pre-conditions	User is authenticated
Post-conditions	Profile data is retrieved and displayed
Nominal Flows	1. User opens their profile page 2. System fetches user data excluding password 3. Based on user role : - If influencer : fetch creator data - If brand : fetch advertiser data 4. Display the complete profile
Exceptional Flows	None explicitly shown in this flow.

Table 13: Use Case - Consult Profile

Use Case - Update Profile

Use Case Name	Update Profile
Actor(s)	User (Brand, Creator)

Pre-conditions	The user is authenticated.
Post-conditions	The user's general and role-specific profile information is updated.
Nominal Flows	1. The user accesses the profile update form. 2. The user edits general and/or role-specific fields and optionally uploads a new profile photo. 3. The system validates the input. 4. The system updates the general user information. 5. The system updates role-specific information based on the user type (Brand or Creator). 6. The system saves all changes and confirms the update with a success message.
Exceptional Flows	1. Missing fields : One or more required fields are left blank; the system prompts the user to complete them. 2. Invalid data : The user enters incorrectly formatted data (e.g., invalid email); the system displays an error. 3. Unsupported image format : The uploaded profile photo is in an invalid format; the user is asked to choose a valid file type.

Table 14: Use Case - Update Profile

Use Case - Search for Creators

Use Case Name	Search for Creators
Actor(s)	Brand
Pre-conditions	The brand user is authenticated.
Post-conditions	A list of creators matching the search and filter criteria is displayed.
Nominal Flows	1. The brand accesses the creator search interface. 2. The brand enters a search keyword or selects filters (e.g., name, category). 3. The system processes the search criteria. 4. The system displays a list of matching creators.

Exceptional Flows	1. Empty search input : The brand submits the search without entering any criteria; the system displays all creators. 2. No matches found : The filters yield no results; the system shows a message such as "No creators found matching your criteria."
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Table 15: Use Case - Search for Creators

2.3.3 Admin Panel

Use Case - Consult Users

Use Case Name	Consult Users
Actor(s)	Admin
Pre-conditions	The admin is authenticated and has access to the admin panel.
Post-conditions	The selected list of users (advertisers or creators) is displayed in the admin interface.
Nominal Flows	1. The admin opens the user management section. 2. The admin selects either the "Advertisers" or "Creators" tab. 3. The system retrieves and displays the corresponding list of users.
Exceptional Flows	1. No users available : The selected category has no users; the system displays a message such as "No users available."

Table 16: Use Case – Consult Users

Use Case - Suspend User

Use Case Name	Suspend User
Actor(s)	Admin
Pre-conditions	The admin is authenticated and has access to the user list.

Post-conditions	The user's suspension status is updated and reflected in the interface.
Nominal Flows	1. The admin selects a user from the list. 2. The admin clicks the "Suspend" button. 3. The system updates the user's status to suspended. 4. A confirmation message is shown.
Exceptional Flows	1. User already suspended : The system notifies the admin that the user is already suspended.

Table 17: Use Case - Suspend User

Use Case - Verify User

Use Case Name	Verify User
Actor(s)	Admin
Pre-conditions	The admin is authenticated and the user is already registered.
Post-conditions	The user's account is marked as verified in the system.
Nominal Flows	1. The admin selects a user from the list. 2. The admin clicks the "Verify" button. 3. The system updates the user's verification status. 4. A confirmation message is shown to the admin.
Exceptional Flows	1. User already verified : The system notifies that the user is already verified.

Table 18: Use Case - Verify User

Use Case - Delete User

Use Case Name	Delete User
Actor(s)	Admin
Pre-conditions	Admin is authenticated and user exists in the system
Post-conditions	User account is deleted from the database
Nominal Flows	1. Admin initiates deletion 2. System verifies user exists by ID 3. If found, system deletes the user 4. System confirms deletion 5. Admin receives alert "User deleted with success"

Exceptional Flows	None explicitly shown in this flow.
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Table 19: Use Case - Delete User

Use Case - Add Category

Use Case Name	Add Category
Actor(s)	Admin
Pre-conditions	The admin is authenticated.
Post-conditions	A category is added if it doesn't already exist.
Nominal Flows	1. The admin clicks the "Add Category" button. 2. The admin fills in the category name and submits. 3. The system checks whether the category already exists. 4. If it does not, the system adds the new category. 5. A confirmation is shown and the category list is updated.
Exceptional Flows	1. Category already exists : The system displays an error message and prevents duplication. 2. Empty field : The admin submits without a category name; the system requests input.

Table 20: Use Case - Add Category

Use Case - Delete Category

Use Case Name	Delete Category
Actor(s)	Admin
Pre-conditions	Authenticated admin accesses the category list.
Post-conditions	The selected category is removed from the system.
Nominal Flows	1. The admin selects the category to delete. 2. The system verifies that the category exists. 3. The system deletes the category. 4. A confirmation message and the category list is updated.
Exceptional Flows	1. Category in use : The system blocks deletion and alerts that the category is linked to existing content.

Table 21: Use Case - Delete Category

2.4 Class Diagram

Figure 10 illustrates the class diagram of the first sprint. In this diagram, the colored classes represent the entities developed during this sprint, including core components such as **Admin**, **User** (**Creator**, **Brand**). These classes form the foundation of user management within the Colab platform, defining essential attributes and relationships to handle authentication, roles, and permissions.

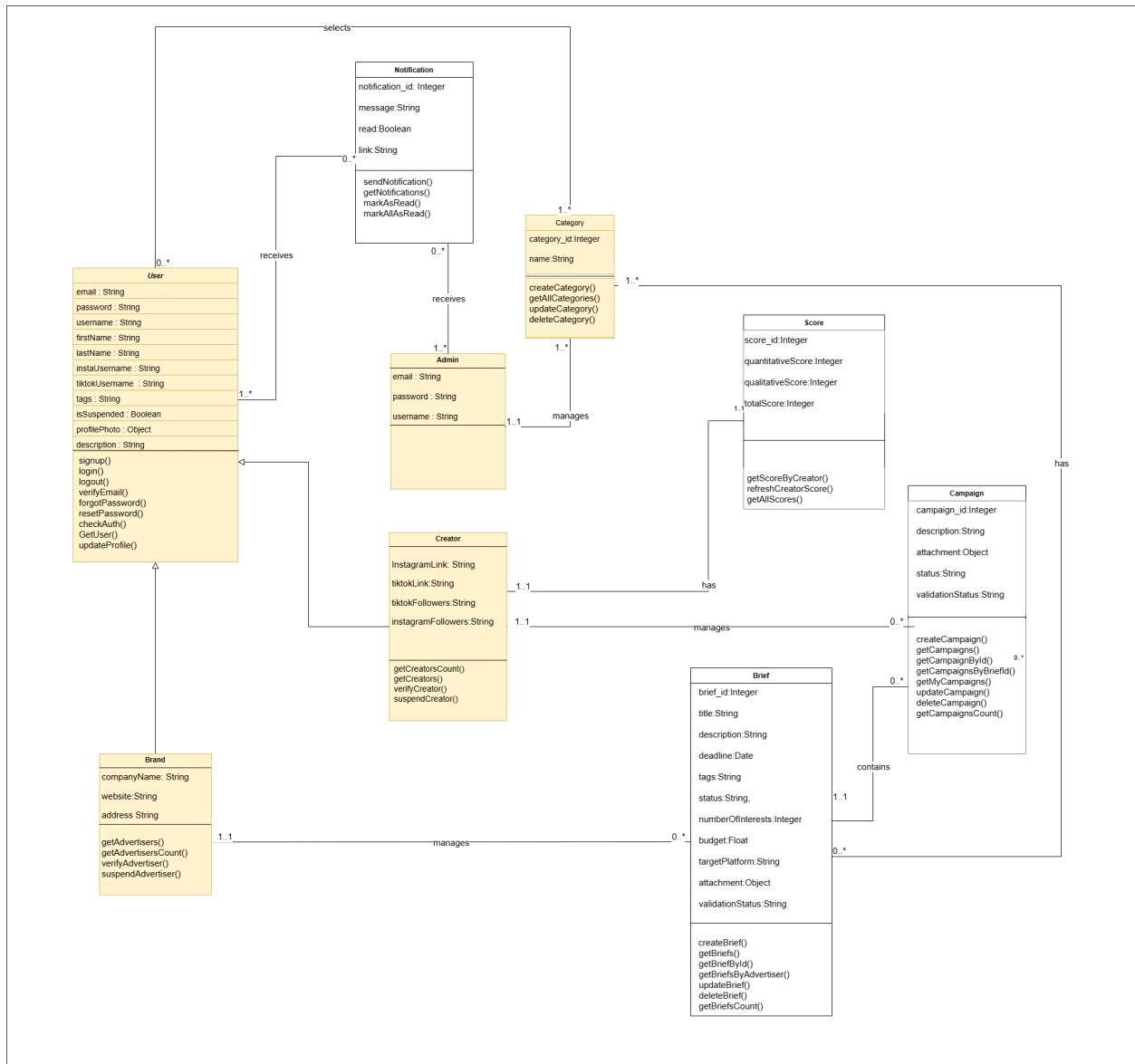


Figure 10: Class Diagram of Sprint 1

3 Detailed sequence diagrams

In this section, we present detailed sequence diagrams categorized by the main feature groups defined in the Sprint 1 backlog. These diagrams illustrate the dynamic interactions between actors and system components to fulfill the core functionalities developed in this sprint.

3.1 User Authentication

This section covers the authentication flow for users, including account creation, login, logout, and password reset functionalities. These flows ensure secure access and session management across the platform.

3.1.1 Sign Up

Figure 11 shows the sign-up process. The system checks if the user exists, hashes the password, creates the user, sets the role-specific profile (creator or advertiser), generates a token, and sends a verification email.

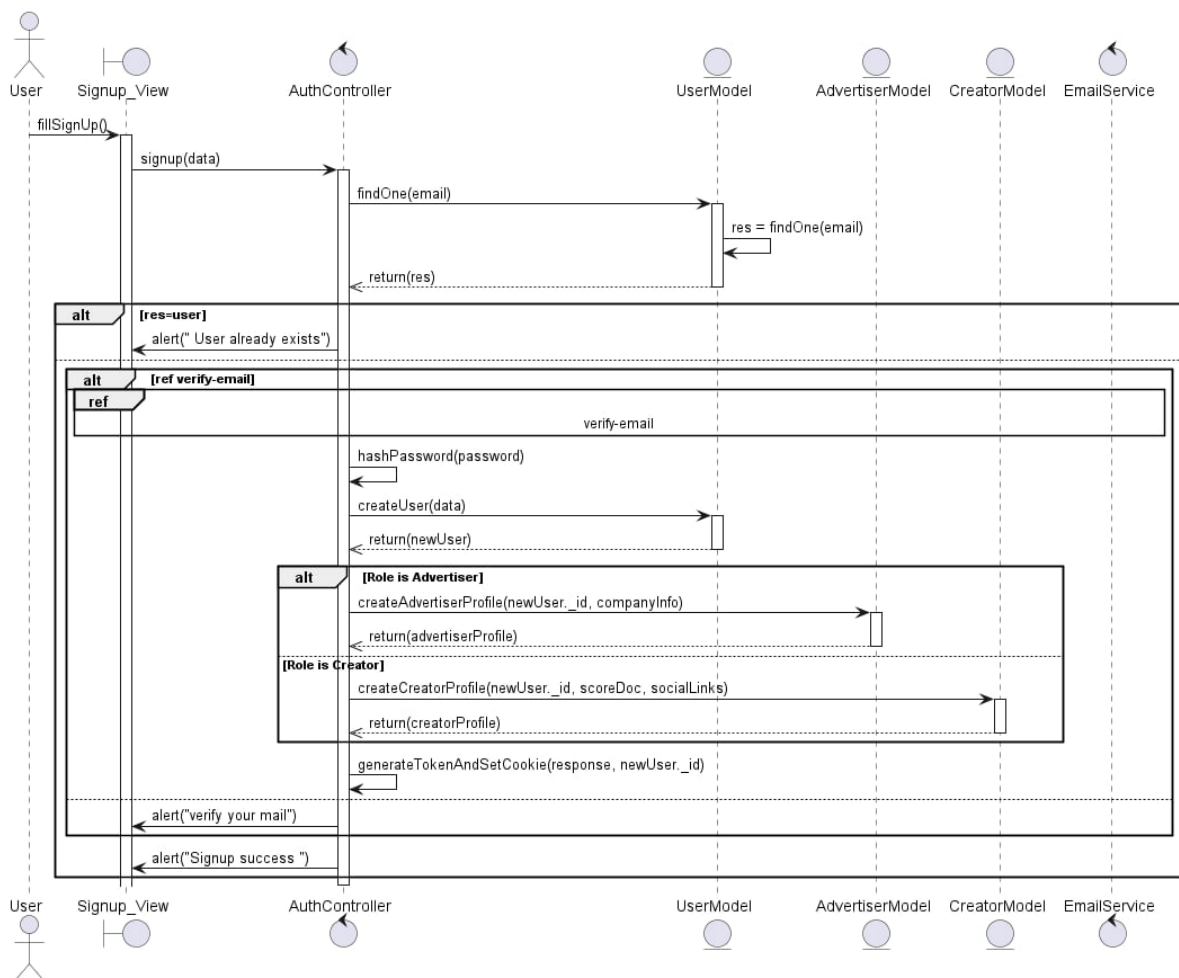


Figure 11: Sequence Diagram – Sign Up

3.1.2 Log In

Figure 12 illustrates the login flow. The user submits credentials, the system verifies email and password, checks account verification, and redirects based on role.

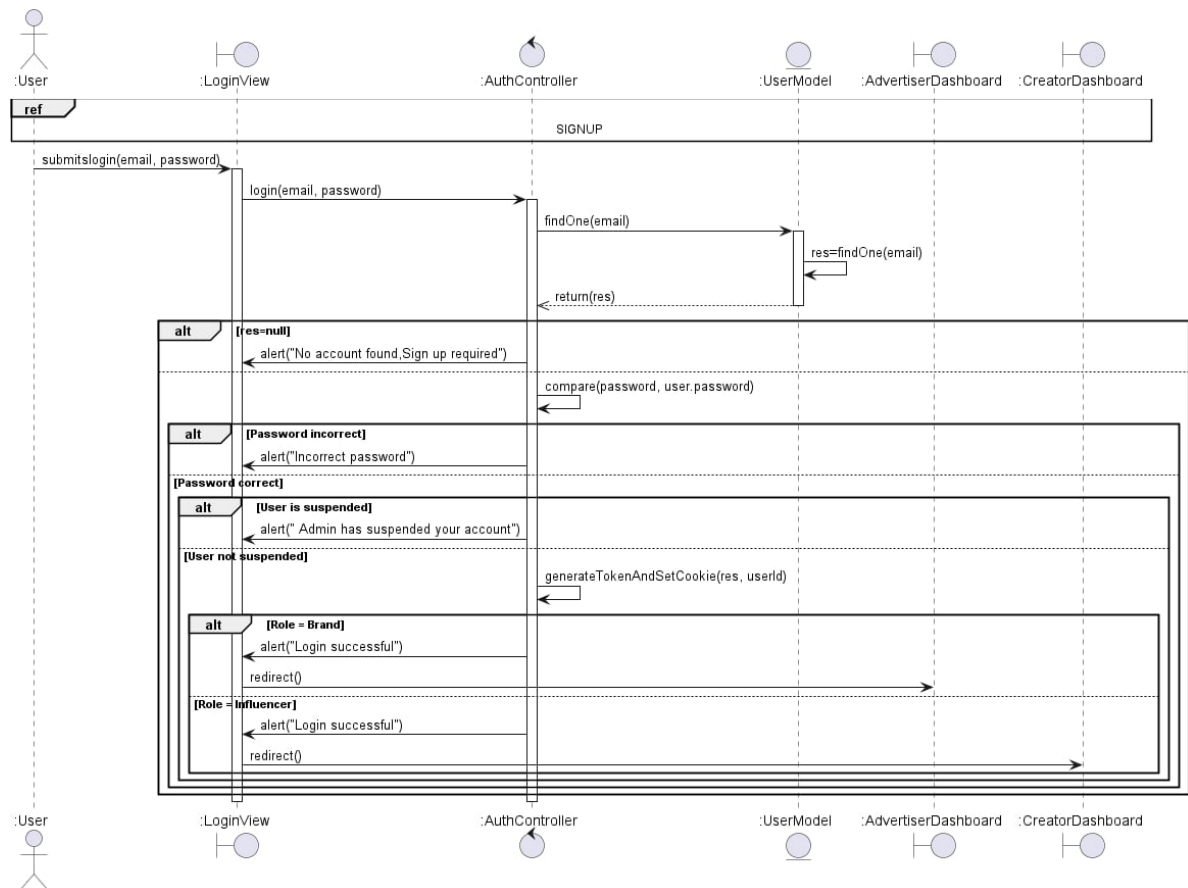


Figure 12: Sequence Diagram – Log In

3.1.3 Log Out

Figure 13 depicts the logout process. The user clicks the logout button, the system clears the token, and redirects the user to the login page.

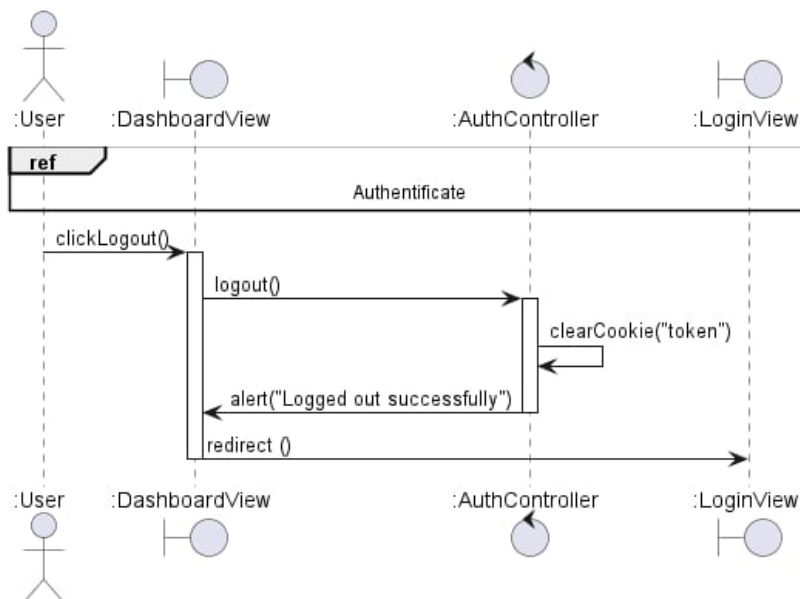


Figure 13: Sequence Diagram – Log Out

3.1.4 Password Reset

Figure 14 shows the process for resetting a forgotten password. The system verifies the user's email, generates a reset token, updates the user record, and sends a reset link.

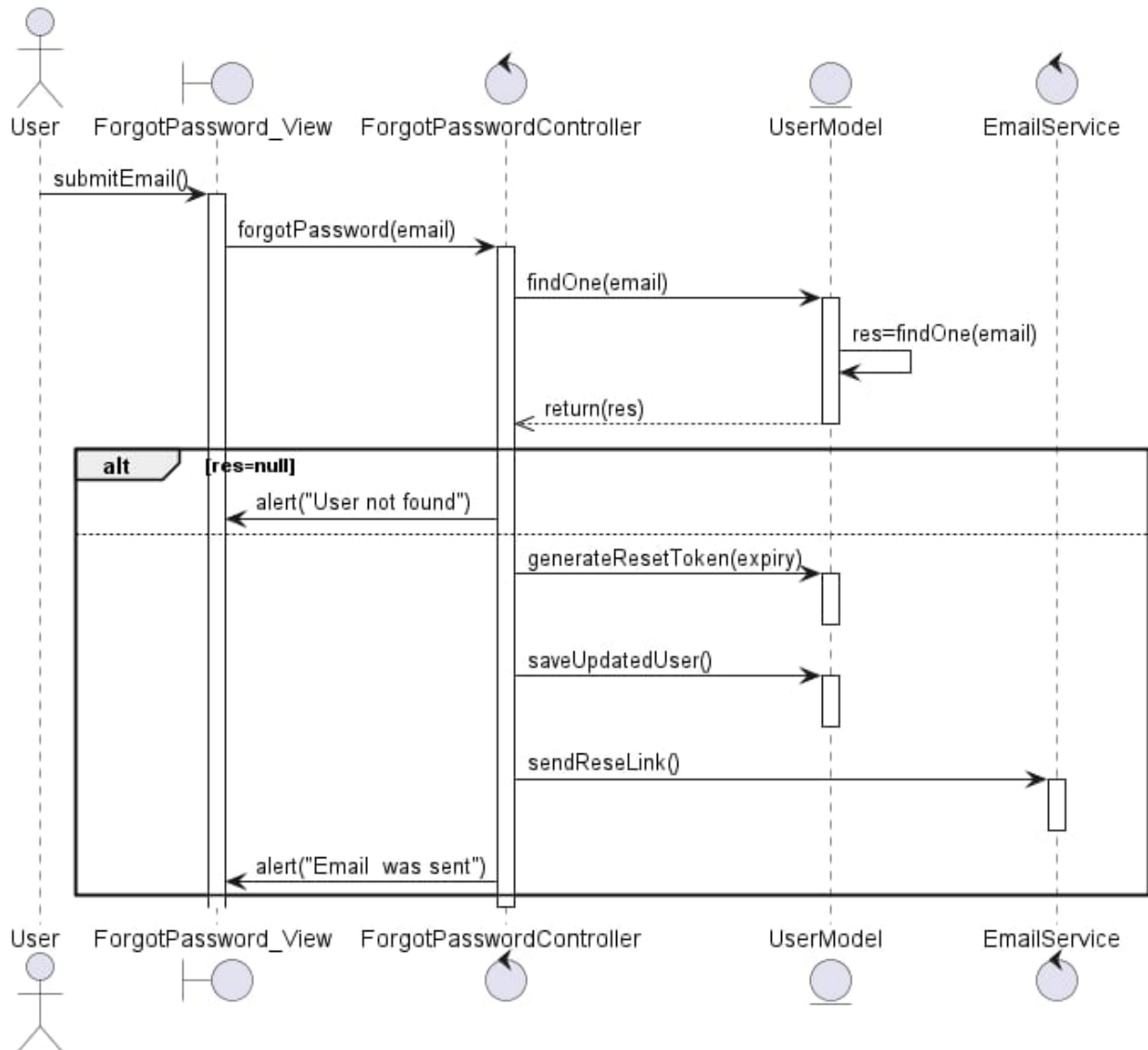


Figure 14: Sequence Diagram – Reset Password

3.2 User Profile

This section presents the sequence flows related to user profile management. These include updating profile information, handling profile images and categories, and managing role-specific updates for creators and brands.

3.2.1 Update Profile

Figure 15 shows the profile update process. The system verifies the user, updates general profile data, and applies role-specific updates, such as influencer details or brand info.

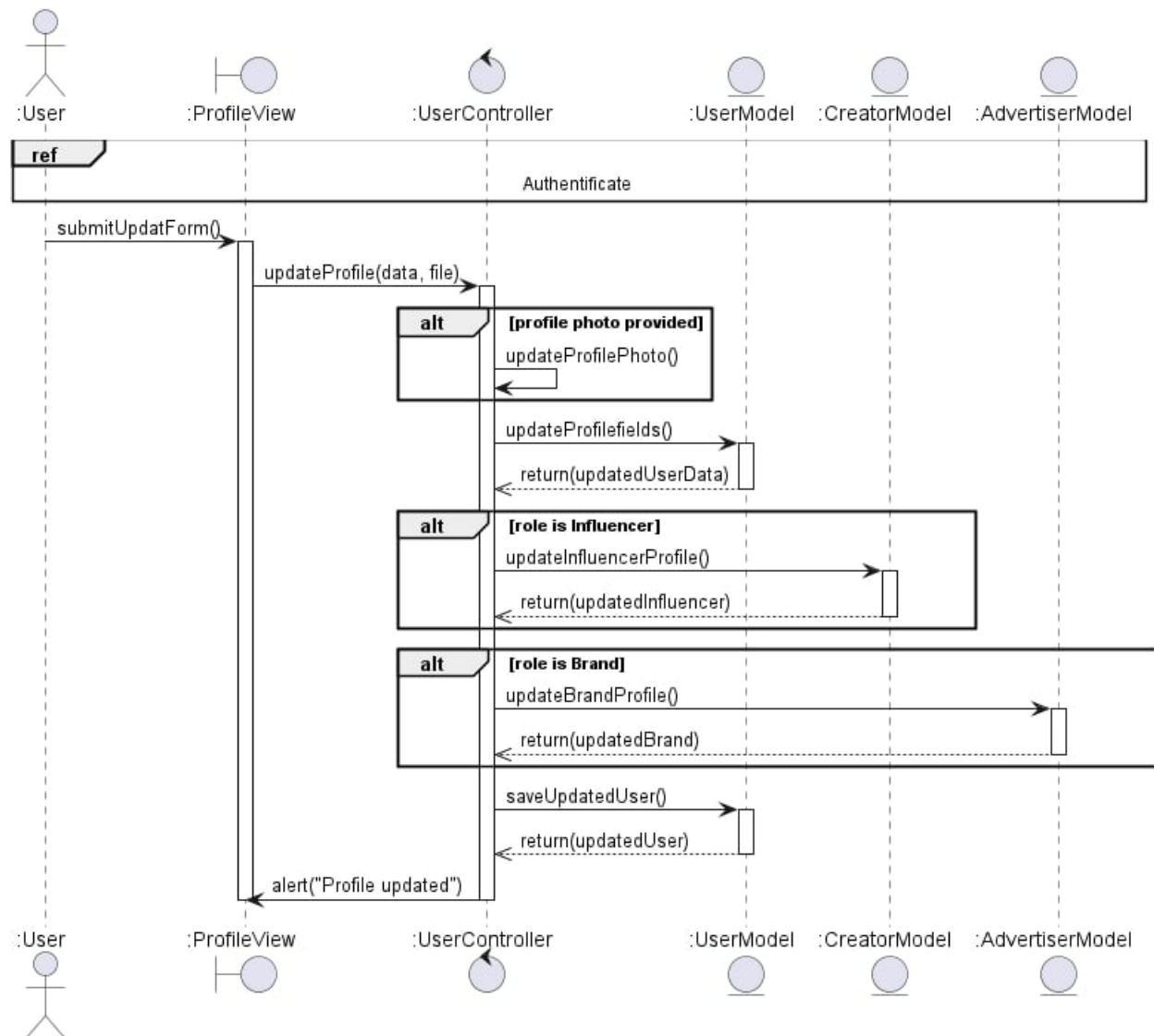


Figure 15: Sequence Diagram – Update Profile

3.2.2 Consult Profile

Figure 16 illustrates how a user can consult their profile. After verifying the user's identity, the system retrieves merged profile information based on their role (creator or brand).

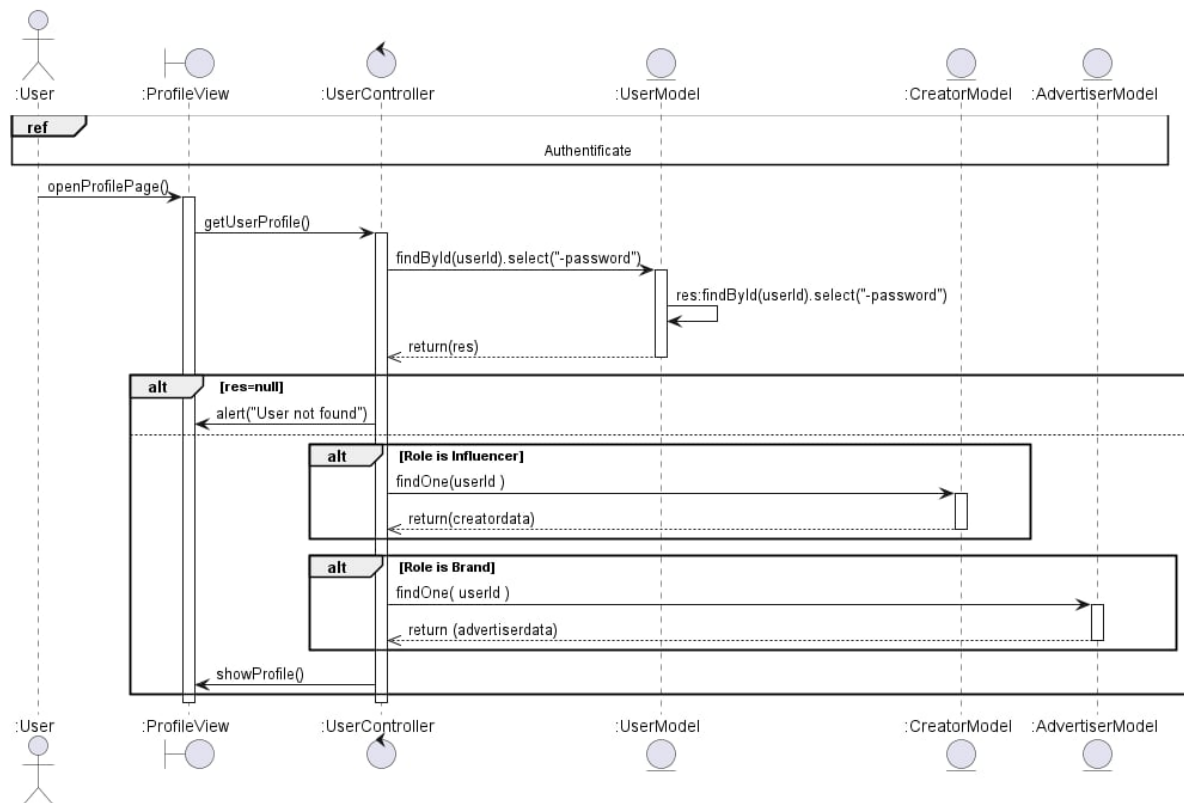


Figure 16: Sequence Diagram – Consult Profile

3.2.3 Search for Creators

Figure 17 shows how a brand searches for creators. The system applies optional filters (like name or category), fetches data, and returns a list of matching creator profiles.

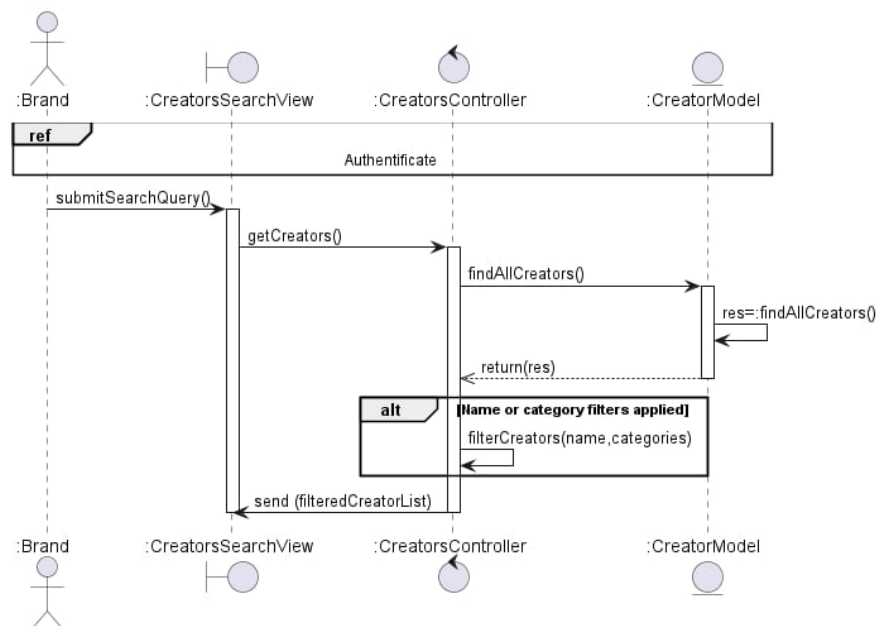


Figure 17: Sequence Diagram – Search Creators

3.3 Admin Panel

This section outlines the admin's capabilities for managing users and categories. Admins can view, verify, suspend, or delete users, and create or delete content categories.

3.3.1 Consult Users

Figure 18 shows how the admin consults the list of users. The system retrieves all user accounts and returns the data for display in the admin panel.

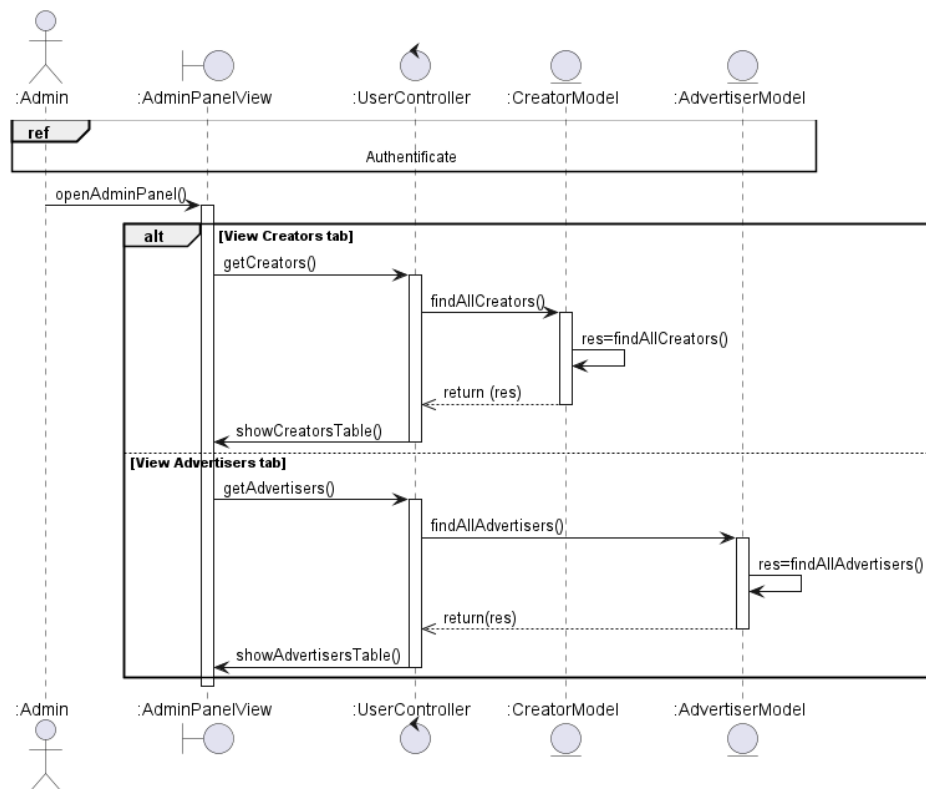


Figure 18: Sequence Diagram – Consult Users

3.3.2 Suspend User

Figure 19 illustrates how an admin suspends or unsuspends a user. The system updates the user's 'isSuspended' status accordingly.

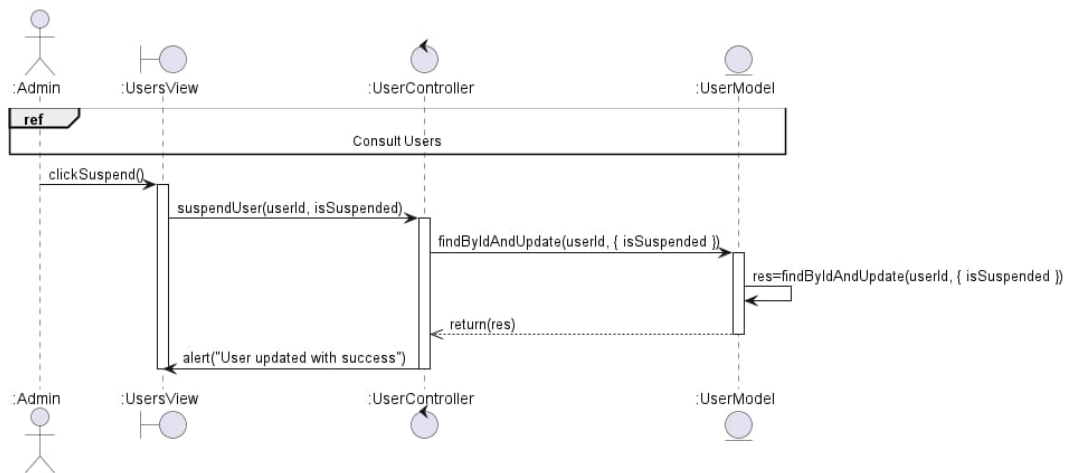


Figure 19: Sequence Diagram – Suspend User

3.3.3 Verify User

Figure 20 shows the verification of a user account by the admin. The system sets the 'isVerified' flag to true and confirms the update.

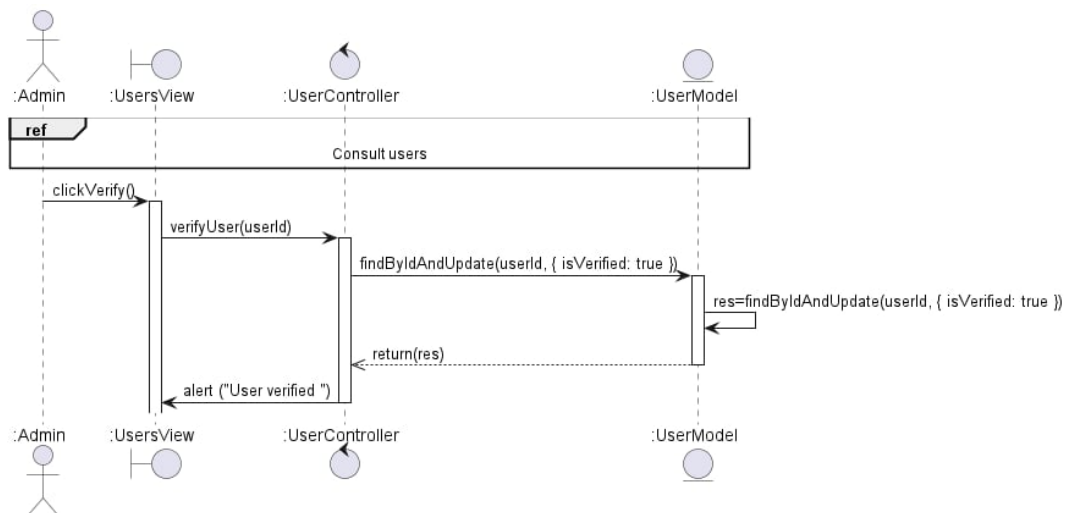


Figure 20: Sequence Diagram – Verify User

3.3.4 Delete User

Figure 21 shows how an admin deletes a user account. The system verifies the user exists, then removes the user.

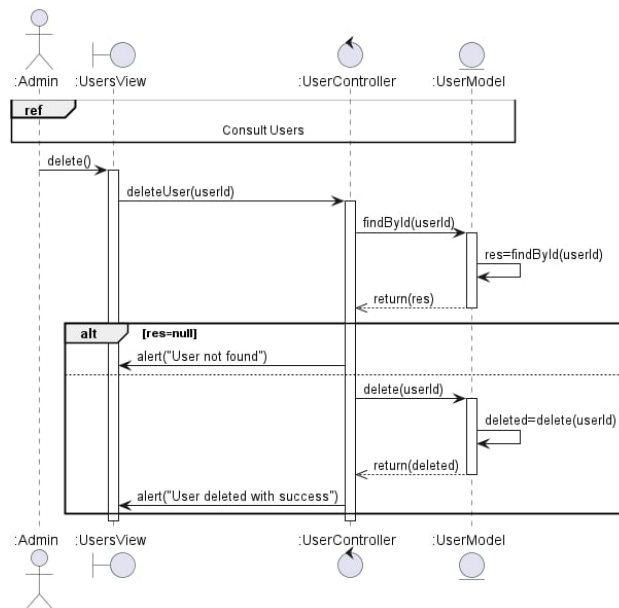


Figure 21: Sequence Diagram – Delete User

3.3.5 Add Category

Figure 22 shows how an admin adds a new category. The system first checks if the category already exists, then saves or returns an error.

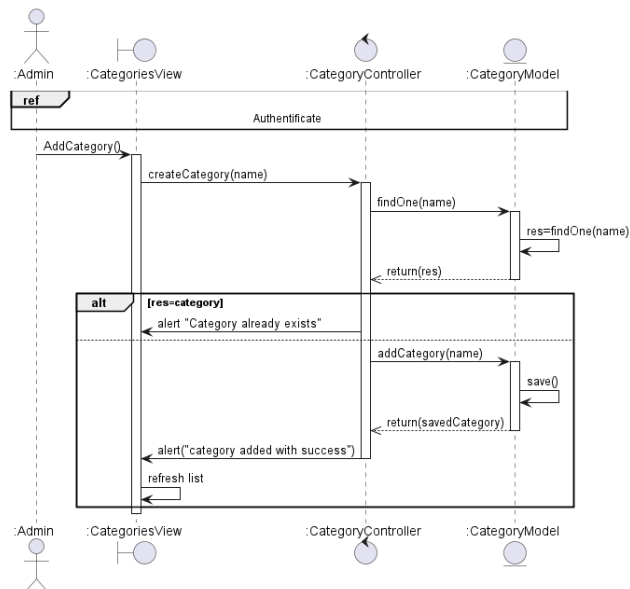


Figure 22: Sequence Diagram – Add Category

3.3.6 Delete Category

Figure 23 describes how a category is deleted by the admin. The system checks if it exists and removes it from the database.

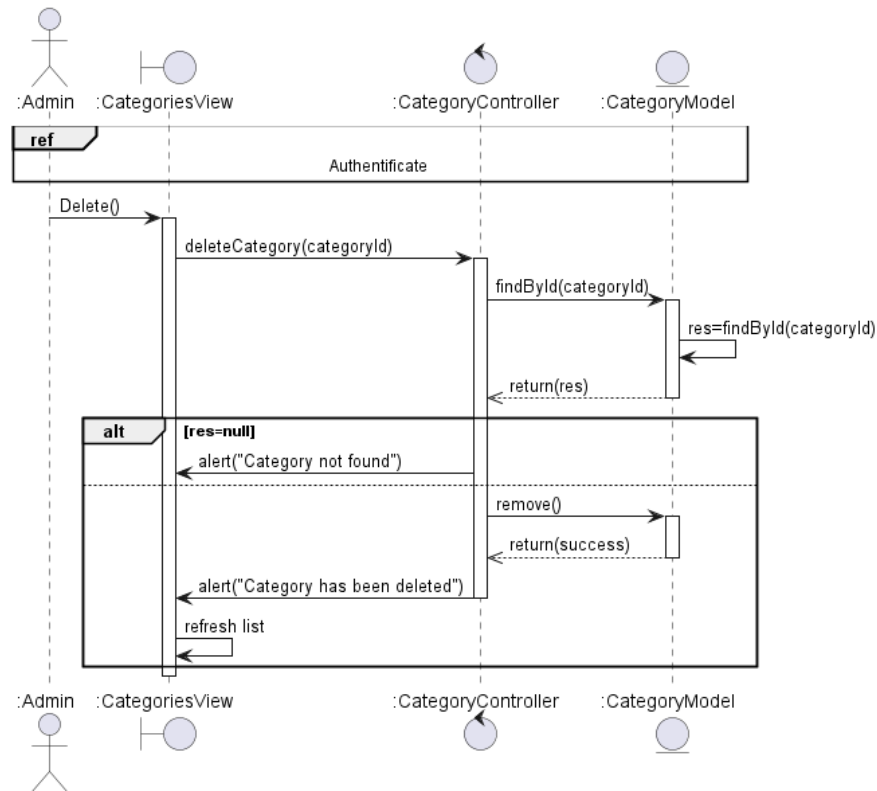


Figure 23: Sequence Diagram – Delete Category

4 Implementation

This section presents the user interfaces implemented during Sprint 1 of the Colab platform.

4.1 Homepage – Overview

The homepage presents Colab's vision with animated visuals, a slogan, a call-to-action button, client logos, and contact access.

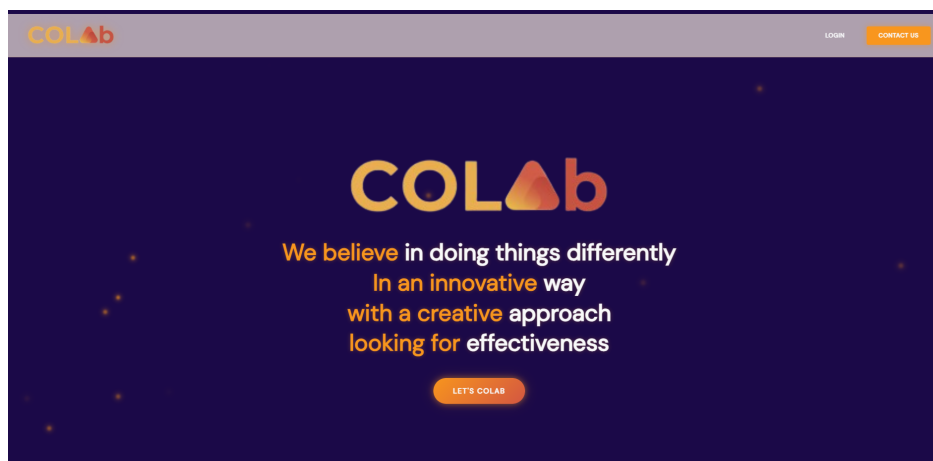


Figure 24: Homepage

4.2 User Authentication

Sign-up Choice Page

This interface allows new users to select whether they want to sign up as an influencer or a brand.

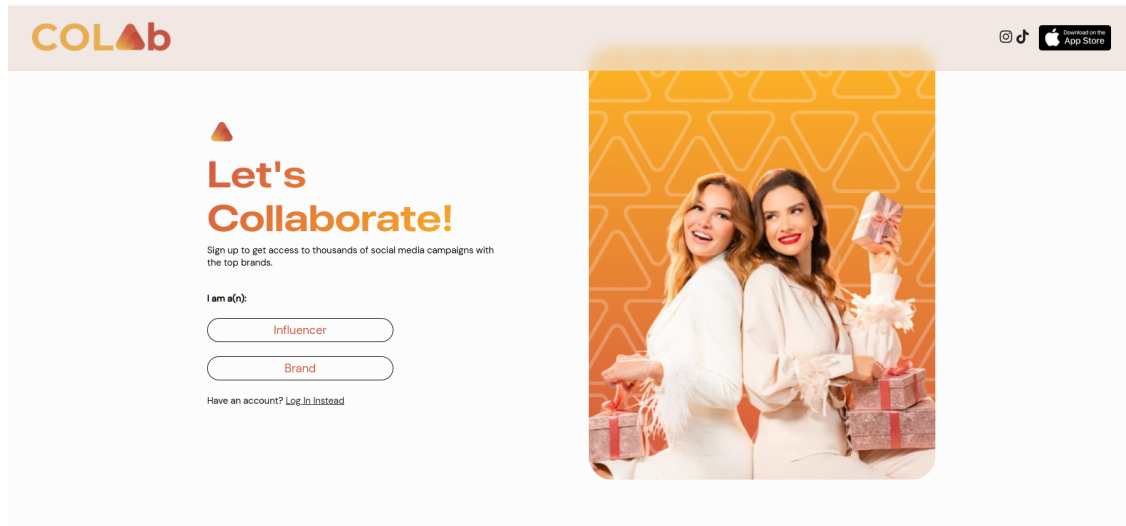


Figure 25: Sign-up Choice

Login Page

A simple login screen for existing users to authenticate and access their dashboards.

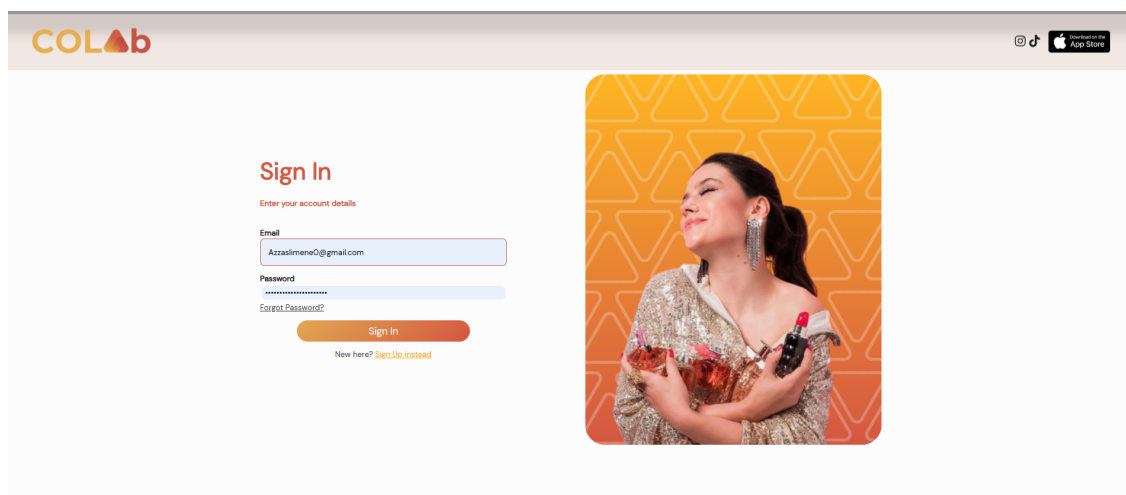
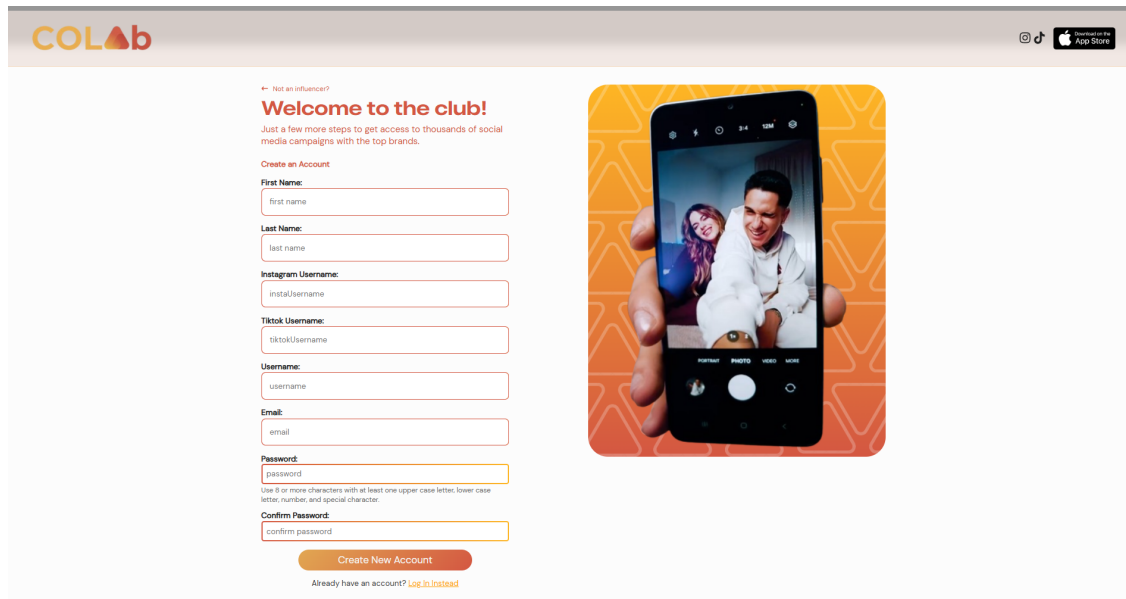


Figure 26: Login

Influencer Sign-up

This form gathers influencer details such as social media handles and basic personal information.



The COLAb Influencer Sign-up form is titled "Welcome to the club!" and includes a link for non-influencers. It features a form with fields for First Name, Last Name, Instagram Username, TikTok Username, Username, Email, Password, and Confirm Password. A "Create New Account" button is at the bottom, with a link to "Log in instead" for existing users. To the right is an image of a hand holding a smartphone displaying a photo of a man and a woman.

COLAb

← Not an influencer?

Welcome to the club!

Just a few more steps to get access to thousands of social media campaigns with the top brands.

Create an Account

First Name:

Last Name:

Instagram Username:

TikTok Username:

Username:

Email:

Password:

Confirm Password:

Use 8 or more characters with at least one upper case letter, lower case letter, number, and special character.

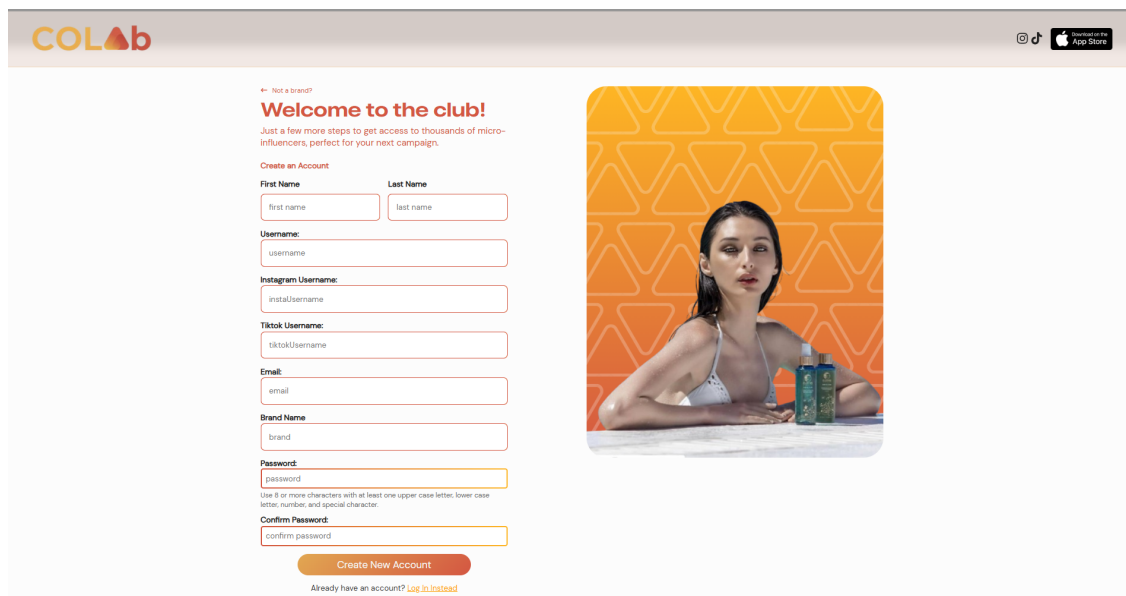
Create New Account

Already have an account? [Log in instead](#)

Figure 27: Influencer Sign-up

Brand Sign-up

Brands register here by providing company information and contact credentials.



The COLAb Brand Sign-up form is titled "Welcome to the club!" and includes a link for non-brands. It features a form with fields for First Name, Last Name, Username, Instagram Username, TikTok Username, Email, Brand Name, Password, and Confirm Password. A "Create New Account" button is at the bottom, with a link to "Log in instead" for existing users. To the right is an image of a woman sitting at a table with a bottle of water.

COLAb

← Not a brand?

Welcome to the club!

Just a few more steps to get access to thousands of micro-influencers, perfect for your next campaign.

Create an Account

First Name: Last Name:

Username:

Instagram Username:

TikTok Username:

Email:

Brand Name:

Password:

Confirm Password:

Use 8 or more characters with at least one upper case letter, lower case letter, number, and special character.

Create New Account

Already have an account? [Log in instead](#)

Figure 28: Brand Sign-up

Account Created – Message

Upon registration, users are notified that a verification email has been sent to them.

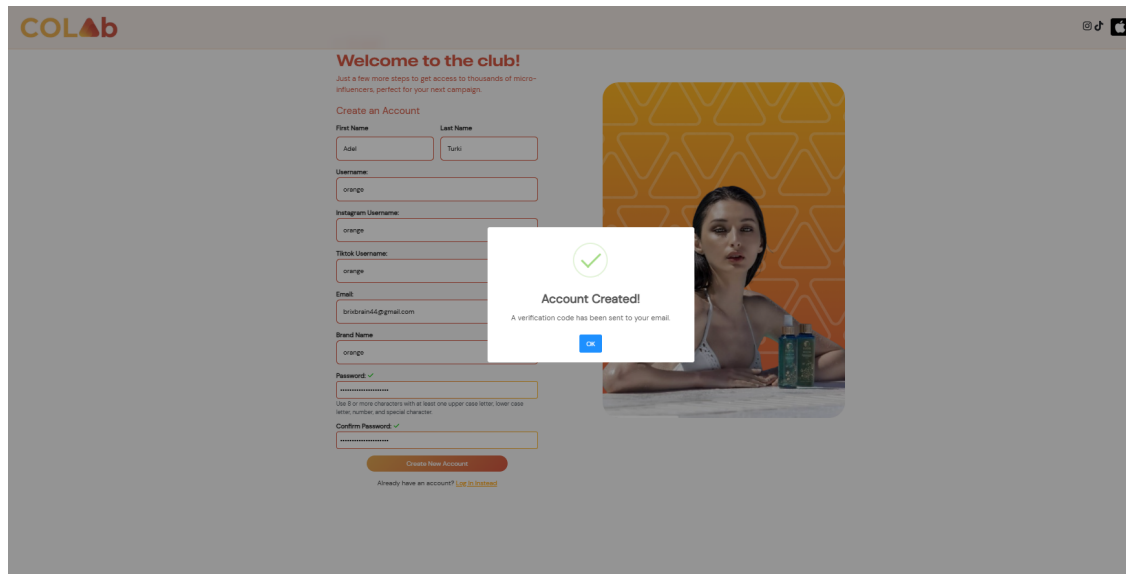


Figure 29: Sign-up Confirmation Message

Email Verification

To confirm identity, users must enter a verification code sent to their inbox.

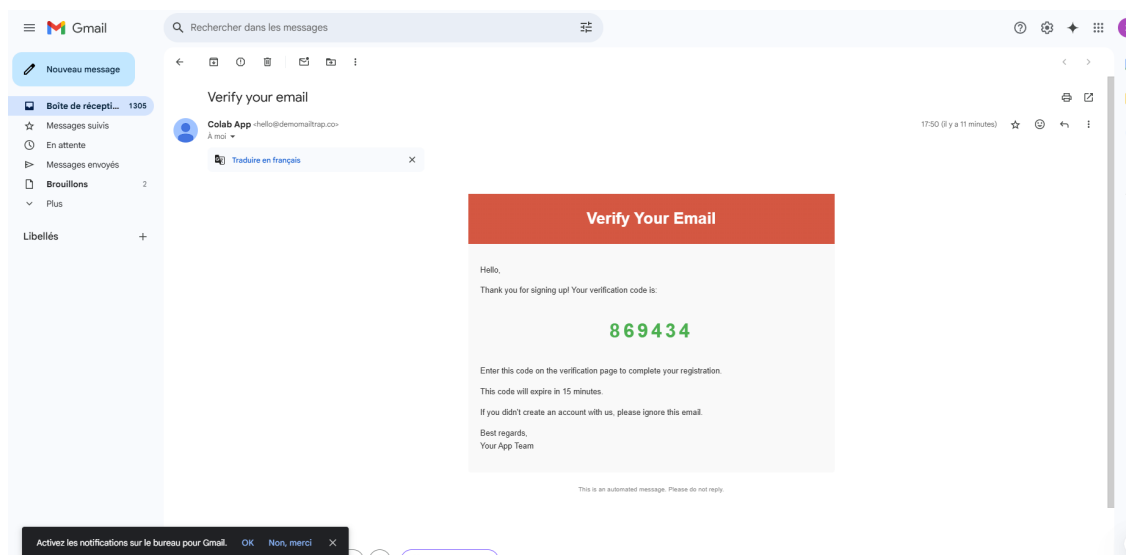


Figure 30: Verify Email with Code

Verify Email

To confirm identity, users must enter a verification code sent via email.

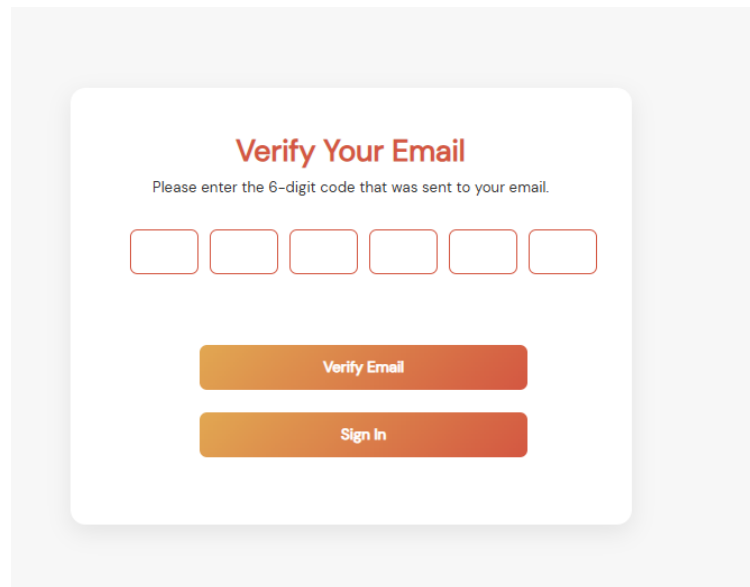
A white rectangular form with rounded corners is centered on a light gray background. At the top, the text "Verify Your Email" is written in a bold, dark red font. Below it, in a smaller, gray font, is the instruction "Please enter the 6-digit code that was sent to your email." Underneath the text are six empty, rounded square input boxes arranged horizontally. Below the input boxes are two orange buttons with rounded corners. The top button is labeled "Verify Email" and the bottom button is labeled "Sign In" in white text.

Figure 31: Verify Email

Email Verification Confirmation

Once the verification code is submitted correctly, users receive a success message confirming their email is verified.

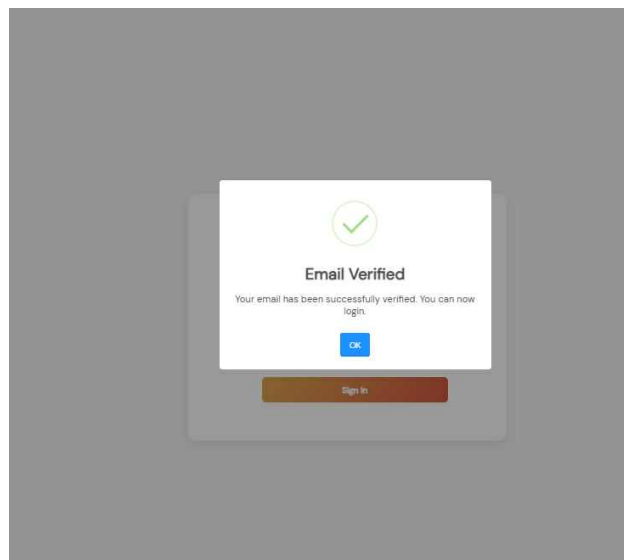


Figure 32: Email Successfully Verified

Reset Password Request

Users who forgot their password can request a reset link by email.

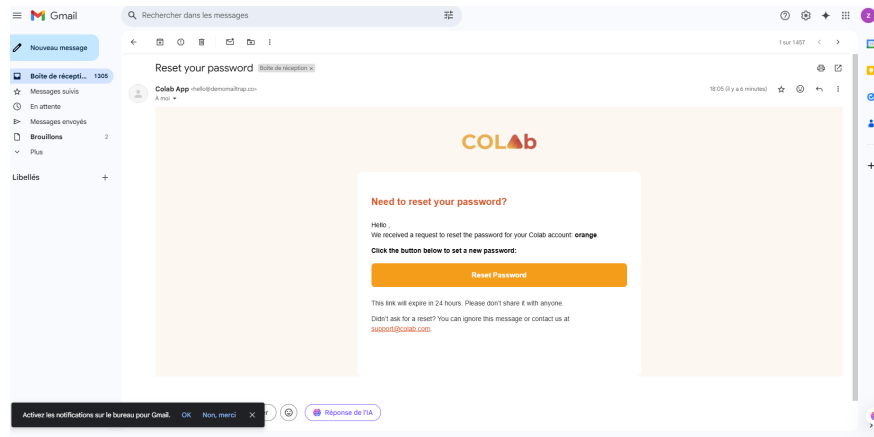


Figure 33: Password Reset Request Email

Reset Password Page

After clicking the reset link, users are prompted to create a new password.

Figure 34: Reset Password Entry Form

4.3 User Profile Update

Update Profile Information

Users can update their profile at any time to reflect changes in personal or professional details. This form allows editing of the username, full name, description, tags, and categories.

COLAB

Hey, **Azza**

Please update your company profile.

Build Your Profile

Username
azza@smene

First Name
Azza

Last Name
slimene

Description
a Tunisian model and actress who made her international fashion debut by walking the runway for Chanel during their

Tags (comma separated)
Enter tags separated by commas

model acting

Categories
Social Media Branding

Save Profile

Skip Photo Upload

Figure 35: Update Profile Information

Update Profile Picture

Users also have the option to update their profile picture. Supported file types are .png, .jpg, and .jpeg with a maximum size of 2MB.

COLAB

Hey, **Azza**

Please update your company profile.

Update Profile Picture

Upload Profile Picture

Upload File

Profile Picture

Go to Dashboard

Figure 36: Update Profile Picture

4.4 Brand Profile View

Profile Overview

The brand profile shows the company name, industry, logo, notifications, and a quick-access button to create a new brief

orange AdelTurki_Orange

Company: Orange Industry: Telecommunication

Create a New Brief

Figure 37: Brand Profile – Orange

Search Creators

This interface allows brands to search for and explore influencer profiles using name, bio, or category filters. Each card displays the creator's handle, bio snippet, follower count, and score.

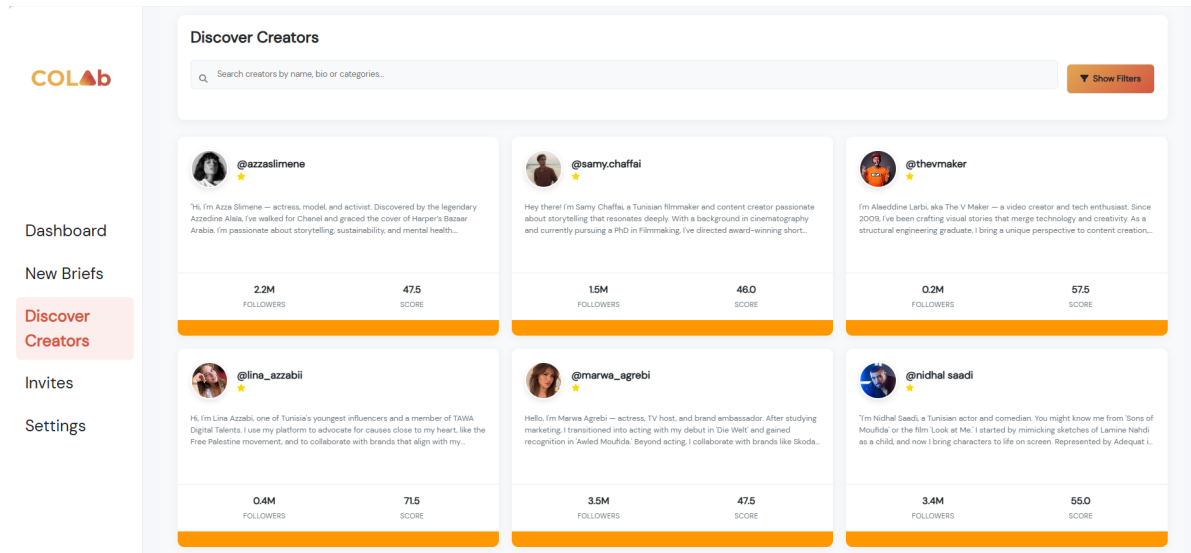


Figure 38: Discover Creators Page

4.5 Admin Panel

Dashboard Overview

The admin dashboard provides a summary of platform activity, including total briefs, campaigns, active creators, and registered advertisers. It also features a panel for recent system updates.

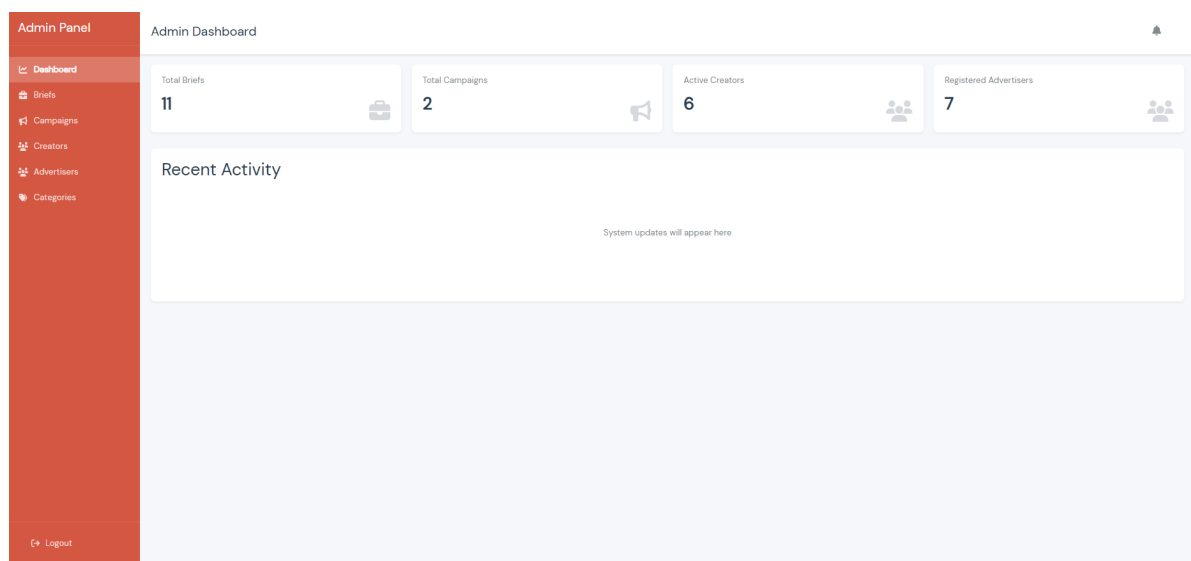


Figure 39: Admin Dashboard

Manage Creators

Admins can view and manage creator accounts with details such as username, email, status, follower counts, and platform scores. Actions include viewing, editing, verifying, assigning roles, or deleting profiles.

Username	Email	Status	Followers	Score	Joined At	Actions
azzaslimene	AzzaslimeneO@gmail.com	Active	2167.461K (IG) / 1468K (TT)	47.5	6/5/2025	View Edit Verify Assign Role Delete
samy.chaffai	Samy.chaffai@gmail.com	Active	1549.303K (IG) / 1200K (TT)	46	6/5/2025	View Edit Verify Assign Role Delete
thevmaker	TthevmakerO@gmail.com	Active	167.512K (IG) / 389K (TT)	57.512695	6/5/2025	View Edit Verify Assign Role Delete
lina_azzabii	Lina_azzabii@gmail.com	Active	430.098K (IG) / 785K (TT)	71.451365	6/5/2025	View Edit Verify Assign Role Delete
marwa_agrebi	Mmarwa_agrebi@gmail.com	Active	3496.731K (IG) / OK (TT)	47.5	6/5/2025	View Edit Verify Assign Role Delete
nidhal saadi	NsaadiO@gmail.com	Active	3369.389K (IG) / OK (TT)	55	6/5/2025	View Edit Verify Assign Role Delete

Figure 40: Admin – Manage Creators

Manage Advertisers

Advertiser management enables admins to monitor company accounts, check status, and view information like company name, industry, email, and join date. Each row includes quick actions to view, edit, verify, manage company data, or remove entries.

Company	Username	Email	Status	Industry	Joined At	Actions
APPLE	brixbrain	brixbrain44@gmail.com	Active	mobile	4/24/2025	View Edit Verify Manage Data Delete
SVR	mohsenO1	MohsenO1@gmail.com	Active	beauty	4/30/2025	View Edit Verify Manage Data Delete
Orange	AdelTurki_Orange	AdelO1@gmail.com	Active	Telecommunication	6/5/2025	View Edit Verify Manage Data Delete
Maestro	FatmaRiahiMaestro	FatmariahiO1@gmail.com	Active	food	6/5/2025	View Edit Verify Manage Data Delete
Skoda	MohammedOSkoda	MohammedOSkoda@gmail.com	Active	cars	6/5/2025	View Edit Verify Manage Data Delete

Figure 41: Admin – Manage Advertisers

Manage Categories

Admins can add, edit, or delete content categories used to classify creators and campaigns. Each entry includes the category name, creation date, and action icons for quick updates.

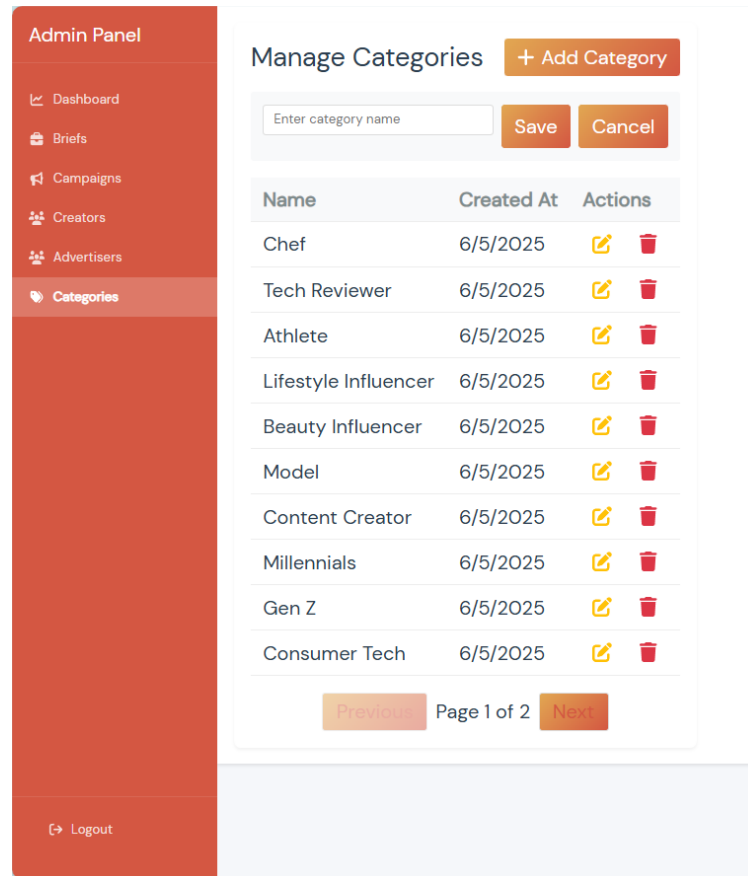


Figure 42: Admin – Manage Categories

5 Conclusion

Sprint 1 successfully implements the core user management functionalities, including authentication, profile management, and administrative operations. These essential features establish a secure and structured foundation for user access and control, paving the way for the integration of more advanced collaboration tools in the upcoming sprints.

Chapter 4 : Sprint 2 – User Management Essentials

1 Introduction

While following the previously defined project structure, Sprint 2 focuses on delivering core functionalities for brief management and brand dashboards. This sprint includes the implementation of brief creation, update, consultation, and application features, along with dashboards for monitoring brief performance, supporting strategic planning and collaboration.

2 Specification of Requirements

In this section, we present the Sprint 2 backlog, the corresponding use case diagram, and a selection of user story descriptions relevant to this sprint. These elements focus on the implementation of brief management workflows and dashboard functionalities for brands and creators.

2.1 Sprint Backlog

ID	Feature	Us ID	User Story	Priority	Complexity
3	Brief Management	3.1	As a brand, I want to create briefs to define brief requirements.	High	13
		3.2	As a brand, I can update my brief.	Medium	8
		3.3	As a brand, I can delete my brief.	Medium	5
		3.4	As a brand, I can consult my own briefs.	Medium	8
		3.5	As a user, I can consult all briefs.	High	8
		3.6	As a creator, I can express interest in briefs.	Medium	8
		3.7	As a creator, I can apply for briefs.	High	13
6	Notification System	6.1	As a user, I want to receive notifications to stay informed.	High	8
7	Admin Panel	7.4	As an admin, I can change the status of briefs.	High	13
8	Dashboard Views	8.1	As a brand, I can consult a dashboard with brief statistics and performance.	High	8

Table 22: Sprint 2 Backlog

2.2 Use Case Diagram

Figure 43 shows the use case diagram for Sprint 2. It highlights the main functionalities developed in this sprint, including brief creation, consultation, application, and administrative oversight. Additionally, it illustrates dashboard access for brands to monitor brief performance.

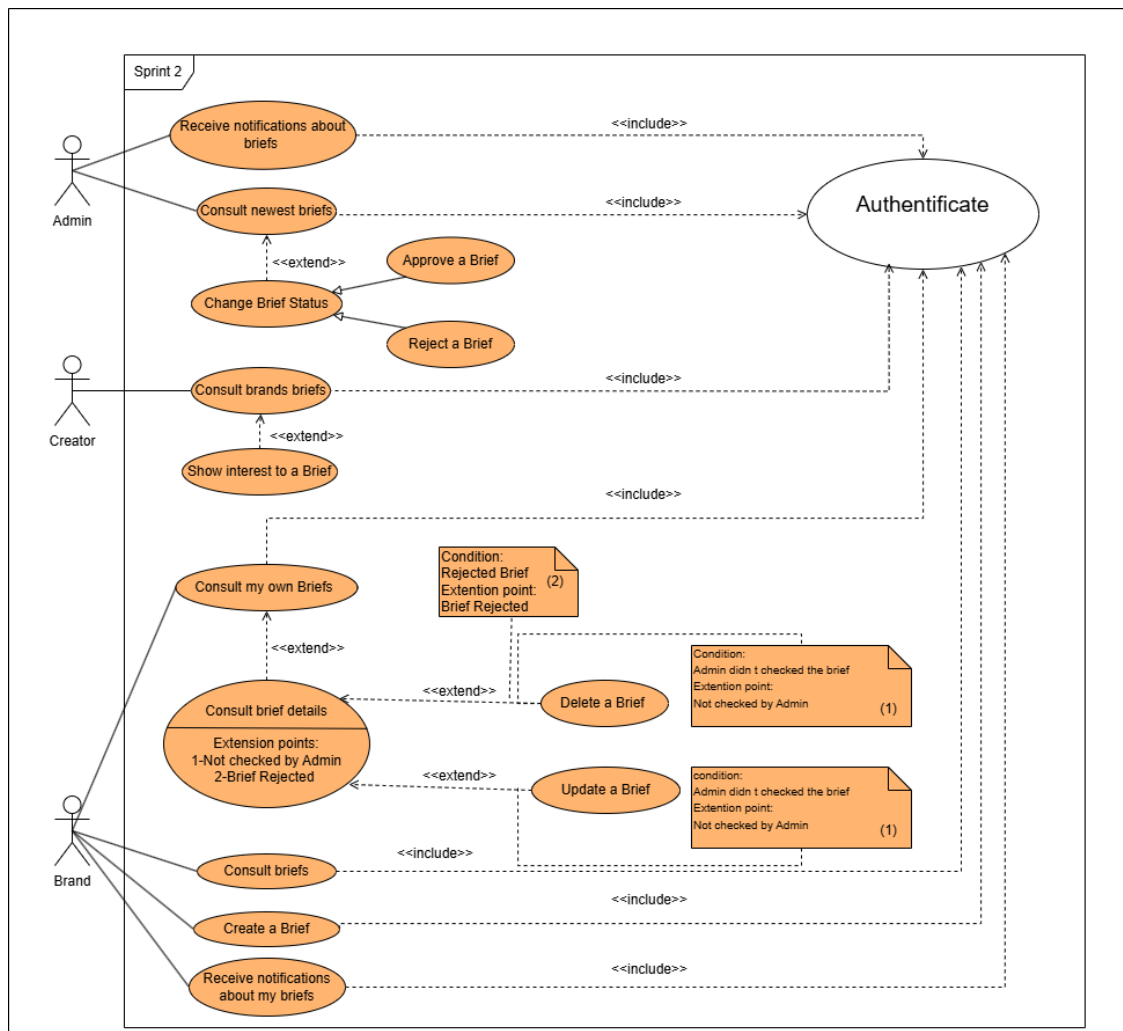


Figure 43: Use Case Diagram – Sprint 2

2.3 Class Diagram

Figure 44 illustrates the class diagram of the second sprint. In this diagram, the colored classes represent the newly developed or updated entities for this sprint, particularly focusing on expanding the campaign and scoring functionalities. These include components such as **Brief** and **Notification**. This sprint builds on the initial user management features and introduces mechanisms for campaign management, performance scoring, and notifications within the Colab platform.

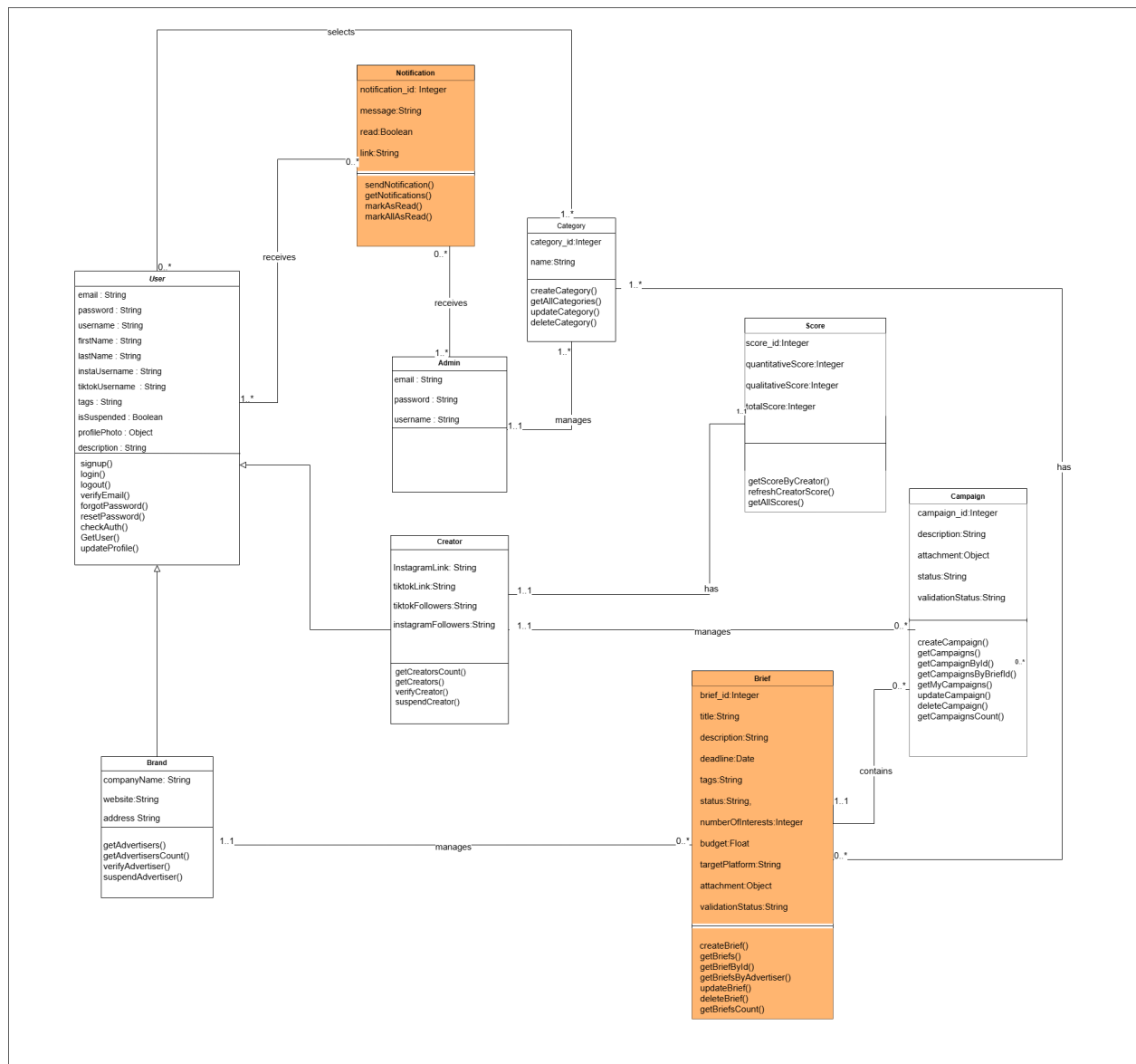


Figure 44: Class Diagram of Sprint 2

2.4 Textual description of use cases

In this section, we present textual descriptions of key use cases specific to Sprint 2 of the Colab platform. These use cases focus on brief management interactions between brands, creators, and admins, along with supporting status changes and dashboard views. They highlight practical platform scenarios that improve collaboration and campaign flow.

2.4.1 Brief Management

Use Case - Consult Briefs List

Use Case Name	Consult Briefs List
Actor(s)	User (Brand or Creator)
Pre-conditions	The user is authenticated.
Post-conditions	A list of all public briefs is displayed.
Nominal Flows	1. The user requests to view the list of briefs. 2. The system queries the database for all public briefs. 3. The retrieved data is displayed to the user.
Exceptional Flows	1. No briefs available : The system informs the user that no briefs are currently published.

Table 23: Use Case - Consult Briefs List

Use Case - Consult My Briefs List

Use Case Name	Consult My Briefs List
Actor(s)	Brand
Pre-conditions	The brand user is authenticated.
Post-conditions	A list of briefs created by the logged-in brand is displayed.
Nominal Flows	1. The brand user opens the dashboard. 2. The system identifies the brand's user ID. 3. Fetches briefs linked to that ID from the database. 4. The system returns and displays the list of briefs.
Exceptional Flows	1. No briefs found : The system informs the user that no briefs are currently available.

Table 24: Use Case - Consult My Briefs List

Use Case - Create New Brief

Use Case Name	Create New Brief
Actor(s)	Brand
Pre-conditions	The brand user is authenticated. The user has access to the brief creation form.
Post-conditions	A new brief is created and stored in the database.
Nominal Flows	1. The brand user accesses the brief creation form. 2. The user fills in the required fields. 3. The system validates the input data. 4. The brief is saved in the database. 5. A success message is displayed.

Exceptional Flows	1. Required fields missing : The system alerts the user to complete all mandatory fields. 2. Invalid input format : The system highlights incorrectly formatted data and prompts for correction.
--------------------------	---

Table 25: Use Case - Create New Brief

Use Case - Consult Brief Details

Use Case Name	Consult Brief Details
Actor(s)	User (Brand or Creator)
Pre-conditions	The user is authenticated. The user has access to at least one brief.
Post-conditions	The full details of the selected brief are displayed.
Nominal Flows	1. The user selects a brief to view details. 2. The system receives the brief ID and fetches the corresponding data. 3. The system returns the brief information. 4. The full brief details are displayed to the user.
Exceptional Flows	None explicitly shown in this flow.

Table 26: Use Case - Consult Brief Details

Use Case - Update Brief

Use Case Name	Update Brief
Actor(s)	Brand
Pre-conditions	The brand is authenticated. The brand has access to an existing brief.
Post-conditions	The brief is successfully updated with the new data.
Nominal Flows	1. The brand accesses the edit form for a specific brief. 2. The brand modifies the content and submits the updated form. 3. The system validates and saves the updated information. 4. A confirmation message “Brief updated” is displayed.
Exceptional Flows	1. Invalid input : The system displays an error prompting the brand to correct the data.

Table 27: Use Case - Update Brief

Use Case - Delete Brief

Use Case Name	Delete Brief
Actor(s)	Brand
Pre-conditions	The brand is authenticated. The brand owns the brief to be deleted.
Post-conditions	The selected brief is removed from the system.
Nominal Flows	1. The brand clicks the "Delete" button for a specific brief. 2. The system receives the request with the brief ID. 3. The system deletes the brief from the database. 4. The system displays a confirmation message "Brief deleted."
Exceptional Flows	1. Deletion cancelled : The brand aborts the deletion process before confirmation. 2. Missing confirmation : The system prompts the brand to confirm before deleting.

Table 28: Use Case - Delete Brief

Use Case - Express Interest in Brief

Use Case Name	Express Interest in Brief
Actor(s)	Creator
Pre-conditions	The creator is authenticated. The creator can view available briefs.
Post-conditions	Interest is successfully recorded. Notifications are sent to the advertiser and administrators.
Nominal Flows	1.The creator clicks the "Show Interest" button for a brief. 2. The system retrieves the selected brief using its ID. 3. The system increments the brief's interest counter. 4. The system sends notifications to the advertiser and all admins. 5. The system displays a success confirmation message.
Exceptional Flows	None explicitly shown in this flow.

Table 29: Use Case - Express Interest in Brief

2.4.2 Notification System

Use Case - Receive Notifications

Use Case Name	Receive Notifications
Actor(s)	User (Brand or Creator)
Pre-conditions	The user is authenticated. The user accesses the dashboard.
Post-conditions	All notifications are retrieved. Unread notifications are visually highlighted.
Nominal Flows	1. The user opens the dashboard. 2. The system retrieves the user's notifications. 3. The system displays the notifications list. 4. Unread items are visually emphasized.
Exceptional Flows	1. No notifications available : The system informs the user that there are no new notifications.

Table 30: Use Case - Receive Notifications

2.4.3 Admin Panel

Use Case - Consult Brief Table

Use Case Name	Consult Brief Table
Actor(s)	Admin
Pre-conditions	The admin is authenticated. The admin has access to the admin panel.
Post-conditions	All briefs are retrieved from the system. Data is displayed in a structured table.
Nominal Flows	1. Admin opens the admin panel. 2. System invokes the research of briefs. 3. The model returns the list of briefs. 4. System displays the data in a table view.
Exceptional Flows	1. No briefs available : The system displays a message indicating no data is currently available.

Table 31: Use Case - Consult Brief Table

Use Case - Update Brief Status

Use Case Name	Update Brief Status
Actor(s)	Admin
Pre-conditions	Admin is authenticated. Admin has access to the brief table.
Post-conditions	The brief's status is updated. A notification is sent to the advertiser.
Nominal Flows	1. Admin clicks "Approve" or "Reject" on a brief. 2. System retrieves the selected brief and updates its status. 3. If necessary, a notification is generated and sent to the advertiser. 4. The system displays the updated status in the brief table.
Exceptional Flows	1. Invalid action : Admin attempts to change status of a brief that is already finalized.

Table 32: Use Case - Update Brief Status

3 Detailed Sequence Diagrams

This section presents key sequence diagrams from Sprint 2, highlighting interactions between users and system components for core features such as brief creation, consultation, application, and admin actions. These diagrams illustrate the logical flow of user actions and system responses.

3.1 Brief Management

This section captures core actions related to managing briefs, including creation, listing, updating, and deletion functionalities performed by brand users.

3.1.1 Consult Briefs List

This sequence diagram illustrates the process through which an authenticated user retrieves and views the details of a specific brief.

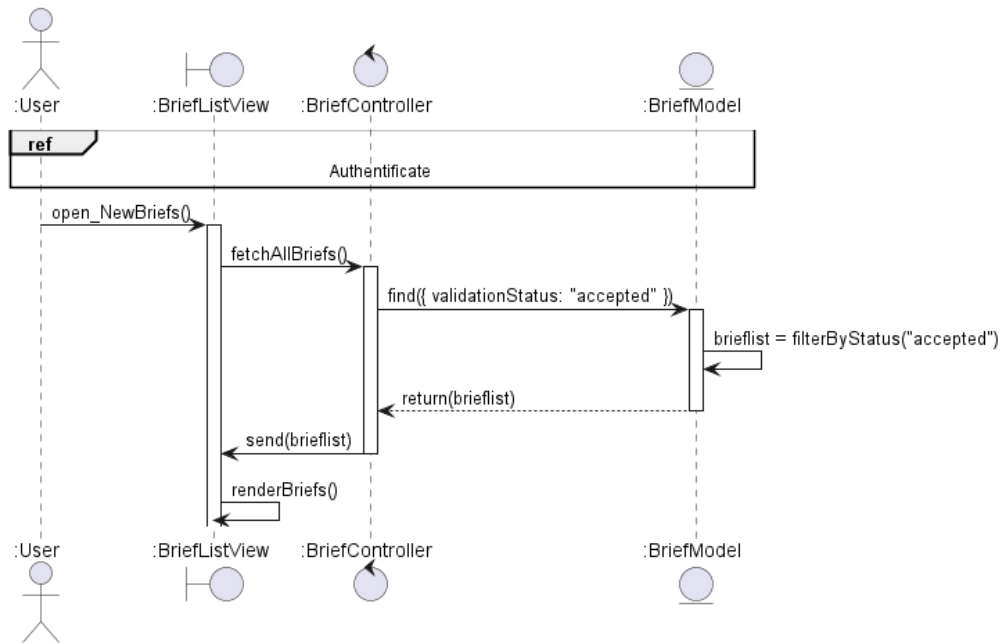


Figure 45: Sequence diagram for consulting list of briefs

3.1.2 Consult My Briefs List

Figure 46 shows how a brand user retrieves and views briefs associated with their account after authentication.

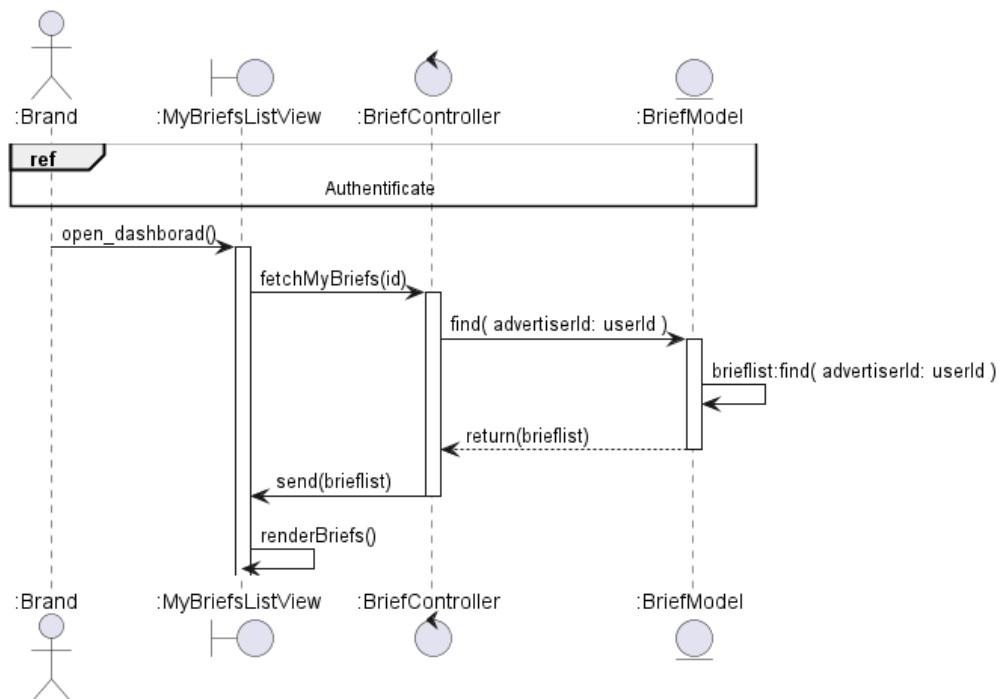


Figure 46: Brand – Consult My Briefs List

3.1.3 Create New Brief

This diagram shows how a brand user submits a new brief, which is saved by the system and confirmed with a success alert.

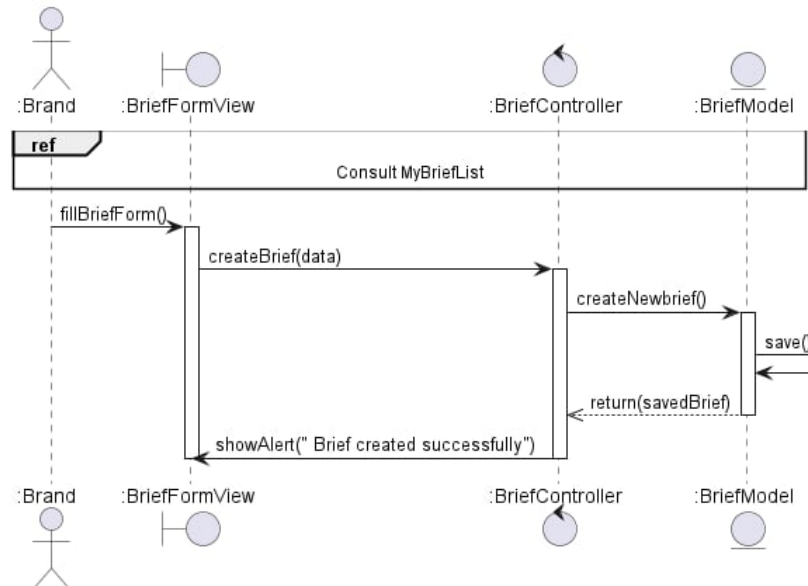


Figure 47: Sequence diagram for creating a new brief

3.1.4 Consult Brief Details

This diagram shows how a user selects a brief from their list and views its details .

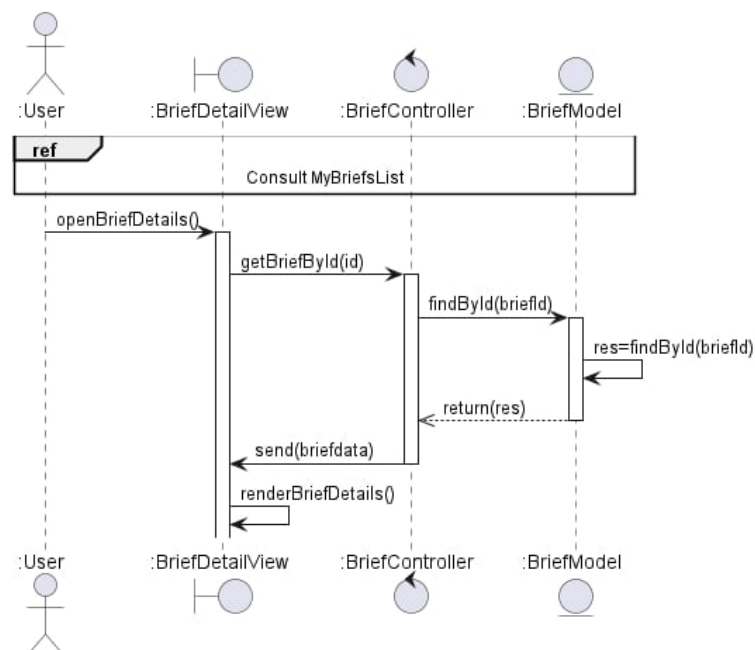


Figure 48: Sequence diagram for consulting brief details

3.1.5 Update Brief

This diagram shows how a brand user submits updated brief information and how the system processes this update.

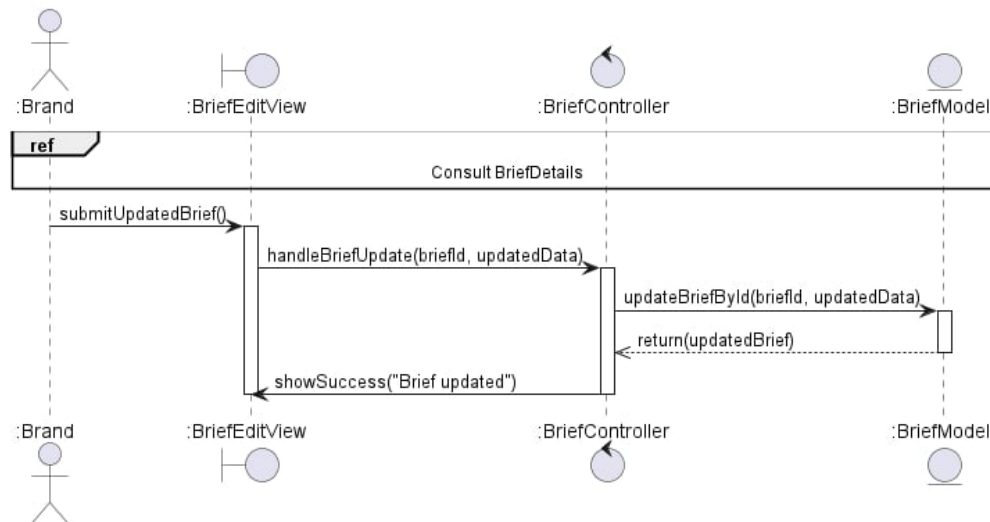


Figure 49: Sequence diagram for updating a brief

3.1.6 Delete Brief

This sequence illustrates the flow for deleting a brief by its identifier.

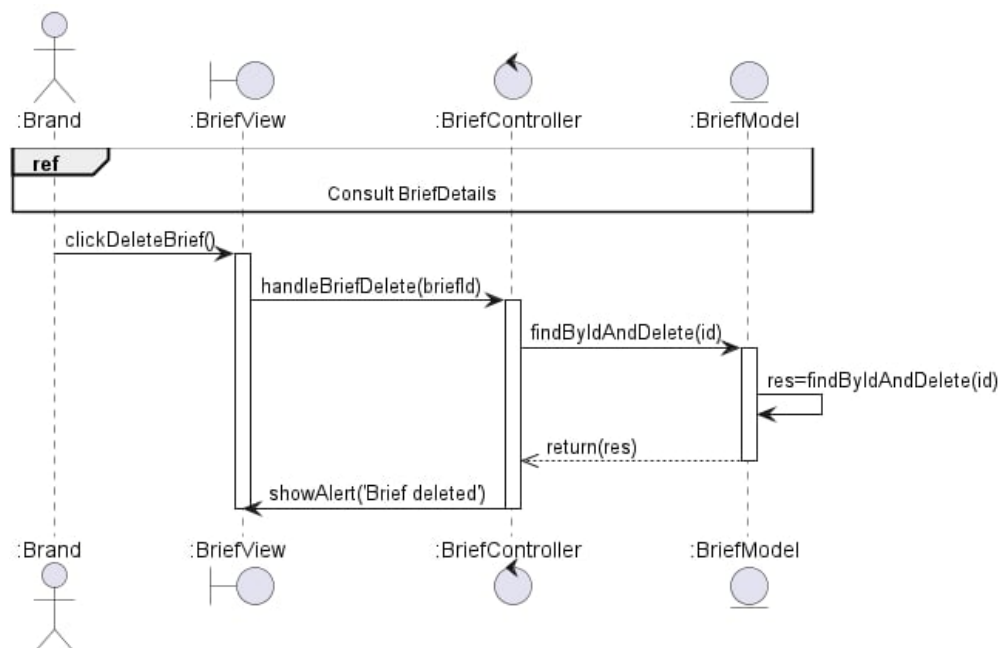


Figure 50: Sequence diagram for deleting a brief

3.1.7 Express Interest

Figure 51 details the flow where a creator expresses interest in a brief. The system verifies the brief and user, checks for an existing campaign, and if none exists, updates the brief and sends notifications to the advertiser and all admins.

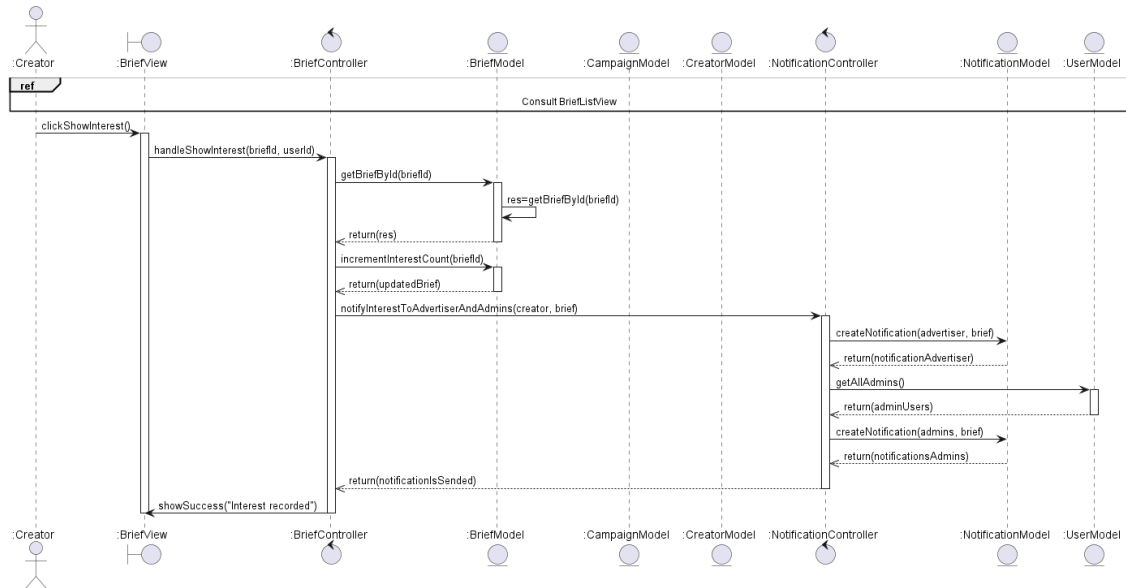


Figure 51: Show Interest in Brief

3.2 Notification System

This section outlines how users receive brief-related notifications when they interact with the dashboard. Figure 52 illustrates the process where a user opens the dashboard and the system fetches all unread notifications. These are displayed with animation to highlight new alerts.

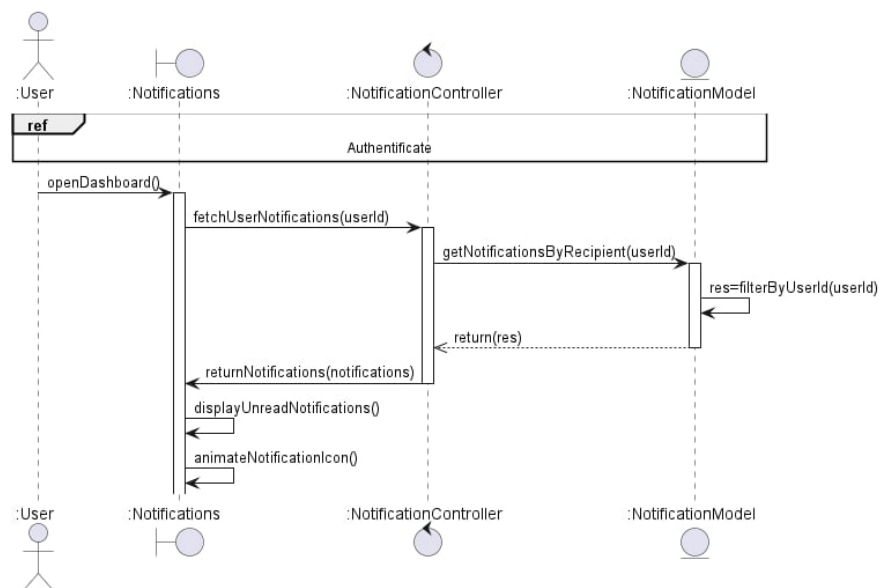


Figure 52: Receive Notifications

3.3 Admin Panel

Figure 54 describes how an admin approves or rejects a submitted brief. The system verifies the admin role, checks the brief status, applies the update, and notifies the brand accordingly.

3.3.1 Consult Brief Table

Figure 53 illustrates how an admin retrieves and views all briefs in the system after authentication.

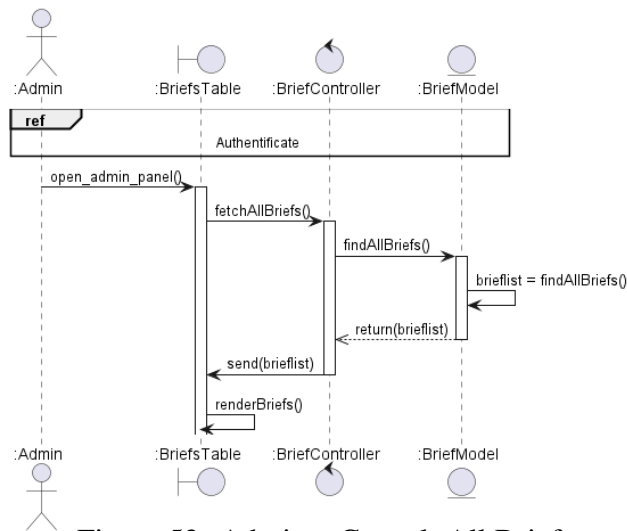


Figure 53: Admin – Consult All Briefs

3.3.2 Update Brief Status

Figure 54 describes how an admin approves or rejects a submitted brief. The system verifies the admin role, checks the brief status, applies the update, and notifies the brand accordingly.

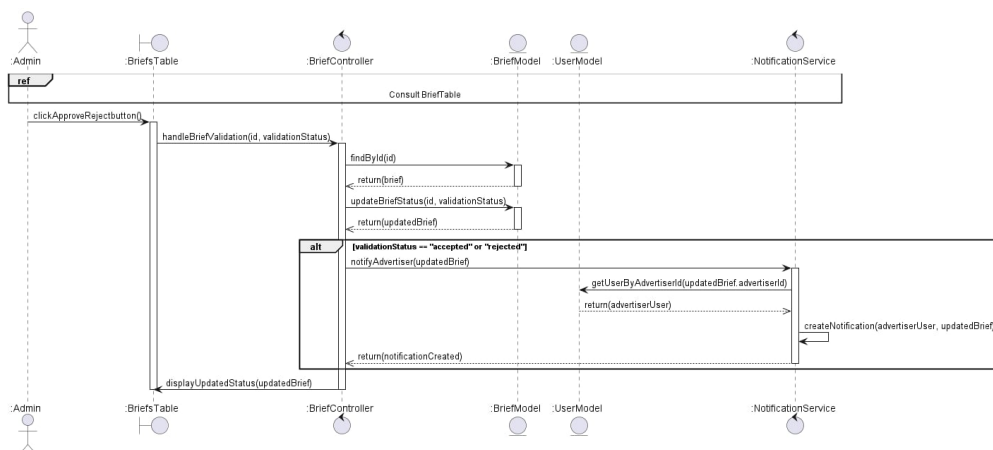


Figure 54: Admin – Brief Status Update

4 Implementation

4.1 Brief Management – Creation Flow

Step 1 : Brief Overview

The brand starts by entering the brief's title, description, and final submission deadline.

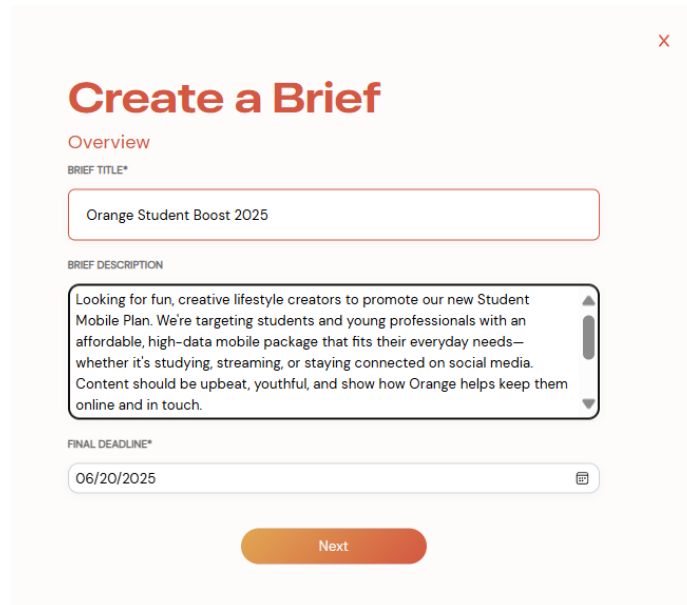
The screenshot shows a web form titled "Create a Brief" with a sub-header "Overview". It contains three main input fields: "BRIEF TITLE*" with the value "Orange Student Boost 2025", "BRIEF DESCRIPTION" with a text area containing a paragraph about targeting students and young professionals, and "FINAL DEADLINE*" with the date "06/20/2025". A "Next" button is at the bottom right. A close button (X) is in the top right corner.

Figure 55: Create a Brief – Overview

Step 2 : Define Deliverables

Brands select the brief's categories, tags, and platforms relevant to the campaign.

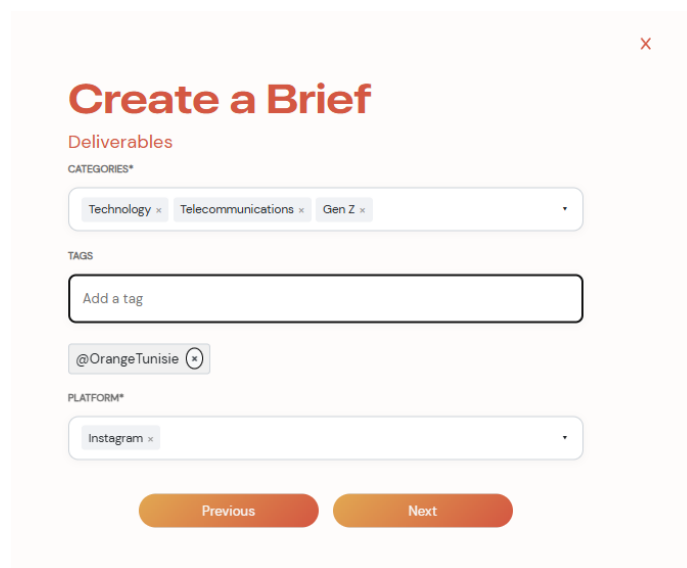
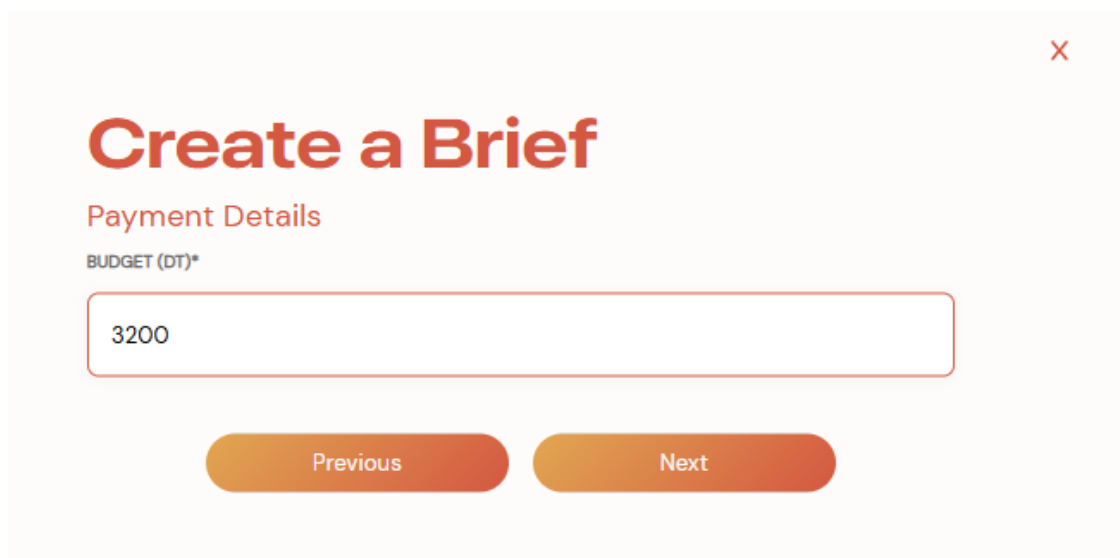
The screenshot shows a web form titled "Create a Brief" with a sub-header "Deliverables". It contains three main input fields: "CATEGORIES*" with a dropdown menu showing "Technology", "Telecommunications", and "Gen Z"; "TAGS" with a text input field containing "@OrangeTunisie"; and "PLATFORM*" with a dropdown menu showing "Instagram". "Previous" and "Next" buttons are at the bottom. A close button (X) is in the top right corner.

Figure 56: Create a Brief – Deliverables

Step 3 : Set Budget

Brands define the financial allocation for the brief in Tunisian Dinar.



Create a Brief

Payment Details

BUDGET (DT)*

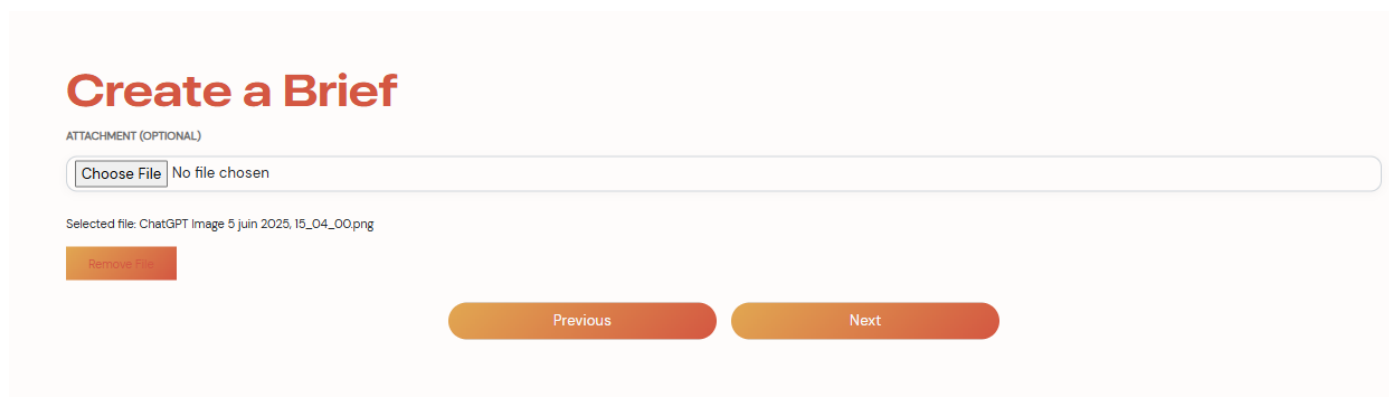
3200

Previous Next

Figure 57: Create a Brief – Budget

Step 4 : Add Attachment (Optional)

Optionally, a file can be uploaded to support the brief content.



Create a Brief

ATTACHMENT (OPTIONAL)

Choose File No file chosen

Selected file: ChatGPT Image 5 juin 2025, 15_04_00.png

Remove File

Previous Next

Figure 58: Create a Brief – Attachment Upload

Step 5 : Preview and Submit

A preview screen displays all brief details before submission, including a summary and contract terms.

Create a Brief

Preview

BRIEF OVERVIEW

Title	Orange Student Boost 2025
Description	Looking for fun, creative lifestyle crea...
Review Deadline	June 19th 2025
Final Deadline	June 20th 2025

CATEGORIES & TAGS

Categories	Technology, Telecommunications, Gen Z
Tags	@OrangeTunisie

PLATFORM & BUDGET

Target Platform	@ Instagram
Budget	3200 Dt

CONTRACT

The creator will provide the following in accordance to the content guidelines:

The creator must upload all content on CoLab by June 19th 2025, 12:00:00 am for review by the brand. All content must be uploaded by June 20th 2025, 12:00:00 am after the Creator receives approval from the brand.

The Creator grants the Brand a worldwide, irrevocable, royalty-free, fully paid-up, transferrable, sub-licensable, and perpetual right and license to reproduce, publish, distribute, display, repost, share and edit all Creator created for or on behalf of the Brand in any and all media now known or developed in the future.

The Creator accepts the terms of the foregoing proposal and agree to the Privacy Policy and Terms & Conditions.

ATTACHMENT

File	ChatGPT Image 5 juin 2025, 15_04_00.png
------	---

[Previous](#) [Create Brief](#)

Figure 59: Create a Brief – Final Preview

View My Briefs

Brands can view all briefs they've created from their dashboard. Each card displays a summary including title, tags, deadline, and budget.

COLab

orange AdelTurki_Orange

Company: Orange Industry: Telecommunication

[+ Create a New Brief](#)

My Briefs [Show Filters](#)

Orange City Pulse

We're partnering with dynamic creators to showcase how Orange powers the modern urban lifestyle. Whe...

Technology Telecommunications

Due: 7/12/2025 2500 DT

Orange Family Moments

We're looking for warm, heartfelt content from creators that shows how Orange Mobile keeps families ...

Lifestyle Influencer Technology Telecommunications

Due: 7/8/2025 2000 DT

Orange Speed Unleashed 2025

We're calling on tech-savvy and lifestyle creators to help us highlight the ultra-fast, reliable mob...

Telecommunications Technology Content Creator

Due: 7/3/2025 4000 DT

Figure 60: Consult My Briefs

Brief Details View

Selecting a brief opens a detailed view with sections for campaign brief, deliverables, categories, budget, and any attached reference materials. Brands can also edit or delete briefs here.

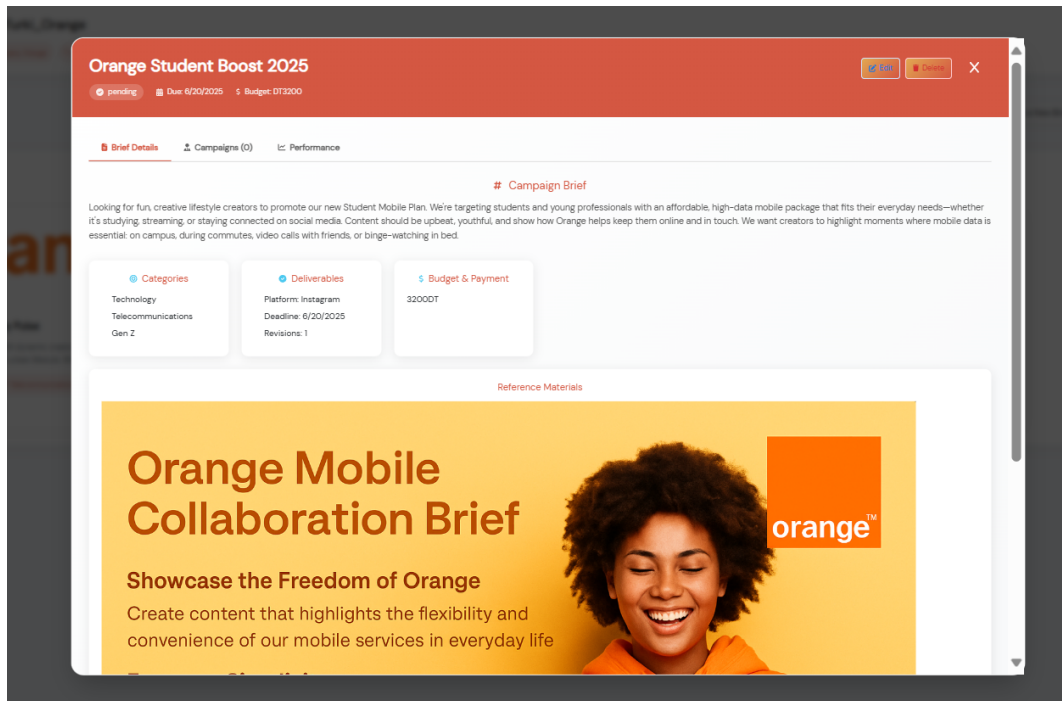


Figure 61: Brief Details View

Update Brief

From the details view, brands may update the campaign title, description, review deadline, and final deadline.

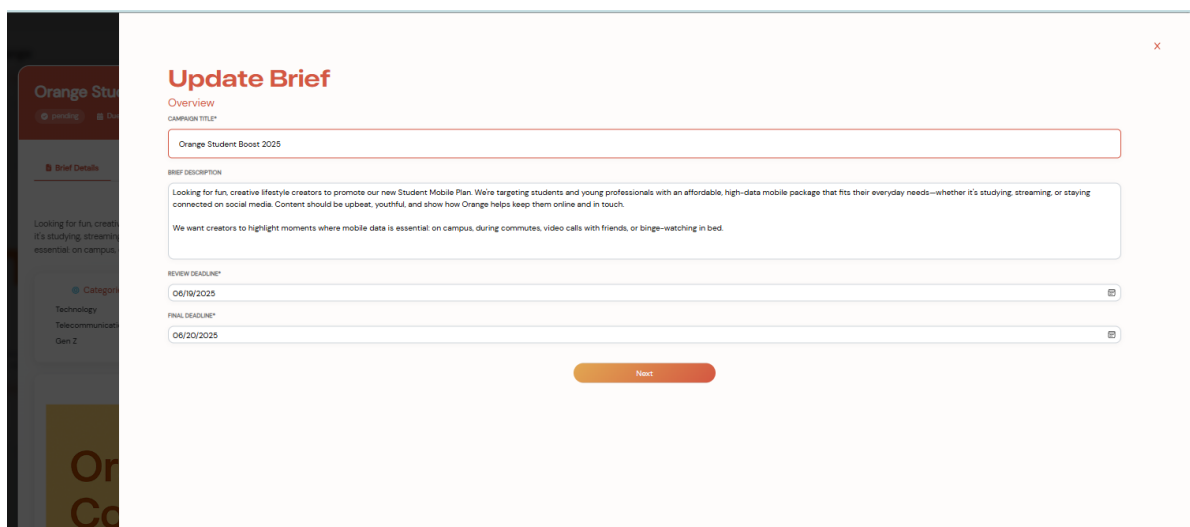


Figure 62: Update Brief

Delete Brief

When a brand chooses to delete a brief, a confirmation dialog appears to prevent accidental data loss. This step ensures deliberate action before permanent deletion.

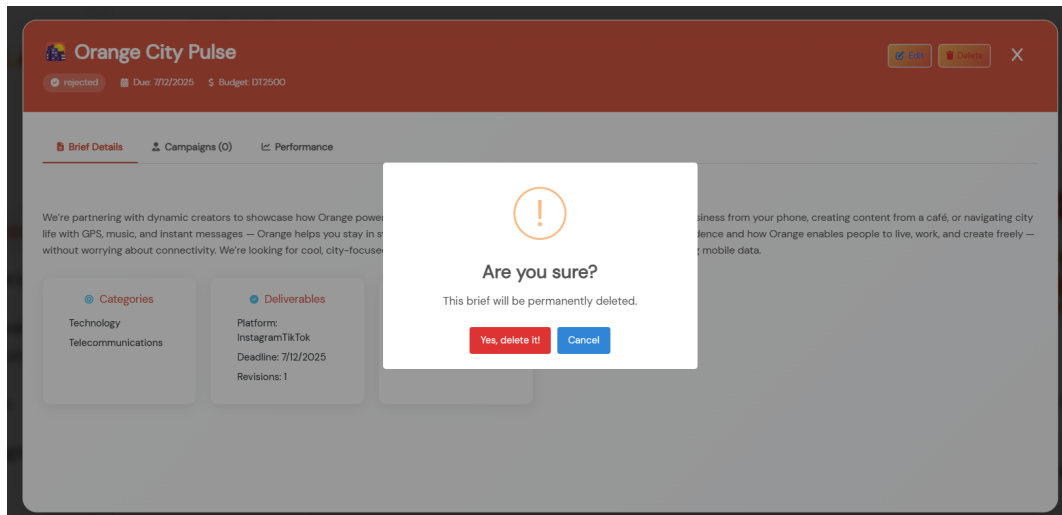


Figure 63: Brief Deletion Confirmation Prompt

Show Interest in a Brief

Creators can view available briefs and express their interest through a dedicated action. This indicates their willingness to participate and be considered for collaboration by the brand.

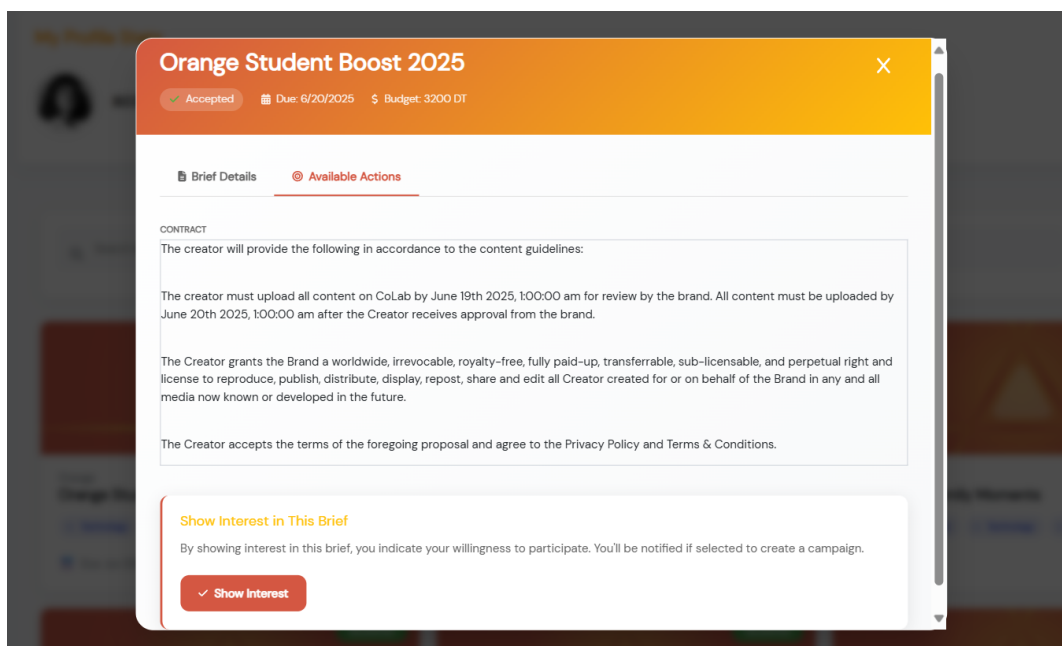


Figure 64: Creator – Express Interest in a Brief

4.2 Notification System

Users receive real-time updates regarding the status of their briefs. Notifications include approvals, rejections, and other relevant changes made by the admin.

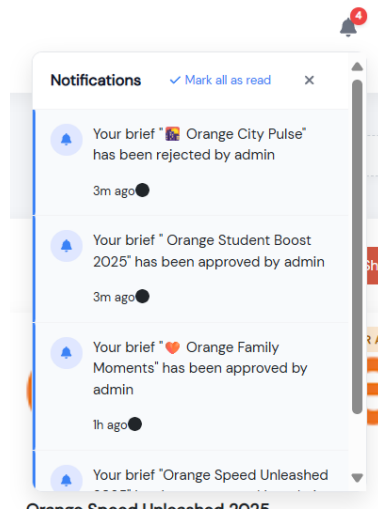


Figure 65: Notification Panel

4.3 Admin Panel – Brief Status Management

Admins can view all submitted briefs and manage their validation status (pending, accepted, or rejected). Each brief entry displays its title, advertiser, budget, deadline, and current validation state. Quick action icons allow status updates, edits, deletions, or detail views.

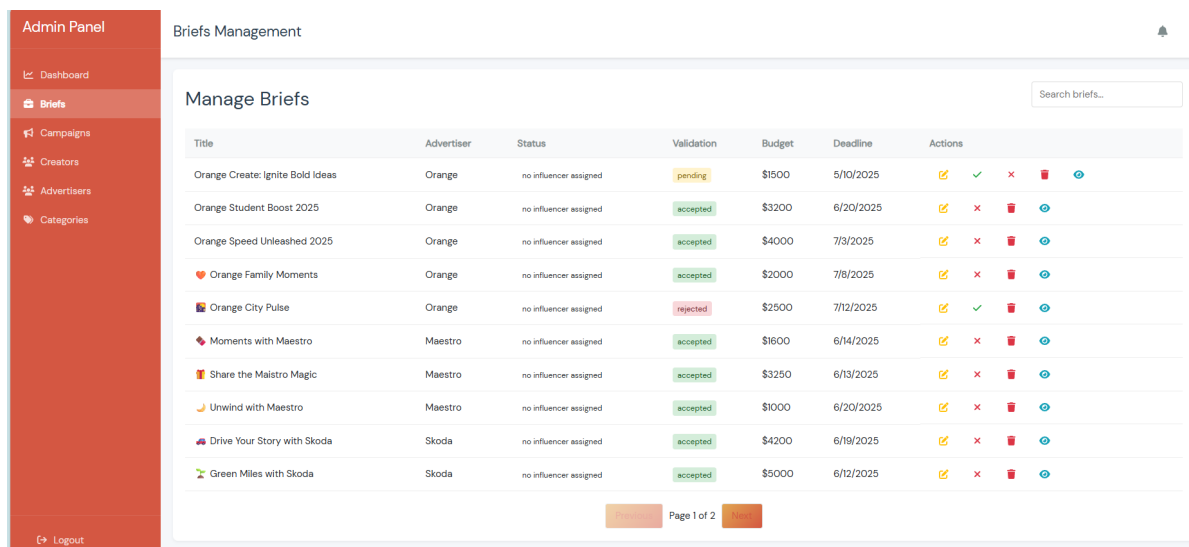


Figure 66: Admin – Briefs Management and Status Change

5 Conclusion

Sprint 2 successfully implements the core functionalities for brief management and dashboard views. Brands can now create, update, consult, and manage briefs, while creators are able to browse, express interest in, and apply to them. Additionally, the admin panel includes oversight capabilities for brief moderation. The integration of brand dashboards allows for performance monitoring, supporting data-driven campaign decisions. These features collectively enhance collaboration, transparency, and planning across the Colab platform.

Chapter 5 : Sprint 3 – Campaign Management, Scoring, Notifications & Creator Dashboards

1 Introduction

While continuing the development of the Colab Platform, Sprint 3 focuses on delivering key functionalities for campaign lifecycle management, creator scoring, user notifications, and creator dashboard views. This sprint enhances creator engagement by enabling them to manage campaigns, view performance metrics, and receive system notifications.

2 Specification of Requirements

In this section, we present the Sprint 3 backlog, the corresponding use case diagram, and a selection of user story descriptions relevant to this sprint. These stories reflect essential features that support creator activity, performance evaluation, and platform communication.

2.1 Sprint Backlog

ID	Feature	Us ID	User Story	Priority	Complexity
4	Campaign Management	4.1	As a creator, I can create campaigns.	High	13
		4.2	As a creator, I can delete campaigns.	Medium	5
		4.3	As a creator, I can update campaigns.	Medium	5
		4.4	As a creator, I can consult my campaigns.	Medium	5
		4.5	As a brand, I can consult the campaigns related to my briefs.	High	13
		4.6	As a brand, I can accept or reject campaigns related to my briefs.	High	8

2.3 Class Diagram

Figure 68 shows the updated class diagram for Sprint 3. New classes such as Campaign and Score were added to support campaign management and social media scoring.

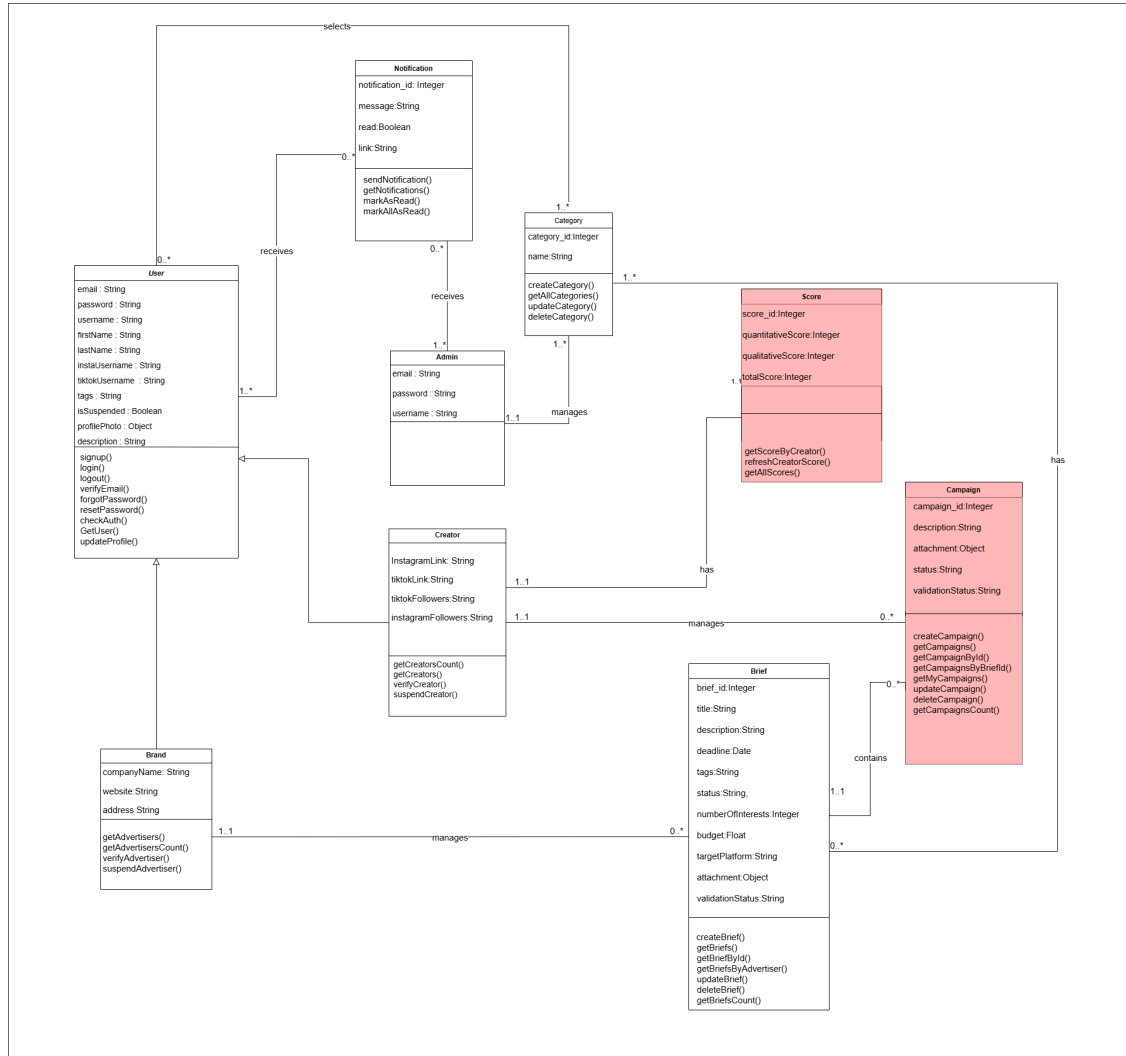


Figure 68: Class Diagram – Sprint 3

2.4 Use Case Sheets

This section presents the main use cases developed during Sprint 3 of the Colab platform. These use cases cover core campaign management activities for creators and brands, including creation, consultation, update, and deletion of campaigns. Additional scenarios also address profile scoring, dashboard statistics, and campaign validation, reflecting enhanced collaboration and evaluation mechanisms.

2.4.1 Campaign Management

Use Case - Consult My Campaigns

Use Case Name	Consult My Campaigns
Actor(s)	Creator
Pre-conditions	Creator is authenticated.
Post-conditions	The creator's campaigns are retrieved and displayed.
Nominal Flows	1. Creator opens the campaign dashboard. 2. System identifies the creator and fetches associated campaigns. 3. The campaign list is displayed on the interface.
Exceptional Flows	1. No campaigns found : The system informs the creator that no campaigns are available to display.

Table 34: Use Case - Consult My Campaigns

Use Case - Create Campaign

Use Case Name	Create Campaign
Actor(s)	Creator
Pre-conditions	Creator is authenticated. A valid brief has been selected.
Post-conditions	A campaign is successfully created. The associated brief is updated. Notifications are triggered.
Nominal Flows	1. Creator fills and submits the campaign form. 2. System fetches the selected brief and creator data. 3. System uploads any included files to Cloudinary. 4. System saves the campaign in the database. 5. System updates the brief's status accordingly. 6. System sends notifications to the advertiser and admins. 7. System confirms with a success message.
Exceptional Flows	1. Incomplete campaign form submission : The system prompts the creator to fill all required fields before submission. 2. Unsupported file type or missing upload : The system alerts the user to upload valid file formats only.

Table 35: Use Case - Create Campaign

Use Case - Consult Campaigns Related to a Brief

Use Case Name	Consult Campaigns Related to a Brief
Actor(s)	Brand
Pre-conditions	Brand is authenticated. Brand has access to a specific brief.
Post-conditions	Approved campaigns related to the brief are displayed.
Nominal Flows	1. Brand opens a brief detail view. 2. System fetches the brief by its ID. 3. System retrieves all campaigns linked to the brief with approved status. 4. System displays the list of campaigns.
Exceptional Flows	1. No approved campaigns : The system informs the brand that no approved campaigns are currently available for this brief.

Table 36: Use Case - Consult Campaigns Related to a Brief

Use Case - Consult Campaign Details

Use Case Name	Consult Campaign Details
Actor(s)	Creator
Pre-conditions	Creator is authenticated and has existing campaigns.
Post-conditions	Detailed information about the selected campaign is displayed.
Nominal Flows	1. Creator selects a campaign to view. 2. System fetches the campaign by ID. 3. Campaign details are rendered on the screen.
Exceptional Flows	None explicitly shown in this flow

Table 37: Use Case - Consult Campaign Details

Use Case - Delete Campaign

Use Case Name	Delete Campaign
Actor(s)	Creator
Pre-conditions	Creator is authenticated and has a campaign to delete.
Post-conditions	Campaign is removed from the database .
Nominal Flows	1. Creator clicks the delete button. 2. System retrieves and deletes the campaign. 3. Success alert is shown.
Exceptional Flows	None explicitly shown in this flow

Table 38: Use Case - Delete Campaign

Use Case - Update Campaign

Use Case Name	Update Campaign
Actor(s)	Creator
Pre-conditions	Campaign exists and creator is authenticated.
Post-conditions	Campaign is updated with new data and notification is sent.
Nominal Flows	1. Creator submits updated campaign data. 2. System fetches campaign by ID. 3. (Optional) File is uploaded. 4. Campaign is updated. 5. Notification is triggered. 6. Success alert is shown.
Exceptional Flows	1. File upload failure : The system shows an error message.

Table 39: Use Case - Update Campaign

Use Case - Approve or Reject Campaign

Use Case Name	Advertiser Campaign Validation
Actor(s)	Advertiser
Pre-conditions	Advertiser is authenticated and reviewing campaigns related to a brief.
Post-conditions	The campaign status is updated and the creator is notified.
Nominal Flows	1. Advertiser clicks the approve/reject button. 2. System retrieves campaign and creator information. 3. Based on decision, updates campaign status. 4. Sends a contextual notification to the creator. 5. Updates the status in the advertiser interface.
Exceptional Flows	1. Notification failure : System logs the issue and alerts the advertiser.

Table 40: Use Case - Advertiser Campaign Validation

Use Case - Consult Profile Statistics

Use Case Name	Consult Profile Statistics
Actor(s)	User (Creator)
Pre-conditions	User is authenticated and has a profile.
Post-conditions	User sees their profile metrics and scoring data.
Nominal Flows	1. User opens dashboard. 2. System fetches user by ID. 3. Creator data with score is retrieved. 4. Additional stats are fetched from external source. 5. Dashboard displays full profile statistics.
Exceptional Flows	1. Score fetch fails : System displays a warning and retries later.

Table 41: Use Case - Consult Profile Statistics

Use Case - Refresh Creator Score Use Case – Validate Campaign

Use Case Name	Refresh Creator Score
Actor(s)	Creator
Pre-conditions	Creator is authenticated and has a valid Instagram username.
Post-conditions	Creator's score is recalculated and updated in the system.
Nominal Flows	1. Creator triggers score refresh. 2. System finds creator by ID. 3. Instagram profile data is fetched. 4. Quantitative and qualitative scores are computed. 5. Scores are saved to the database. 6. System confirms update with alert.
Exceptional Flows	1. Instagram profile not found : System notifies user and aborts update.

Table 42: Use Case - Refresh Creator Score

Use Case Name	Consult Campaigns
Actor(s)	Admin
Pre-conditions	Admin is authenticated and has access to the campaign panel
Post-conditions	The system displays a list of campaigns in the admin interface
Nominal Flows	1. Admin opens the campaign panel 2. System fetches all campaign records from the database 3. The list is rendered in the interface
Exceptional Flows	1. No campaigns found : System displays "No campaigns available"

Table 43: Use Case – Consult Campaigns

Use Case – Validate Campaign

Use Case Name	Validate Campaign
Actor(s)	Admin
Pre-conditions	Admin is authenticated and a campaign exists to validate
Post-conditions	The campaign is marked as approved or rejected, and notifications are sent
Nominal Flows	1. Admin clicks approve or reject 2. System updates the campaign status 3. If rejected : Notify the creator 4. If approved : Notify both the creator and advertiser
Exceptional Flows	None explicitly shown in this flow

Table 44: Use Case – Validate Campaign

3 Sequence Diagrams

In this section, we present three detailed sequence diagrams categorized by the main feature groups defined in the Sprint 3 backlog. These diagrams illustrate the interactions between users (creators, brands, and administrators) and system components during key processes such as campaign management, social media scoring consultation, and dashboard interaction. Each diagram helps visualize the logical flow behind user actions and system responses implemented in this sprint.

3.1 Campaign Management

3.1.1 Consult My Campaigns

Figure 69 describes the sequence that occurs when a creator accesses their list of campaigns. The controller retrieves both the creator's identity and their associated campaigns before rendering them in the view.

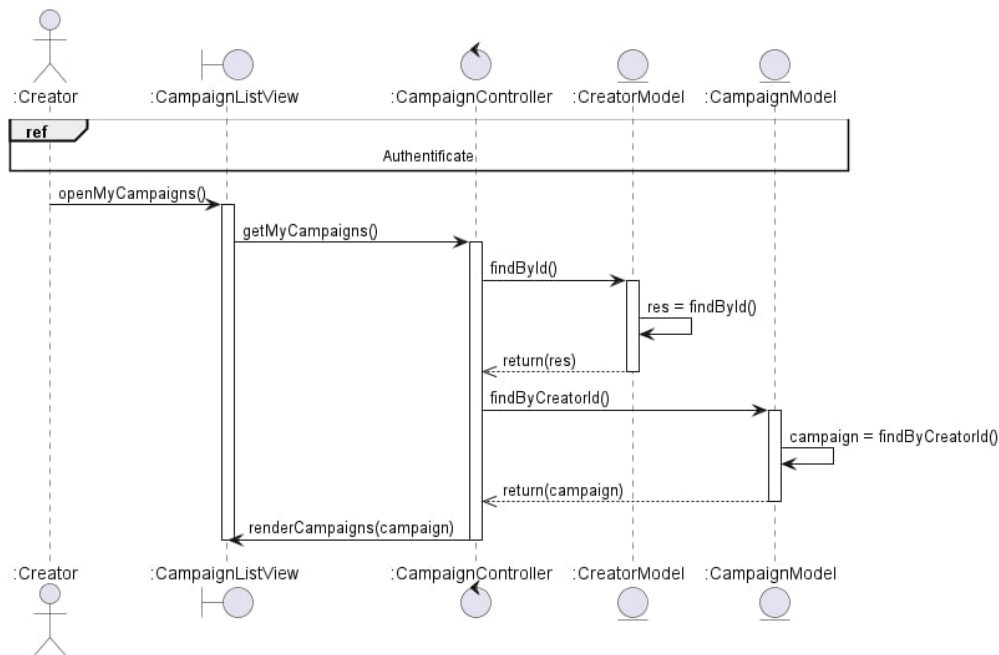


Figure 69: Sequence Diagram – Consult My Campaigns

3.1.2 Create Campaign

Figure 70 illustrates the process of campaign creation by a creator. The interaction starts from the form submission and continues through various system components, including file upload, database operations, brief status update, and notification triggering.

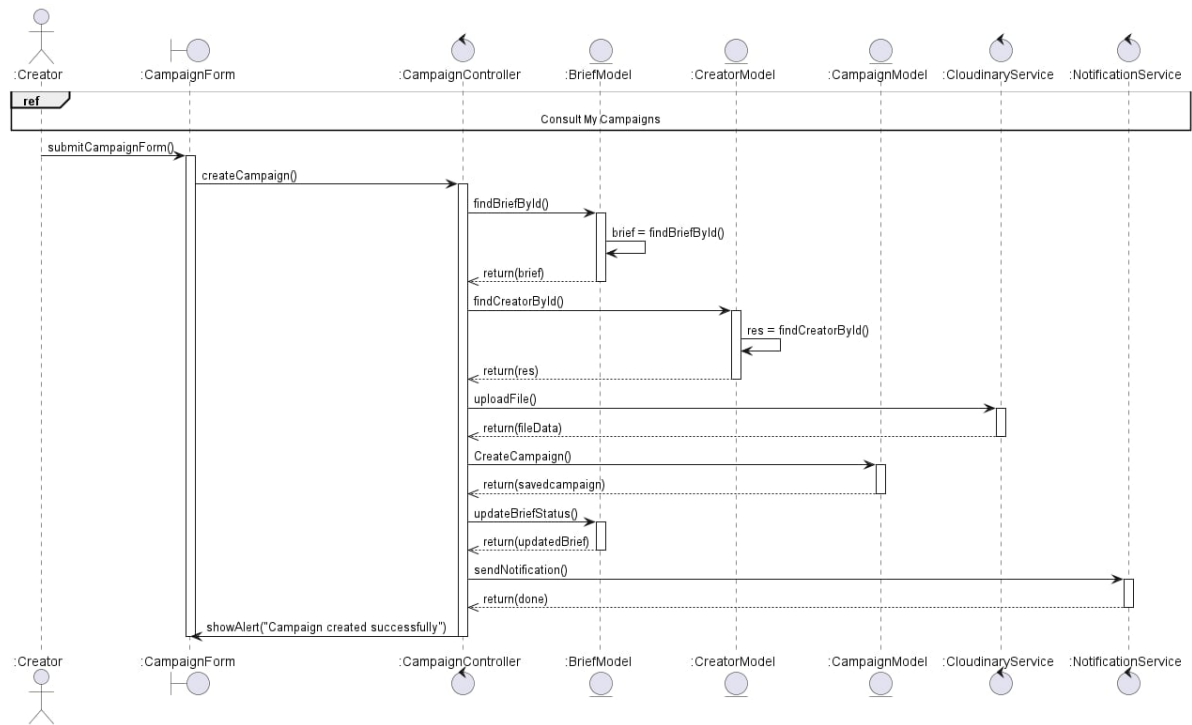


Figure 70: Sequence Diagram – Create Campaign

3.2 Brief-Related Campaign Consultation

Figure 71 illustrates the process through which a brand user consults campaigns that are associated with one of their briefs. After authenticating, the system queries campaigns filtered by brief ID and approval status, then renders the result.

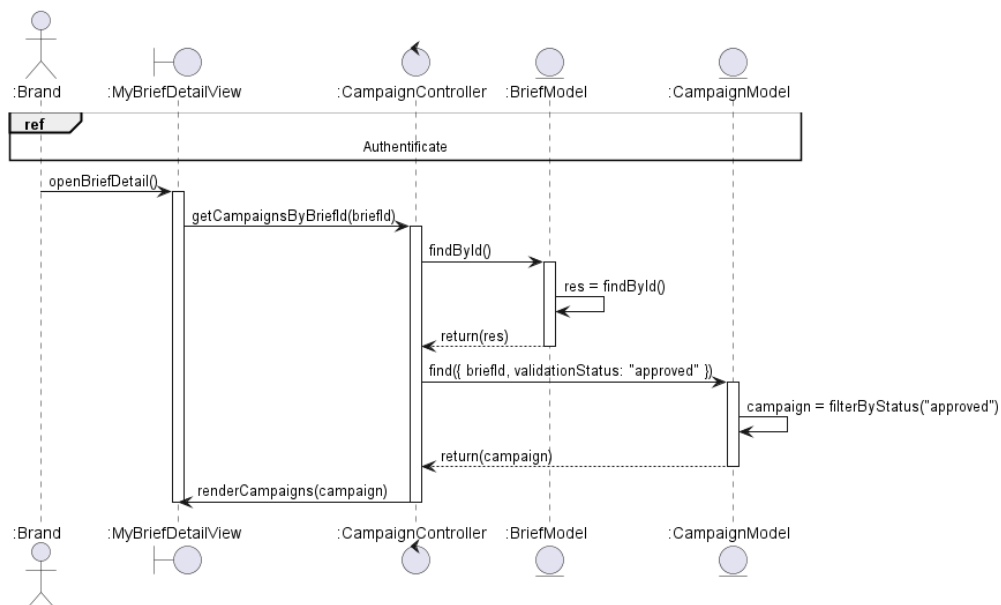


Figure 71: Sequence Diagram – Consult Campaigns Related to a Brief (Brand Perspective)

3.2.1 Consult Campaign Details

Figure 72 depicts the sequence triggered when a creator opens a detailed view of one of their campaigns. The system locates the campaign by its ID and renders the full campaign data for display.

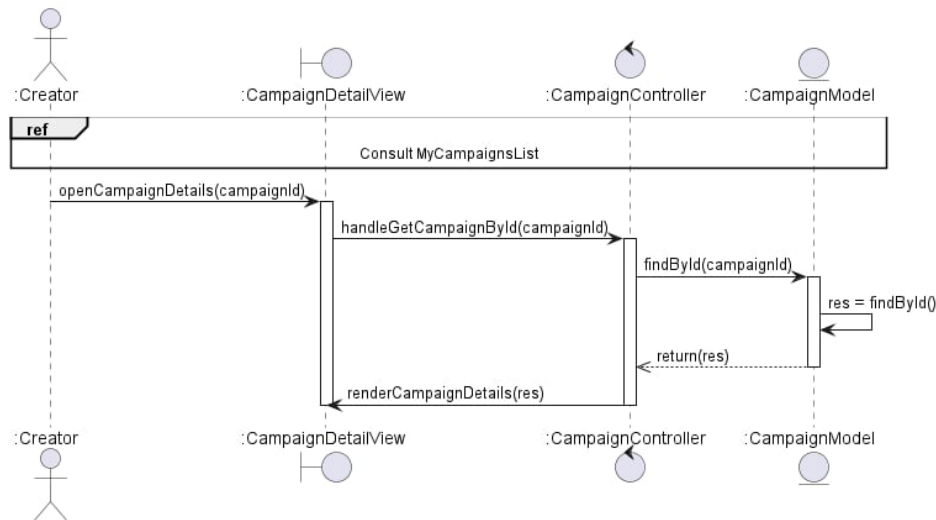


Figure 72: Sequence Diagram – Consult Campaign Details

3.2.2 Delete Campaign

Figure 73 presents the sequence of events when a creator deletes a campaign. After clicking the delete action, the system retrieves the campaign, deletes it from the database, and conditionally removes any associated media files from Cloudinary.

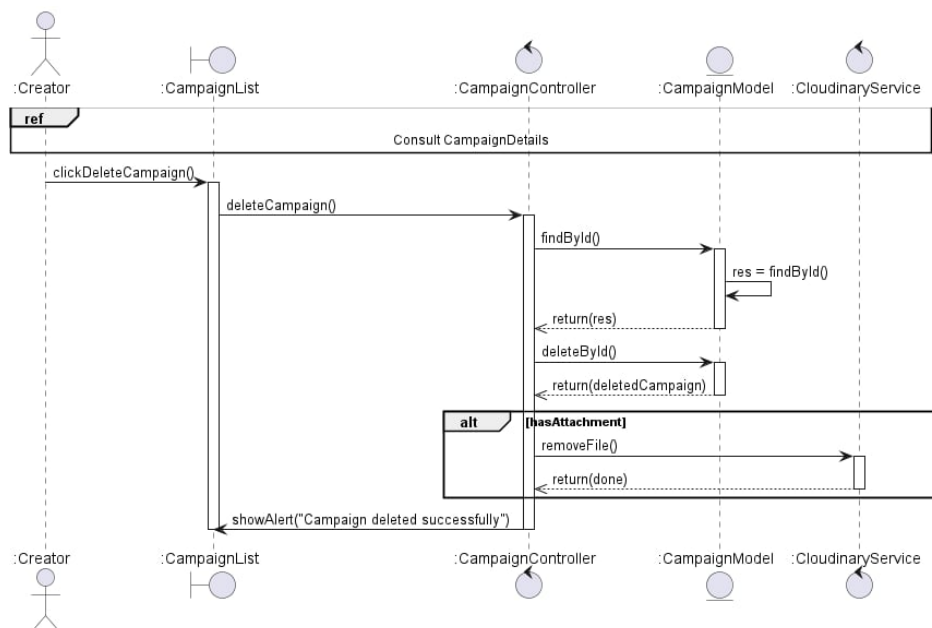


Figure 73: Sequence Diagram – Delete Campaign

3.2.3 Update Campaign

Figure 74 shows the sequence followed when a creator updates a campaign. The controller retrieves the existing campaign data, optionally uploads a new file, updates the campaign record, and triggers a notification to inform relevant stakeholders about the status change.

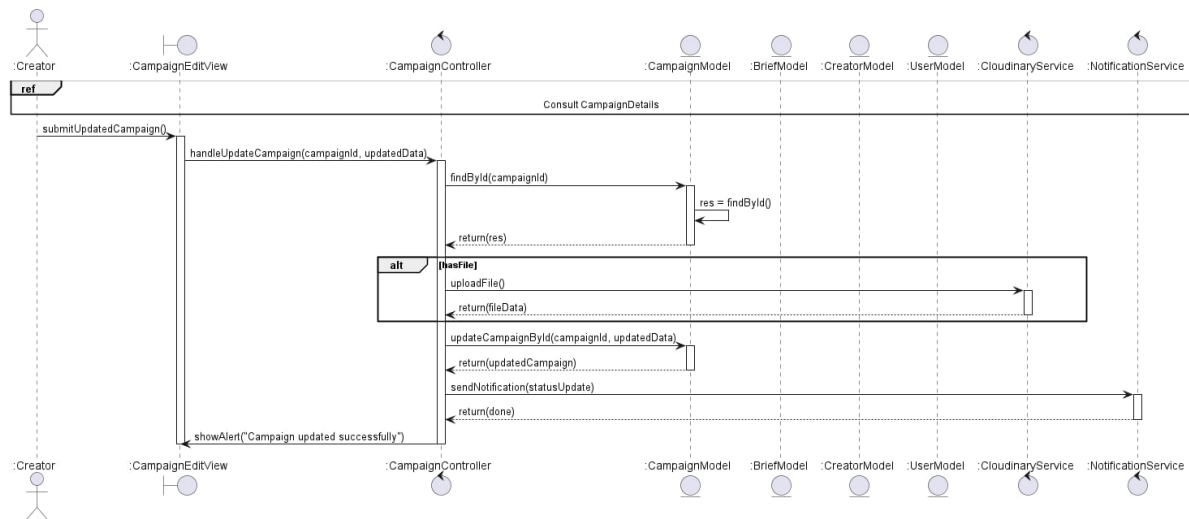


Figure 74: Sequence Diagram – Update Campaign

3.3 Advertiser Campaign Validation

Figure 75 shows how an advertiser can approve or reject a campaign associated with their brief.

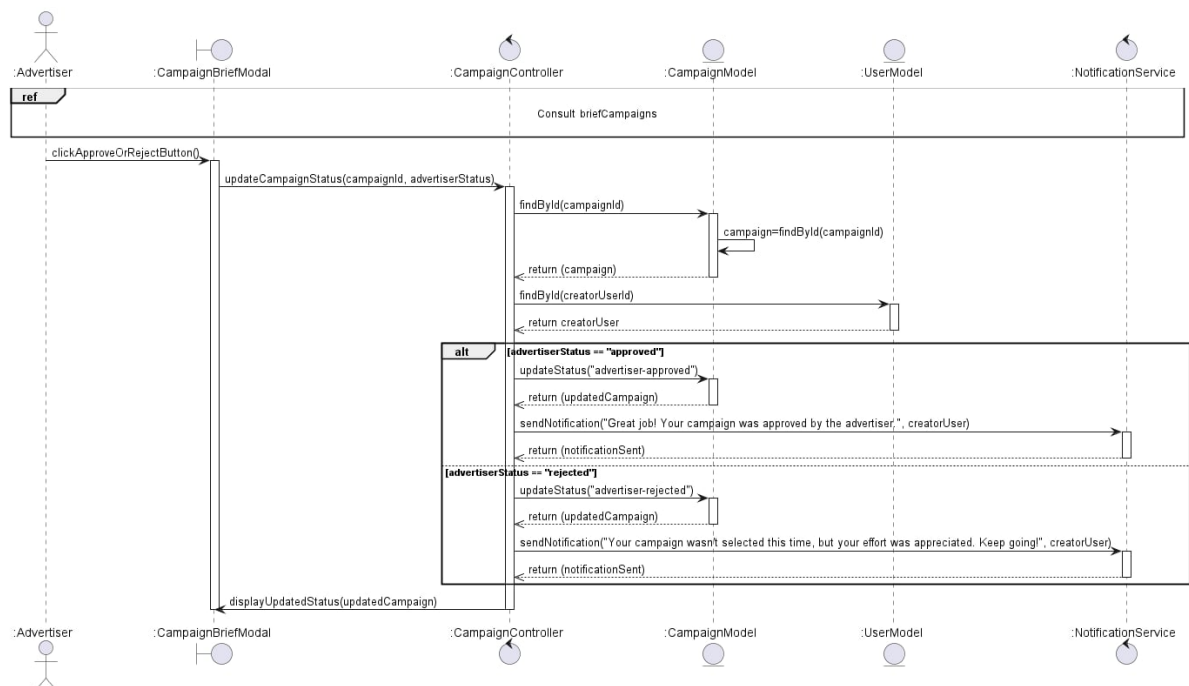


Figure 75: Sequence Diagram – Advertiser Approves or Rejects Campaign

3.4 Scoring System

3.5 Scoring System – Consult Profile Statistics

Figure 76 represents the process through which a user (typically a creator) accesses their profile scoring data. The system retrieves user data, queries the creator model populated with score metrics, and fetches relevant statistics, potentially sourced from external scraping services like Instagram.

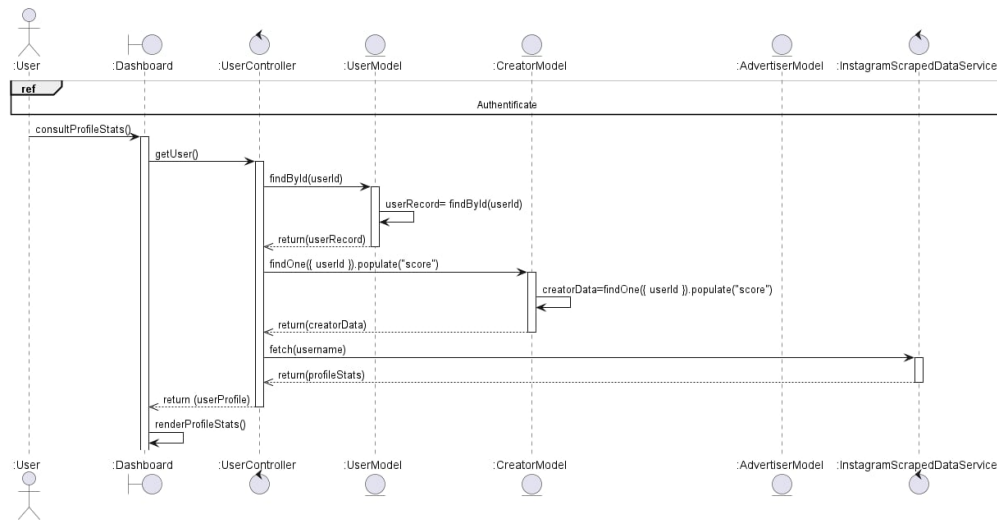


Figure 76: Sequence Diagram – Consult Profile Statistics (Scoring System)

3.5.1 Refresh and Calculate Creator Score

Figure 77 shows the flow triggered when a creator requests a refresh of their score. The system retrieves Instagram data, calculates both quantitative and qualitative scores (potentially via LLM evaluation), and updates the score record in the database.

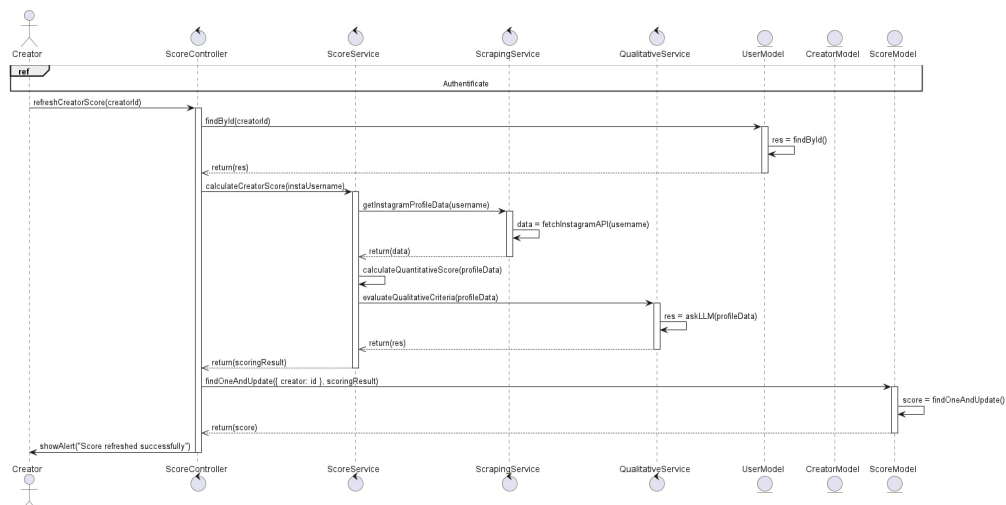


Figure 77: Sequence Diagram – Refresh and Calculate Creator Score

3.6 Admin Panel

The following sequence diagrams describe key administrative functionalities implemented in Sprint 3, including campaign validation and the admin's access to the campaign list.

3.6.1 Change Campaign Status (Approve/Reject)

Figure 78 shows how an admin updates a campaign's status. Depending on whether the campaign is approved or rejected, notifications are sent to the creator and possibly the advertiser, referencing the brief context.

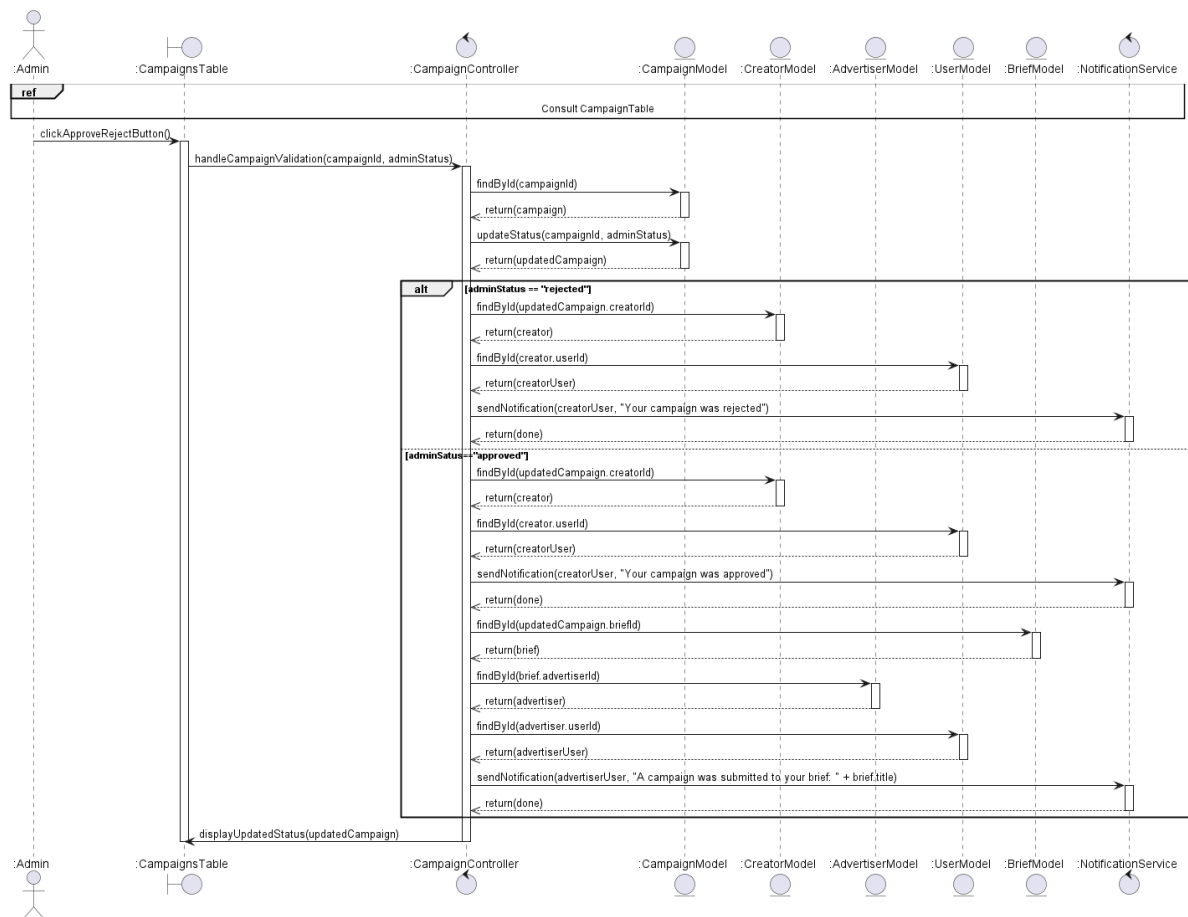


Figure 78: Sequence Diagram – Admin Changes Campaign Status

3.6.2 View Campaigns via Admin Panel

Figure 79 illustrates the process by which an admin logs into the panel and retrieves all campaigns from the system. This enables administrative oversight of campaign activity.

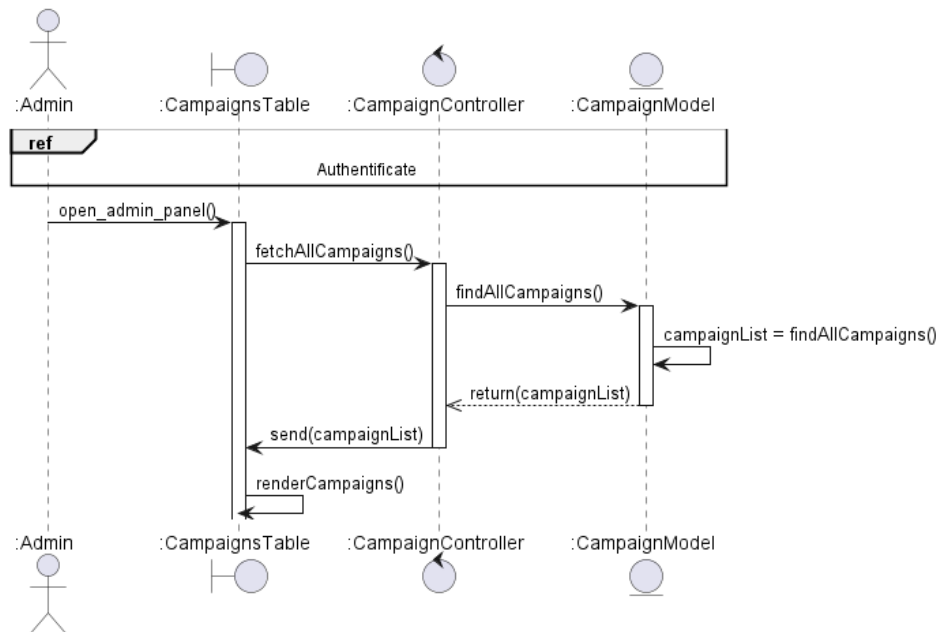


Figure 79: Sequence Diagram – Admin Consults Campaign List

4 Scoring System

The scraping system is built using `Node.js` and is responsible for collecting public data from creators' Instagram profiles. It retrieves key information such as follower count, post engagement (likes, comments), and content captions. This data is then used by two scoring modules: one calculates a quantitative score based on metrics like audience size and posting frequency, while the other uses AI (via OpenAI) to evaluate the quality and influence of the content. Together, these modules generate a global score out of 100 that helps brands assess each creator's value.

4.1 Scraping Layer

To power the scoring system, a scraping service was developed using `Node.js`. This service automates the collection of publicly available data from Instagram profiles, without requiring login credentials. It uses the `node-fetch` library to perform HTTP requests, and `dotenv` to manage configuration and API keys securely.

Data Extracted

Profile-Level Information	Post-Level Information (Latest 12–24 posts)
Username	Number of likes
Follower count	Number of comments
Following count	Caption text
Biography text	Timestamps
Profile picture URL	Thumbnail URLs
Verified badge status	Engagement ratio (likes + comments per post)

Table 45: Scraped Data Categories

The extracted data is structured in JSON format and passed to the scoring modules. It provides the foundation for both quantitative metrics (like engagement rate and posting frequency) and qualitative evaluation (by sending content to the AI-based scoring module).

This scraping layer plays a crucial role in enabling automated, scalable, and objective creator evaluations based on real-time social media activity.

```
PS C:\Users\sgnao\Desktop\ACCESS-PFE\INSTAGRAM-SCRAPESCORE> node index.js cristiano
🔵 Fetching: @cristiano

📁 Profile Info:
  ➤ Full Name: Cristiano Ronaldo
  ➤ Username: @cristiano
  ➤ Followers: 654803081
  ➤ Following: 602
  ➤ Bio: SIUUUsubscribe to my Youtube Channel!
  ➤ Posts Count: 3890
  ➤ Verified: ✅ Yes
  ➤ Private: 🌐 No

🖼️ Recent Posts:
[1] ❤️1272764 💬 25751 📄🔥PT...
[2] ❤️10242145 💬 265374 📄 This chapter is over.
The story? Still being written.
Gratef...
[3] ❤️2792200 💬 65965 📄 Living your best life is all about simplicity. Prioritize qu...
[4] ❤️6604833 💬 164130 📄 One of those nights where the ball just didn't want to go in...
[5] ❤️1527782 💬 142107 📄 You think you know what age I am?
Think again. Watch the rev...
[6] ❤️4396595 💬 648243 📄 Os primeiros SIUUUU de Cristiano Jr. por Portugal! 🇵🇹...
[7] ❤️4296854 💬 437547 📄 Have you heard the news? @uflgame got a big update recently!...
[8] ❤️12582238 💬 97471 📄 Parabéns pela estreia por @portugal, filho. Muito orgulho em...
[9] ❤️1953532 💬 65082 📄 What you do today can add more tomorrows. The all-new WHOOP ...
[10] ❤️4007712 💬 72470 📄 Wake up to a better way—tune in to @whoop Thursday at 10am E...
[11] ❤️12584975 💬 77434 📄 Todos os dias são delas, mas hoje em especial ❤️Feliz dia d...
[12] ❤️6915096 💬 62846 📄 Sometimes the dream has to wait. I'm proud of this team and ...
```

Figure 80: Example Output from the Scraping Script (Instagram Profile: @cristiano)

4.2 Scoring Overview

After scraping, the extracted data is processed to generate a global score for each creator. This score ranges from 0 to 100 and helps brands quickly assess the value and professionalism of a creator. The score is divided into two main components:

4.2.1 Quantitative Scoring

This part of the score evaluates measurable performance metrics using the data collected from the creator's profile and posts. It is calculated using the following three criteria:

- **Audience Metrics (15 pts)** – Based on the number of followers, growth trends, and follower quality. Scores are normalized against a benchmark (e.g., 1 million followers).
- **Engagement Metrics (15 pts)** – Calculated as the average number of likes and comments per post, divided by total followers (engagement rate). Also considers comment richness and interaction velocity.
- **Regularity Metrics (10 pts)** – Measures posting frequency and consistency. An optimal interval receives a higher score. Responding to comments and consistent posting days also contribute.

4.2.2 Qualitative Scoring

The qualitative score evaluates the overall content quality, professionalism, and influence of a creator's social media presence. This component relies on natural language processing and AI analysis, using the OpenAI GPT-3.5-turbo model to interpret post captions and metadata.

Each component is computed separately and then combined to form the final score, which is displayed on the creator's profile page within the platform.

- **Content Quality (20 pts)** – Assesses visual quality, editing, creativity, audio, lighting, and storytelling.
- **Expertise and Authority (15 pts)** – Evaluates demonstrated knowledge, collaborations, awards, and niche credibility.
- **Influence and Impact (15 pts)** – Based on the type of comments received, ability to generate meaningful discussions, and behavioral influence.
- **Professionalism and Branding (10 pts)** – Measures visual consistency, clear messaging, partnership readiness, and adherence to best practices.

Each of these subcomponents is scored individually using AI-generated evaluations and then summed to obtain a final qualitative score out of 60.

4.3 AI Evaluation Layer

The platform uses the OpenAI GPT-3.5-turbo model to analyze creator content and assign scores for qualitative criteria. Captions and metadata are sent to the model using custom prompts in French, depending on the dimension being evaluated (content, expertise, influence, or branding). This ensures consistent and scalable evaluation across all creators.

```
function createPrompt(type, text) {
  const instructions = {
    content: `
Évalue la qualité du contenu sur 20 points.
Critères :
- Qualité technique (résolution, éclairage, montage, stabilité, qualité audio)
- Créativité et originalité
- Capacité de storytelling
Réponds uniquement avec un nombre entre 0 et 20 :`,

    expertise: `
Évalue l'expertise et l'autorité thématique sur 15 points.
Critères :
- Connaissances démontrées dans les publications
- Collaborations notables ou reconnues
- Récompenses, prix ou certifications
Réponds uniquement avec un nombre entre 0 et 15 :`,

    influence: `
Évalue l'impact et l'influence sur 15 points.
Critères :
- Qualité des interactions et des commentaires
- Capacité à générer des conversations
- Influence sur les comportements ou tendances
Réponds uniquement avec un nombre entre 0 et 15 :`,

    branding: `
Évalue le professionnalisme et la marque personnelle sur 10 points.
Critères :
- Cohérence visuelle du contenu
- Communication claire et régulière
- Gestion des partenariats et respect des bonnes pratiques
Réponds uniquement avec un nombre entre 0 et 10 :`
  };

  return `${instructions[type]}\n\nContenu analysé : \n${text.slice(0, 1500)};
}
```

Figure 81: Prompt Construction Logic for Qualitative Scoring

5 Implementation

5.1 Campaign Creation – Creator Workflow

Step 1: Brief Approval and Start Campaign

Once a brief is accepted by the admin, the creator sees a confirmation and gains access to create a campaign for that specific brief.

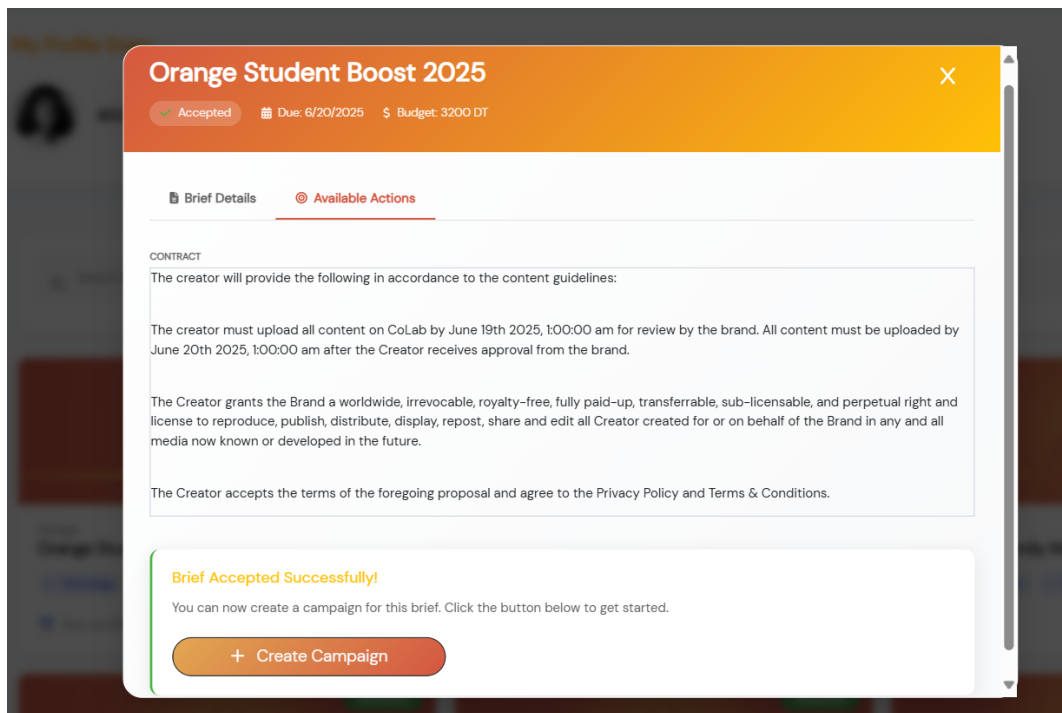


Figure 82: Brief Accepted – Create Campaign Access

Step 2: Fill Campaign Details

Creators can describe their campaign concept and optionally attach a file (e.g., video content or media asset).

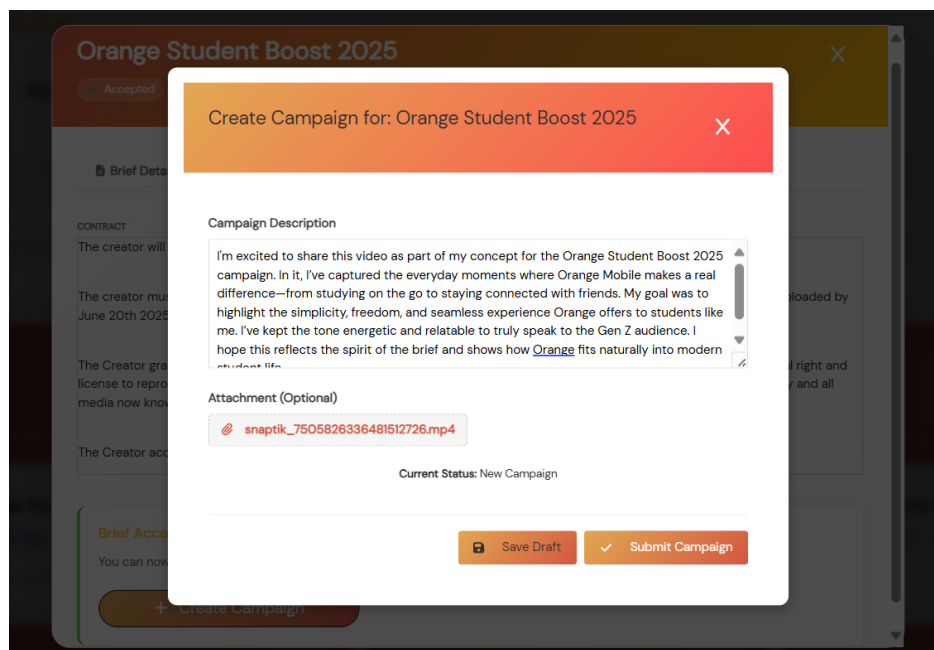


Figure 83: Create Campaign – Description and Attachment

Step 3: Confirm Submission

Before submission, creators are prompted to confirm and warned that no edits will be allowed post-submission.

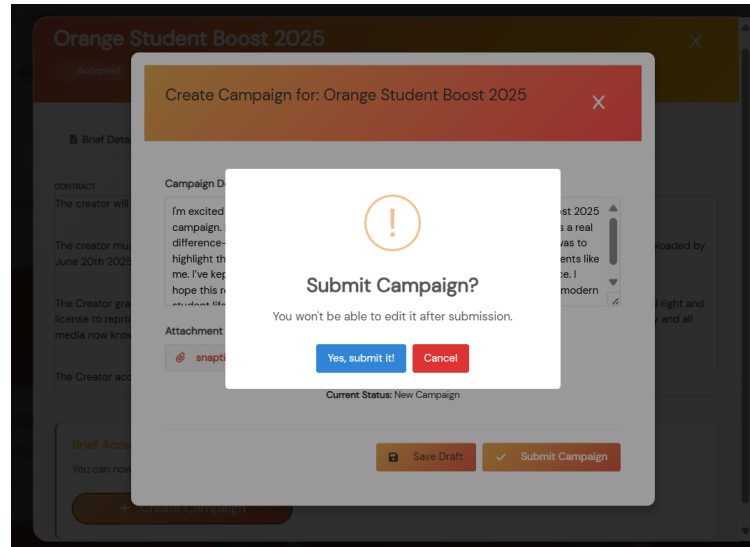


Figure 84: Campaign Submission Confirmation

Step 4: Submission Acknowledgment

A confirmation message notifies the creator that the campaign has been successfully submitted.

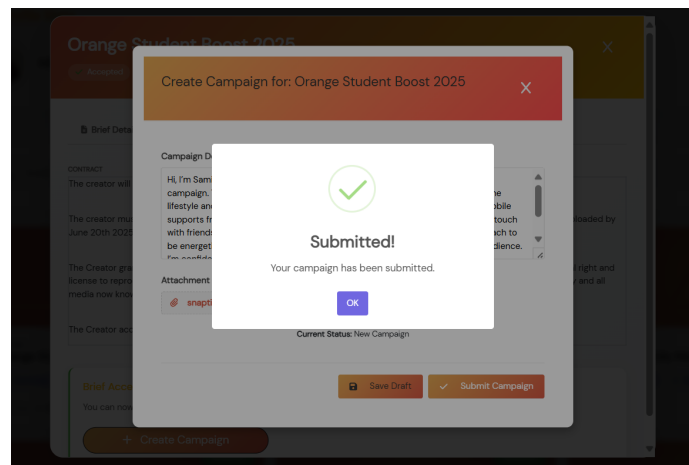


Figure 85: Campaign Submitted Successfully

Creator Campaign Dashboard

Creators can consult a list of their submitted campaigns from their dashboard. Each campaign card includes the title, description snippet, due date, and status (e.g., approved).

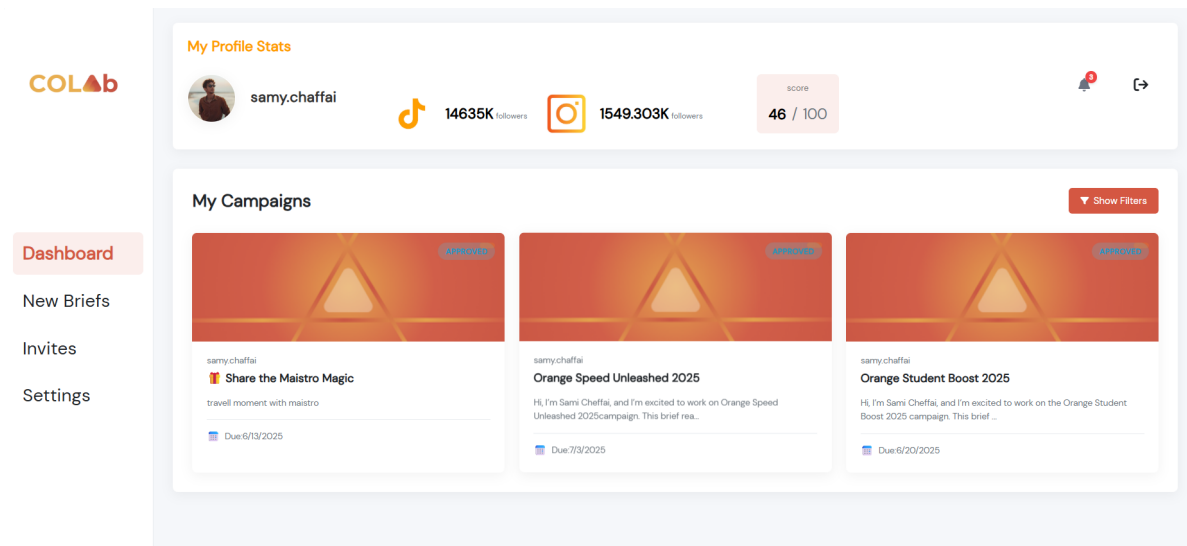


Figure 86: Creator – My Campaigns Overview

View Campaign Submissions

Brands can view all campaign submissions linked to their briefs. Each campaign card displays creator info, video preview, description, and payment status. Campaigns can be reviewed from the “Campaigns” tab within the brief details.

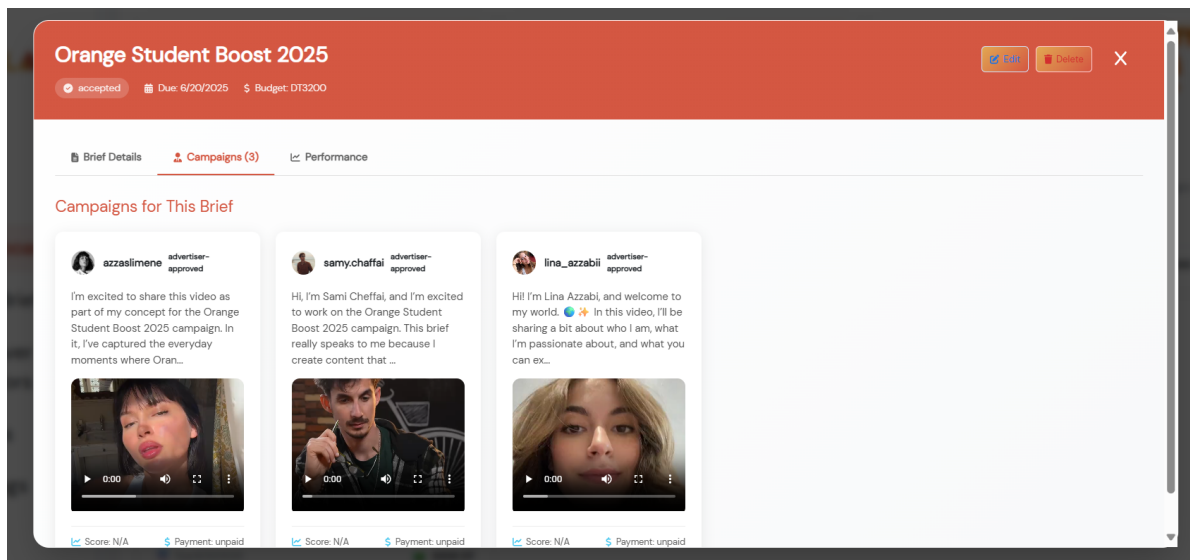


Figure 87: Brand – View Submitted Campaigns for a Brief

Consult Submitted Campaigns

Creators can review the full details of their submitted campaigns. This includes a structured view of the campaign description, brand requirements, associated tags, and deliverables.

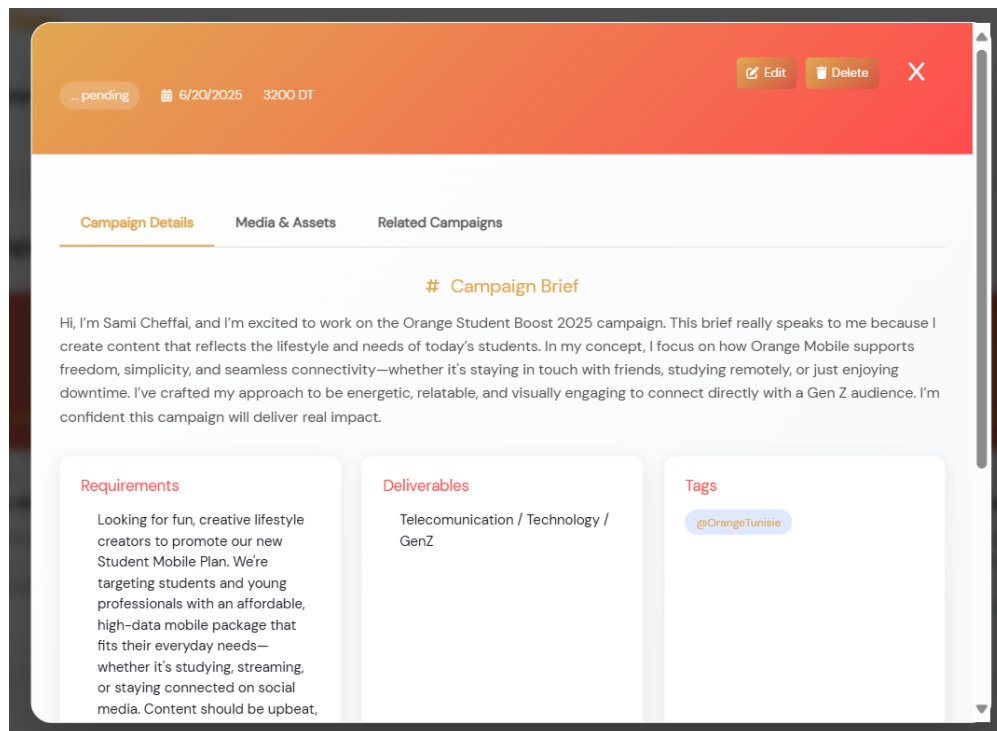


Figure 88: Creator – View Campaign Details

Access Uploaded Media

Each creator can view the video or content they uploaded under the “Media Assets” tab, enabling easy reference or presentation.

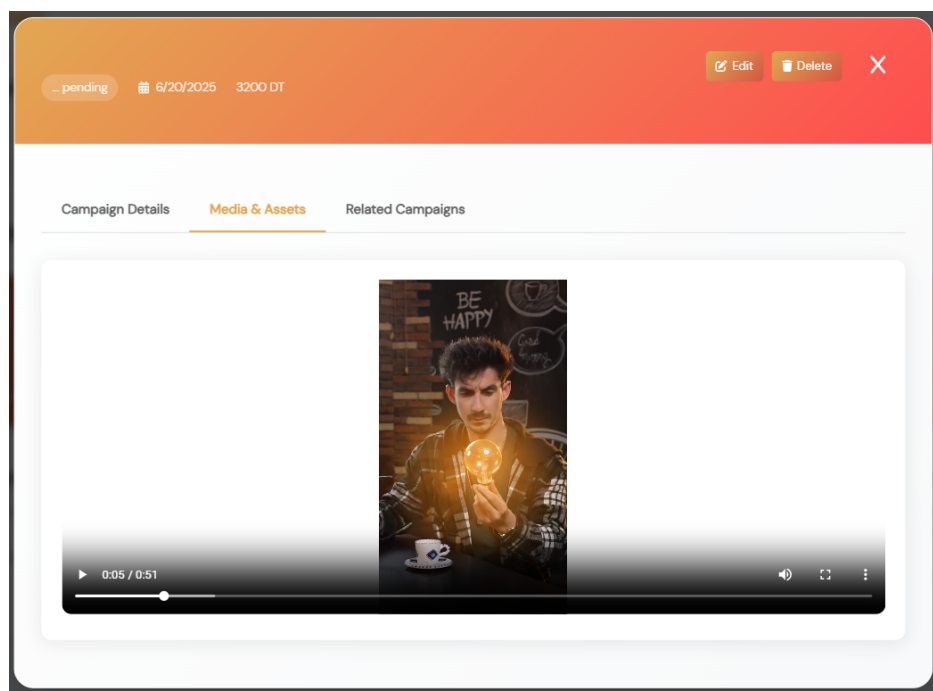


Figure 89: Creator – Uploaded Media Preview

Related Campaigns

A recommendation section shows similar campaigns for inspiration or insight.

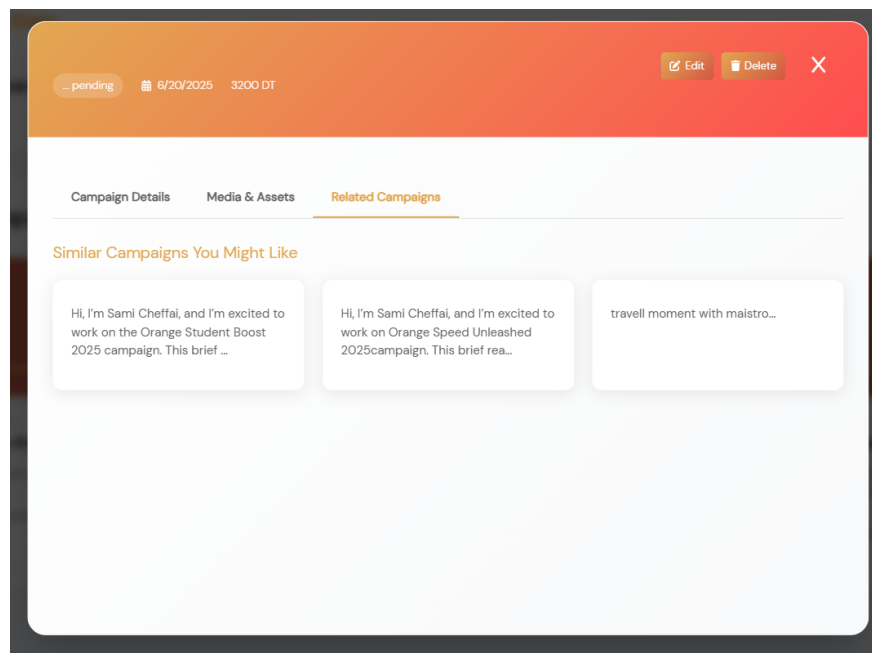


Figure 90: Creator – Related Campaign Suggestions

Update Draft Campaign

Creators can edit their submitted campaigns if the status is still pending. They can revise the campaign description and optionally update the attached media file.

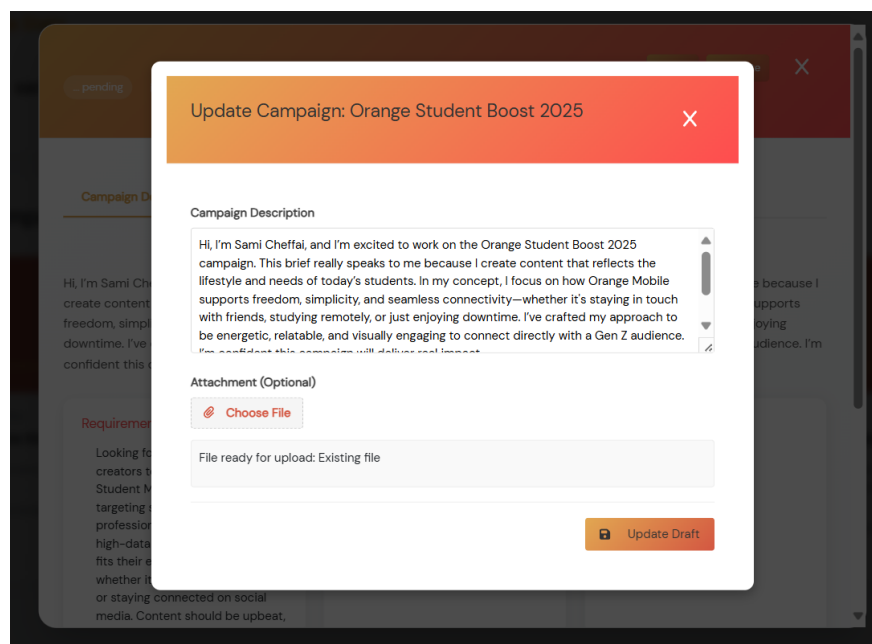


Figure 91: Creator – Update Campaign Form

Delete Campaign Option

If necessary, creators can choose to delete their draft campaigns. A confirmation dialog ensures the user is aware of the irreversible action.

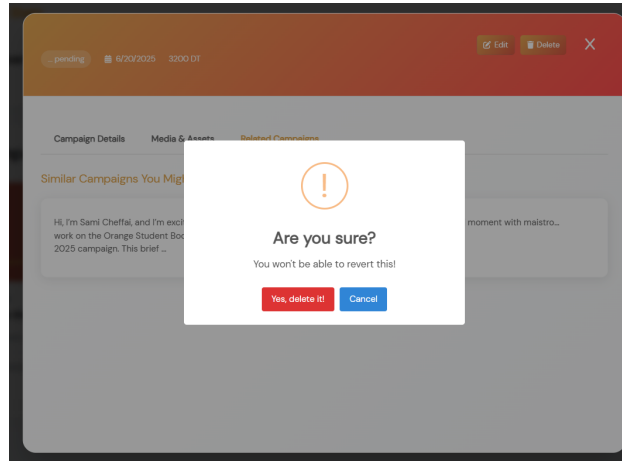


Figure 92: Creator – Campaign Deletion Confirmation

Review Submitted Campaigns

Brands can review the media submitted by creators under the "Media Assets" tab. Once reviewed, they have the option to approve or reject the campaign with a single click.

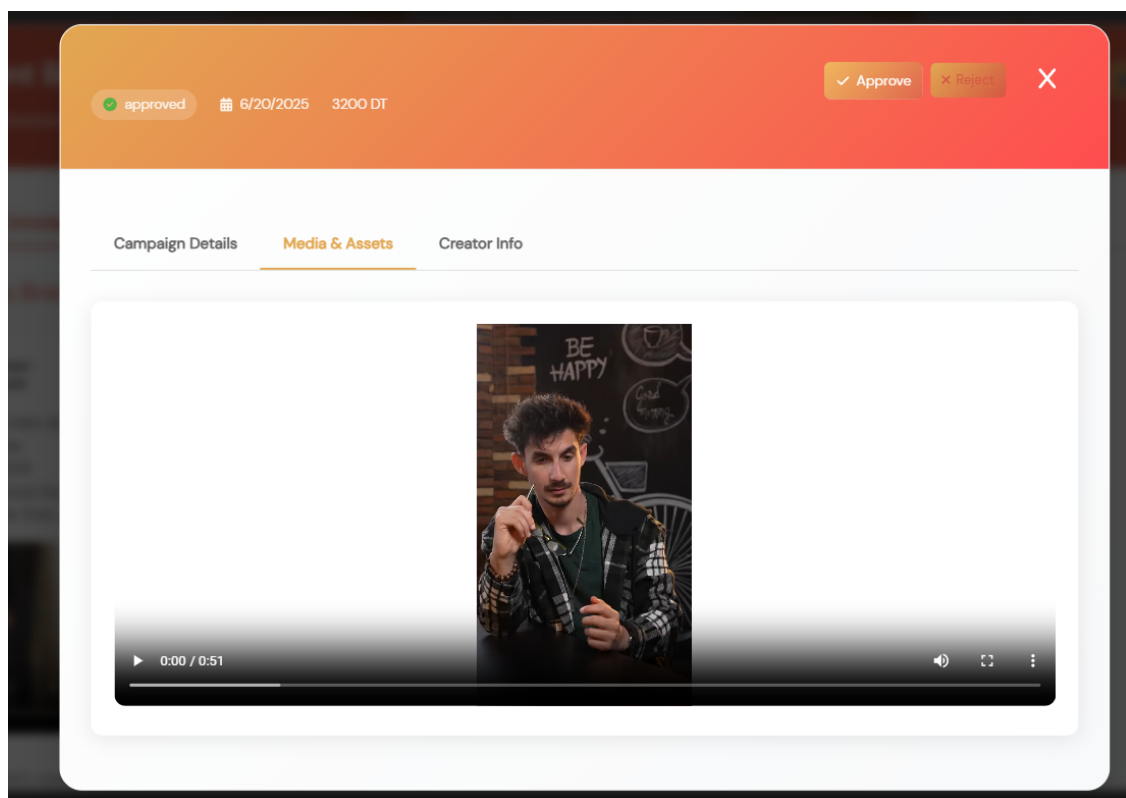


Figure 93: Brand – Approve or Reject Submitted Campaign

Advertiser Feedback Status

Each submitted campaign displays its review status—either approved or rejected—by the advertiser, helping creators track progress.

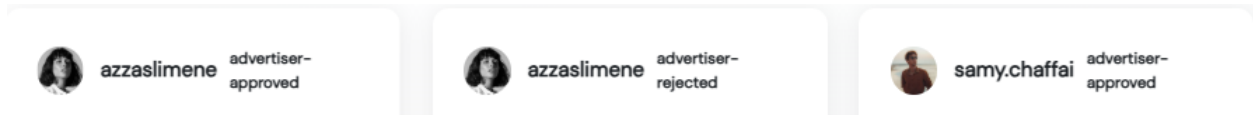


Figure 94: Campaign Status – Approved or Rejected by Advertiser

5.2 Admin Panel – Campaign Management

Admins view all submitted campaigns in a centralized table showing the brief title, creator, status, payment, submission date, and actions (view, edit, approve, reject, delete).

The screenshot shows the 'Admin Panel' on the left with a sidebar menu containing 'Dashboard', 'Briefs', 'Campaigns' (highlighted), 'Creators', 'Advertisers', and 'Categories'. The main area is titled 'Campaigns Management' and 'Manage Campaigns'. It features a search bar and a table with columns: Brief Title, Creator, Status, Score, Payment Status, Submitted At, and Actions. The table lists five campaigns with various statuses (approved, pending) and payment statuses (unpaid). Each row has five action icons: a magnifying glass, a pencil, a checkmark, an 'x', and a trash can. A 'Page 1 of 1' indicator is at the bottom of the table.

Brief Title	Creator	Status	Score	Payment Status	Submitted At	Actions
Orange Student Boost 2025	azzaslimene	approved	N/A	unpaid	6/6/2025	
Orange Student Boost 2025	azzaslimene	pending	N/A	unpaid	6/6/2025	
Orange Student Boost 2025	samy.chaffai	pending	N/A	unpaid	6/6/2025	
Orange Speed Unleashed 2025	samy.chaffai	approved	N/A	unpaid	6/6/2025	
Share the Maestro Magic	samy.chaffai	pending	N/A	unpaid	6/6/2025	

Figure 95: Admin Panel – Manage Campaigns

General Conclusion

At the end of this final-year project, we successfully delivered the first functional version of Colab, a collaborative platform designed to streamline interactions between advertisers and content creators. Guided by the SCRUM methodology, we completed three major sprints that covered user management, brief lifecycle, and campaign handling—while also integrating an automated scoring system based on both quantitative and qualitative criteria.

Throughout this journey, we gained valuable technical skills in full-stack web development (*ReactJS*, *NodeJS*, *MongoDB*), agile project management, and external service integration (e.g., *Cloudinary*, *OpenAI*). More importantly, we learned how to design a solution aligned with real business needs, ensuring performance, scalability, and security.

Despite the progress made, some key features defined in the original project scope remain either partially implemented or in progress:

- Full integration of official **Instagram** and **TikTok APIs** for secure and compliant data collection.
- A **content marketplace** where creators can actively offer their services to advertisers.
- Deployment of **predictive AI algorithms** to anticipate performance trends and optimize scoring.
- A **messaging system** for real-time communication between platform users.
- A **mutual rating mechanism** for structured feedback between brands and creators.

These additions will elevate Colab into a smarter, more interactive, and professional-grade platform for influencer marketing.

We are fully committed to continuing this development **in collaboration with ACCESS**, the host company that supported and mentored us throughout the project. We are eager to work on the next development phases, helping to refine and expand Colab into a fully deployable, production-ready platform. Our shared vision is to bring to market a robust, intelligent, and scalable tool that empowers both advertisers and creators to build more meaningful and successful collaborations.

This project has been a transformative experience for us both professionally and personally. It has shown us what it means to build a real product with purpose, to solve complex challenges, and to work as a team. We are proud of what has been accomplished, and even more excited about what lies ahead.

Webography

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